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Jeon et al.

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(54) **RANDOM ACCESS TYPE SELECTION FOR SMALL DATA TRANSMISSION**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

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Related U.S. Application Data

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(60) Provisional application No. 63/104,218, filed on Oct. 22, 2020.

(51) **Int. Cl.**

H04W 74/08 (2024.01)
H04W 74/0833 (2024.01)
H04W 76/20 (2018.01)

(52) **U.S. Cl.**

CPC ... **H04W 74/0841** (2013.01); **H04W 74/0866** (2013.01); **H04W 76/20** (2018.02)

(58) **Field of Classification Search**

CPC **H04W 74/0841**; **H04W 76/20**; **H04W 74/0866**

See application file for complete search history.

(56)

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Primary Examiner — James P Duffy

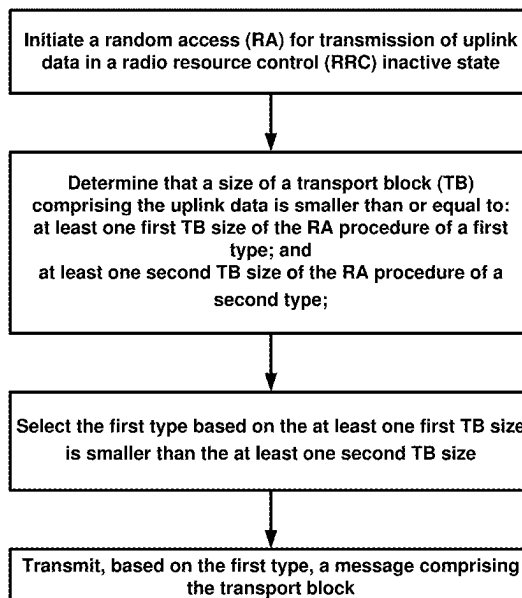
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(57)

ABSTRACT

A wireless device receives configuration parameters indicating a first transport block (TB) size threshold of a first random access (RA) type and indicating a second TB size threshold of a second RA type. The wireless device selects, for transmitting a data unit, an RA type among the first RA type and the second RA type. The RA type is selected based on a comparison among a size of the data unit, the first TB size threshold, and the second TB size threshold. A transport block with the data unit is transmitted using the RA procedure associated with the selected RA type.

20 Claims, 26 Drawing Sheets



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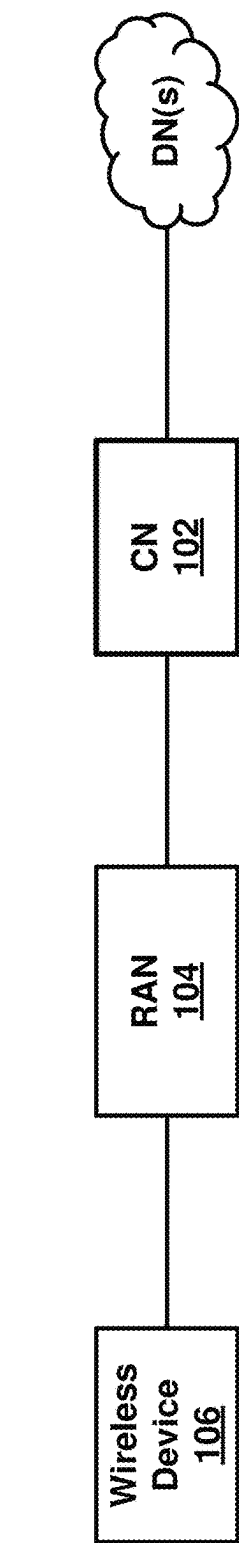


FIG. 1A

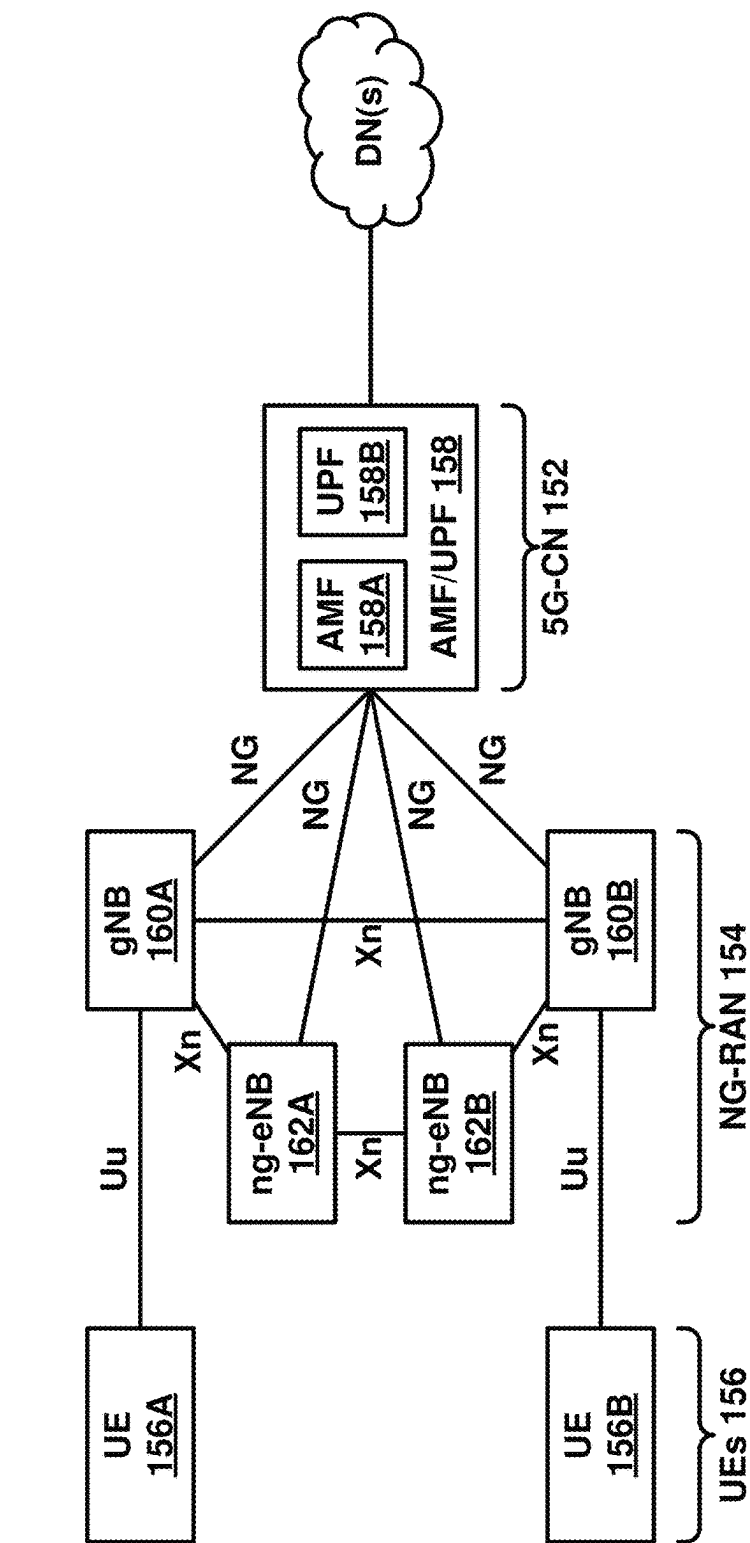


FIG. 1B

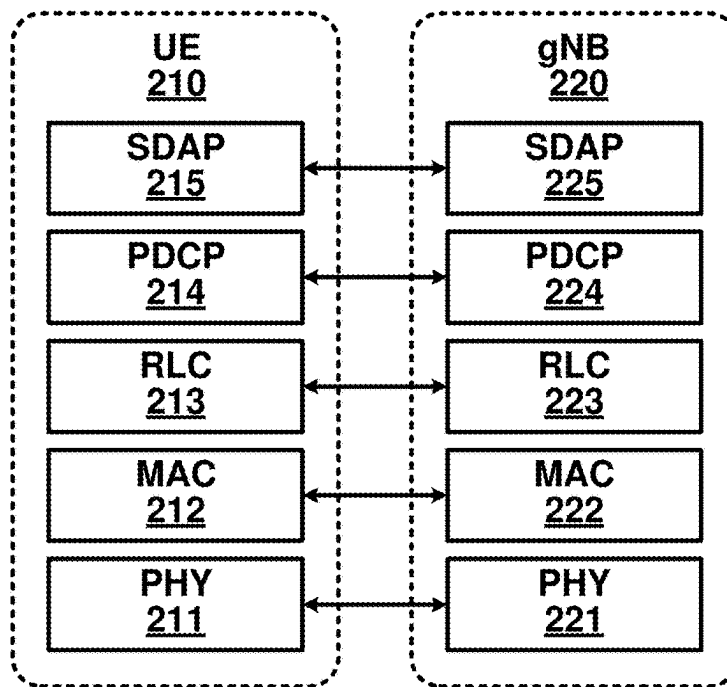


FIG. 2A

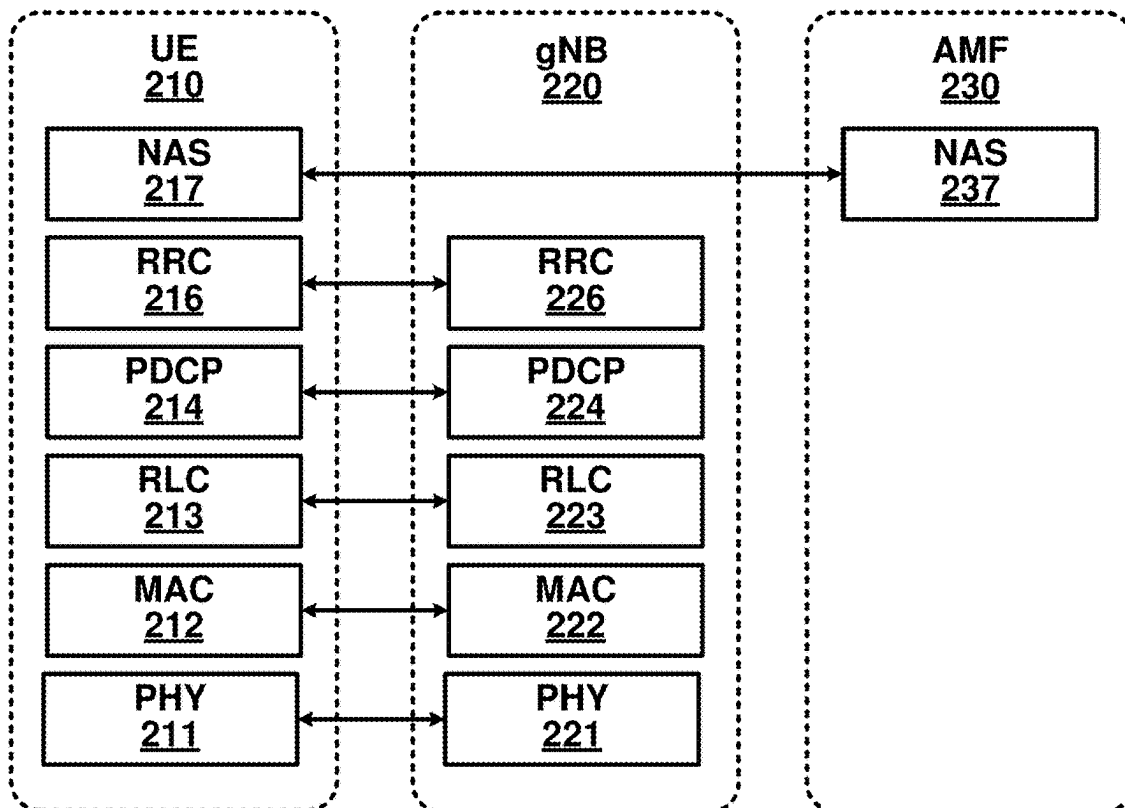


FIG. 2B

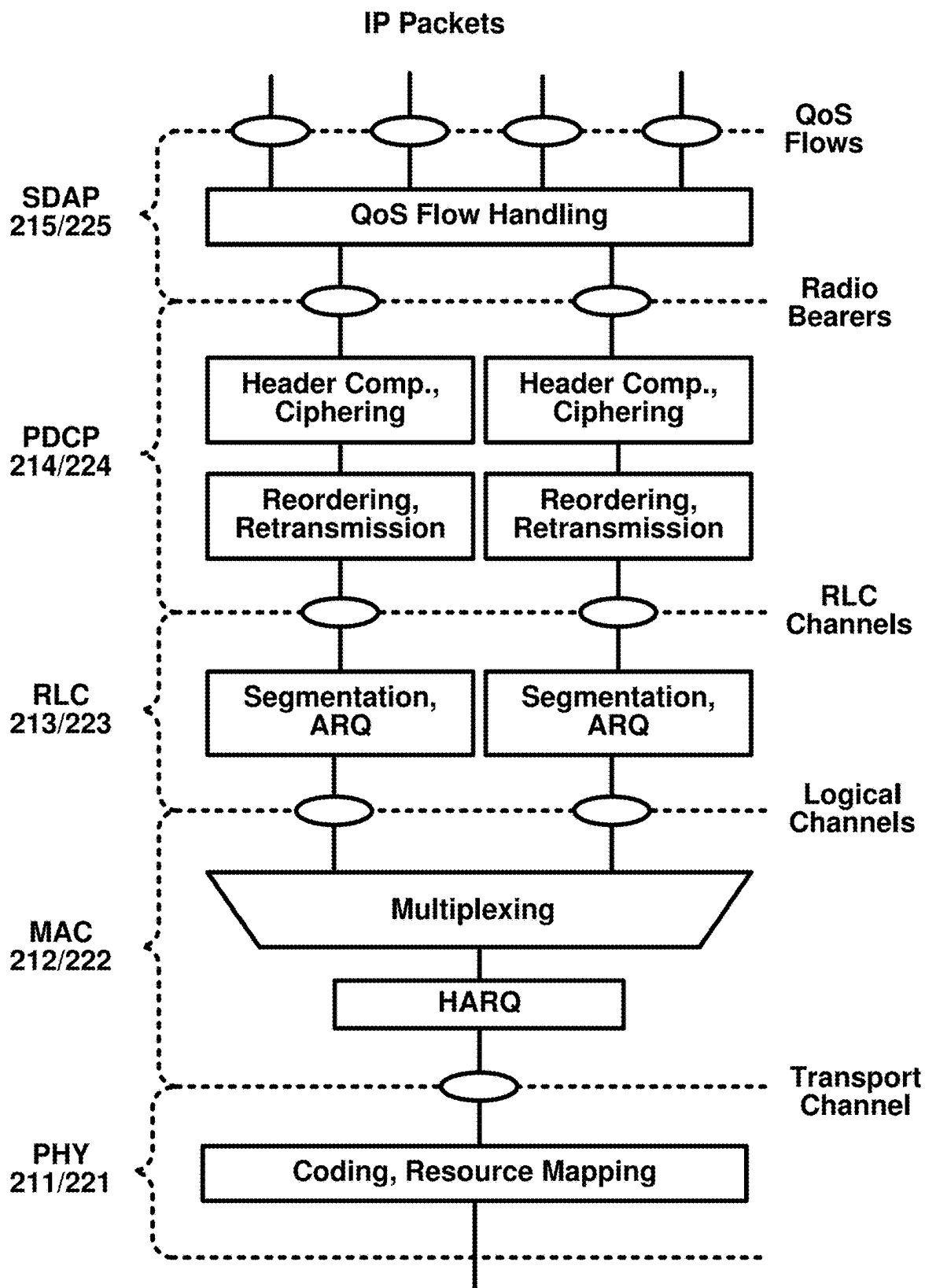


FIG. 3

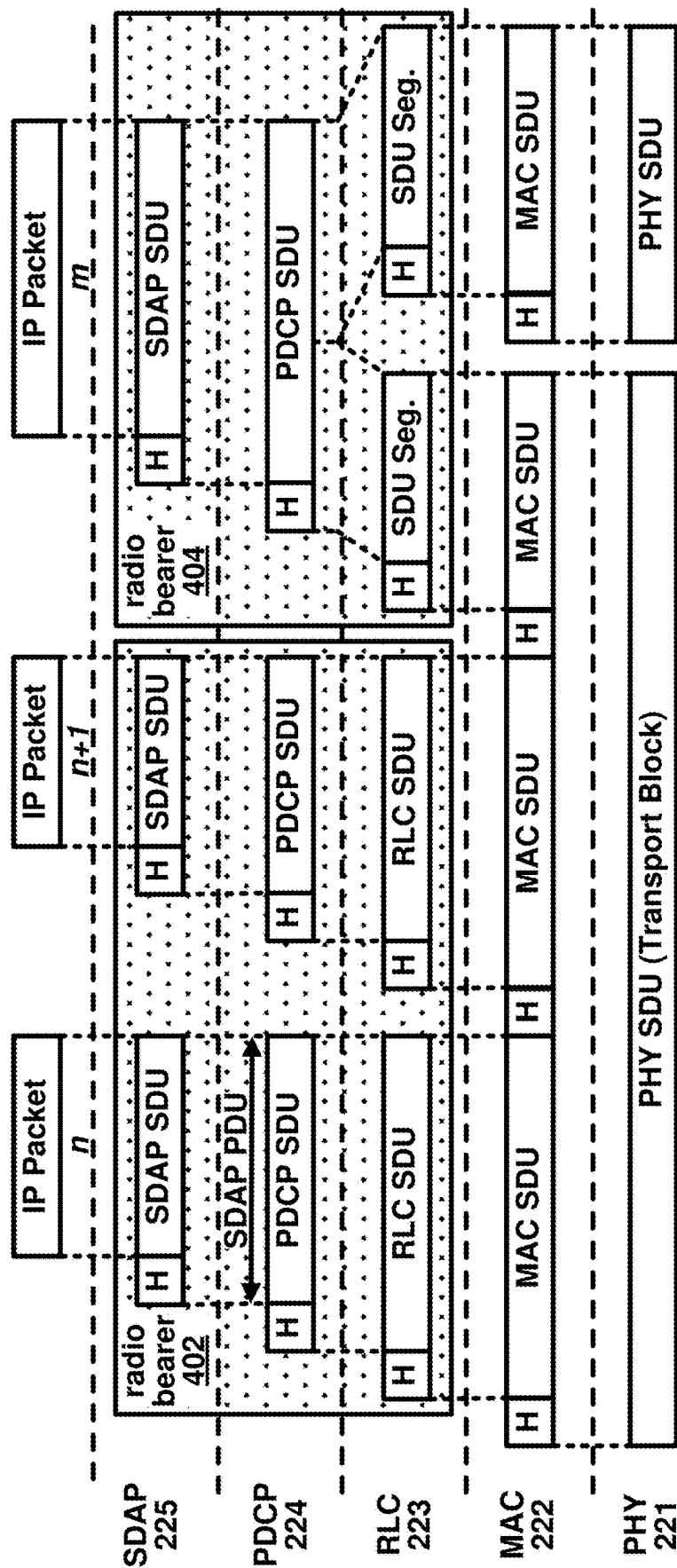


FIG. 4A

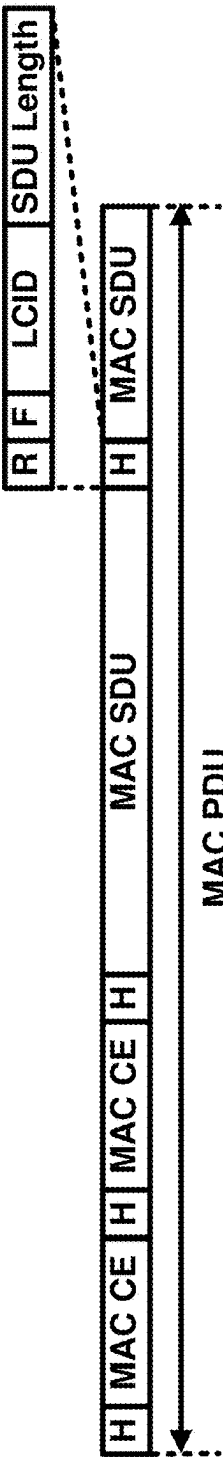
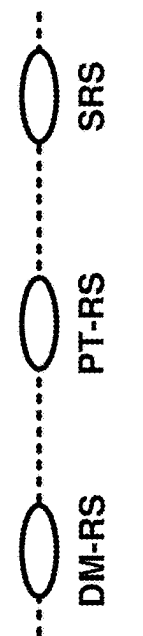
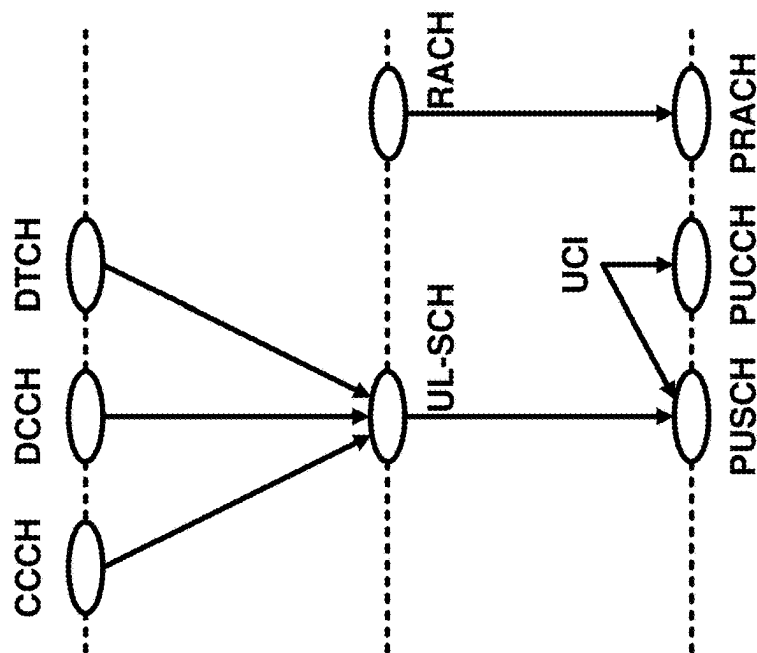
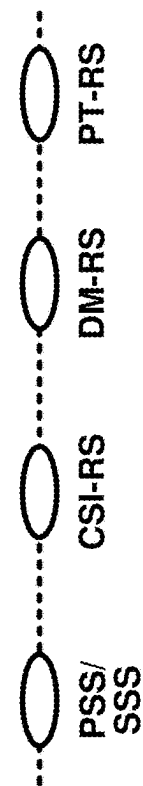
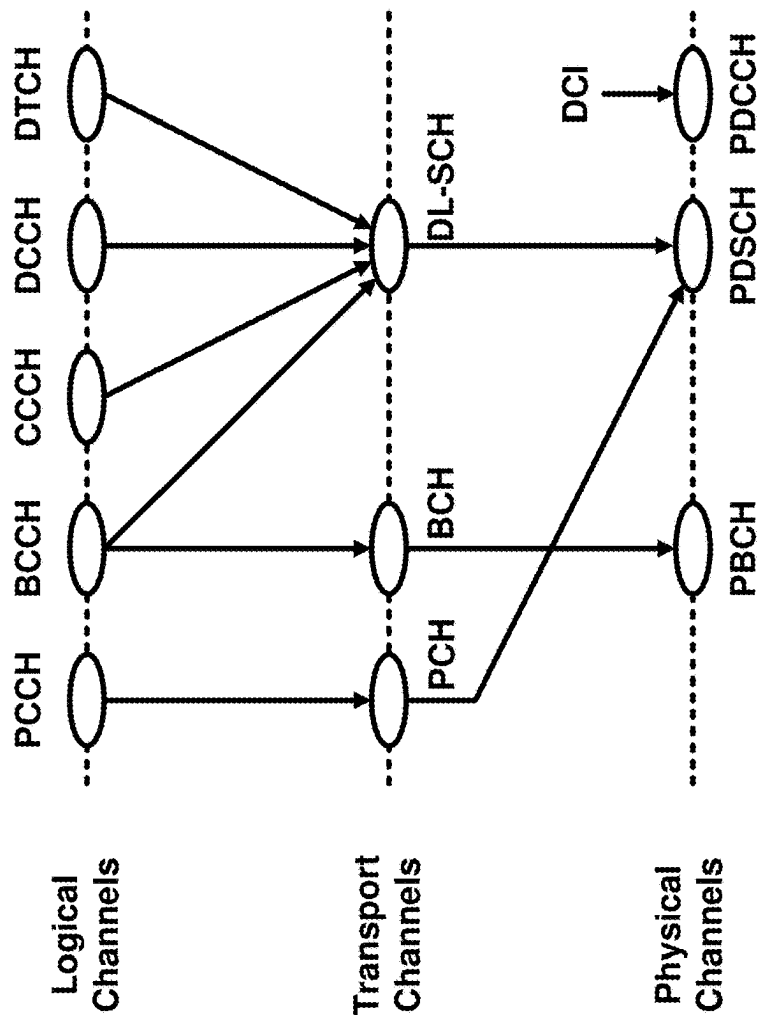


FIG. 4B



Downlink

FIG. 5A



Uplink

FIG. 5B

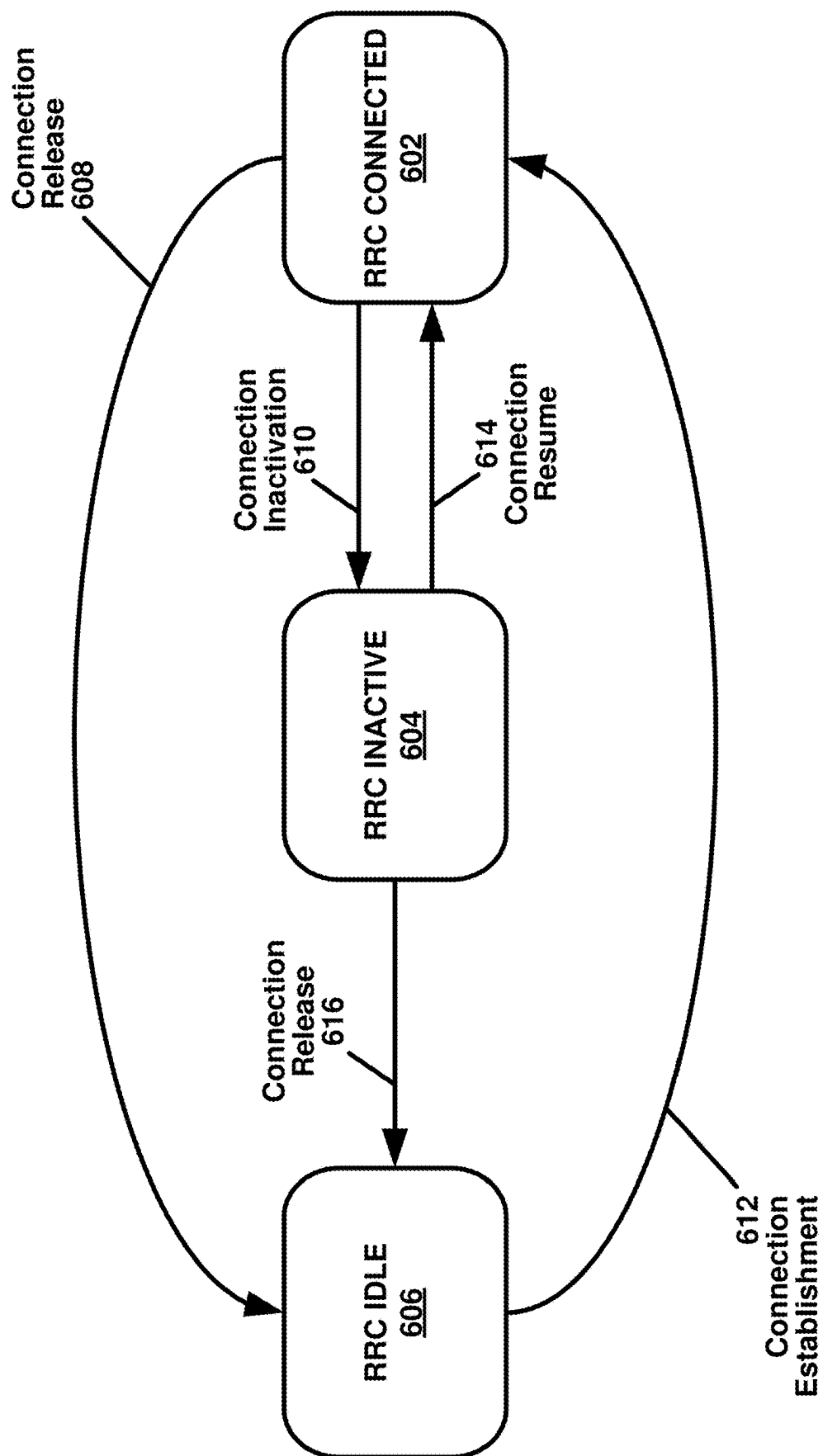


FIG. 6

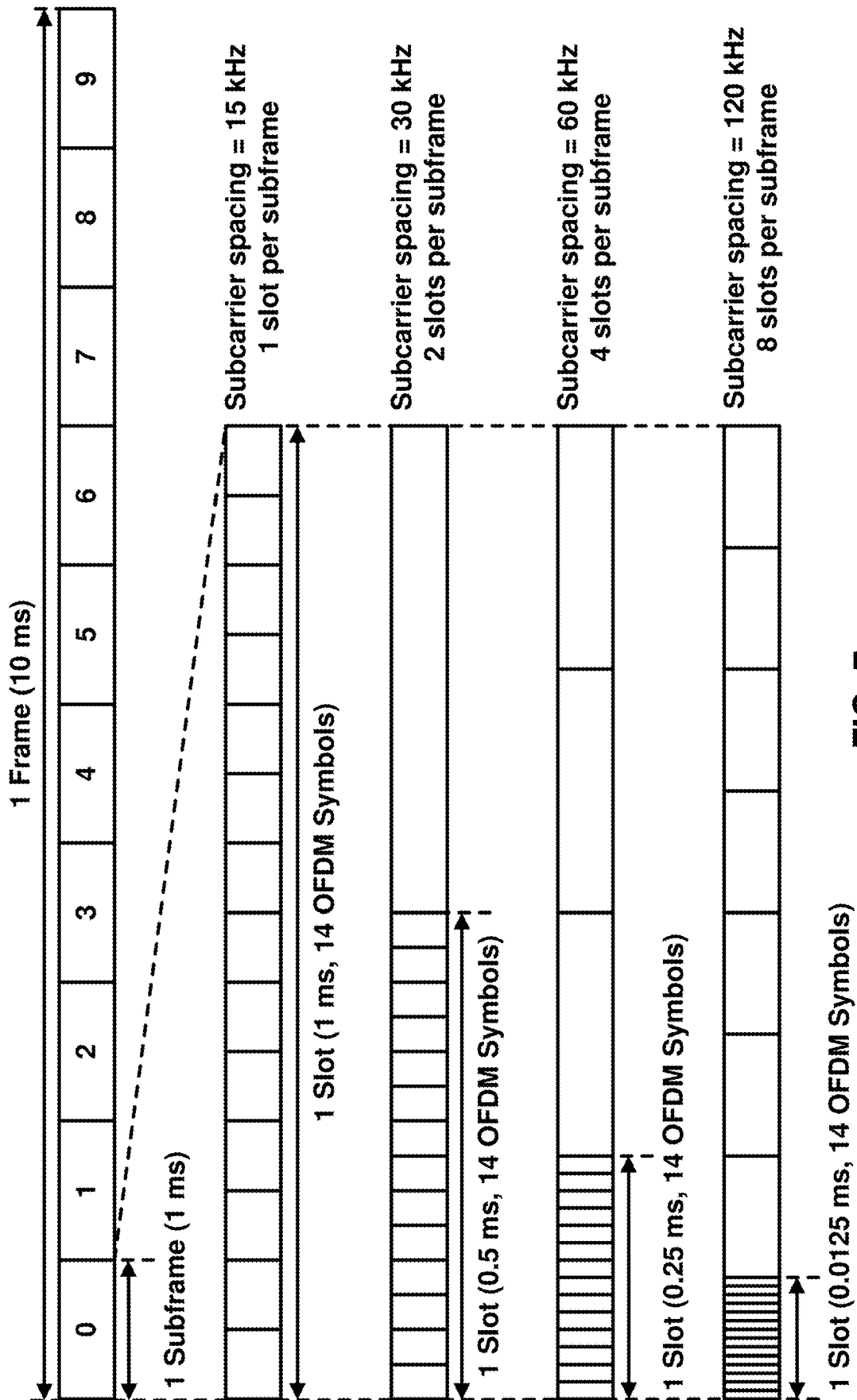


FIG. 7

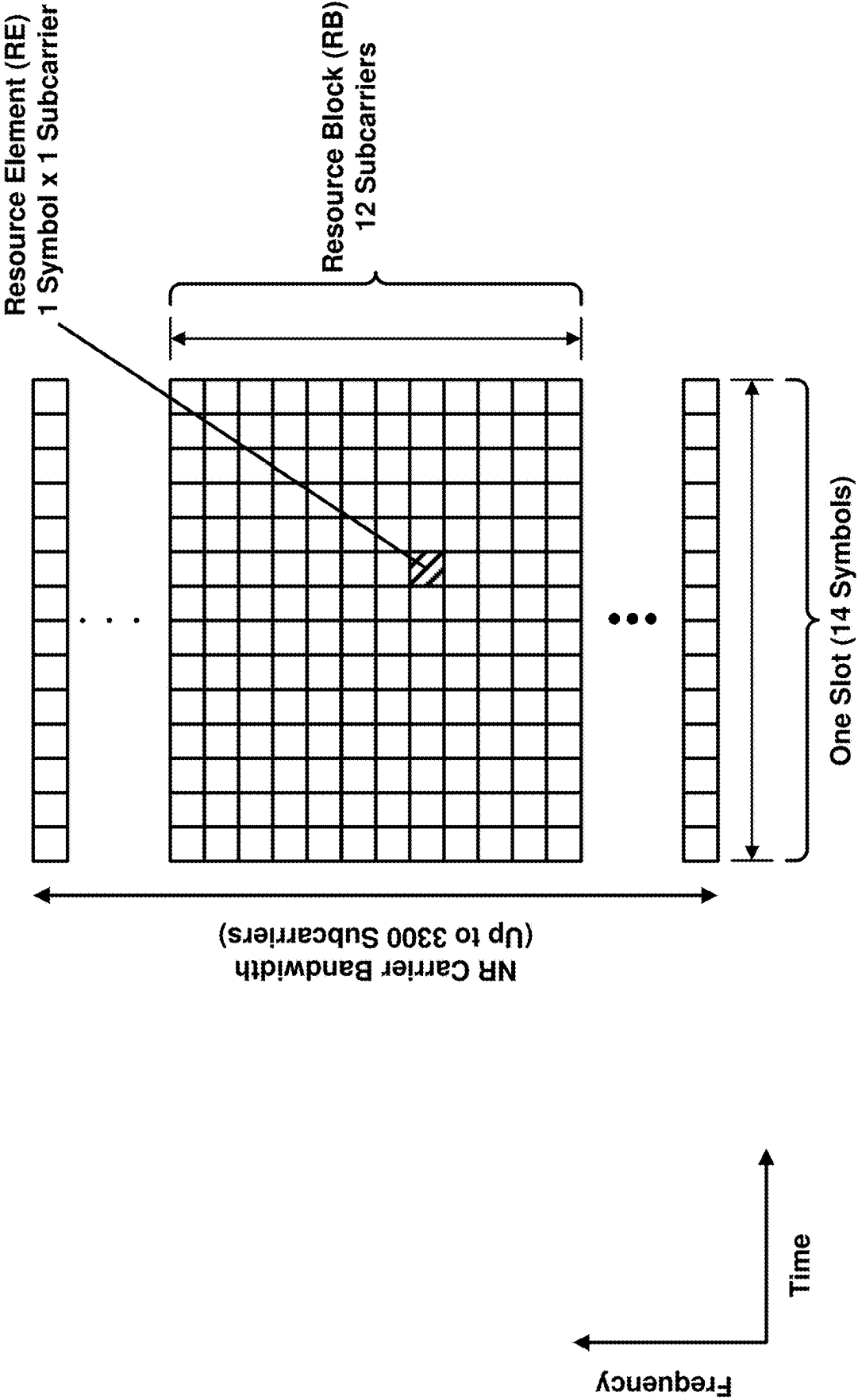


FIG. 8

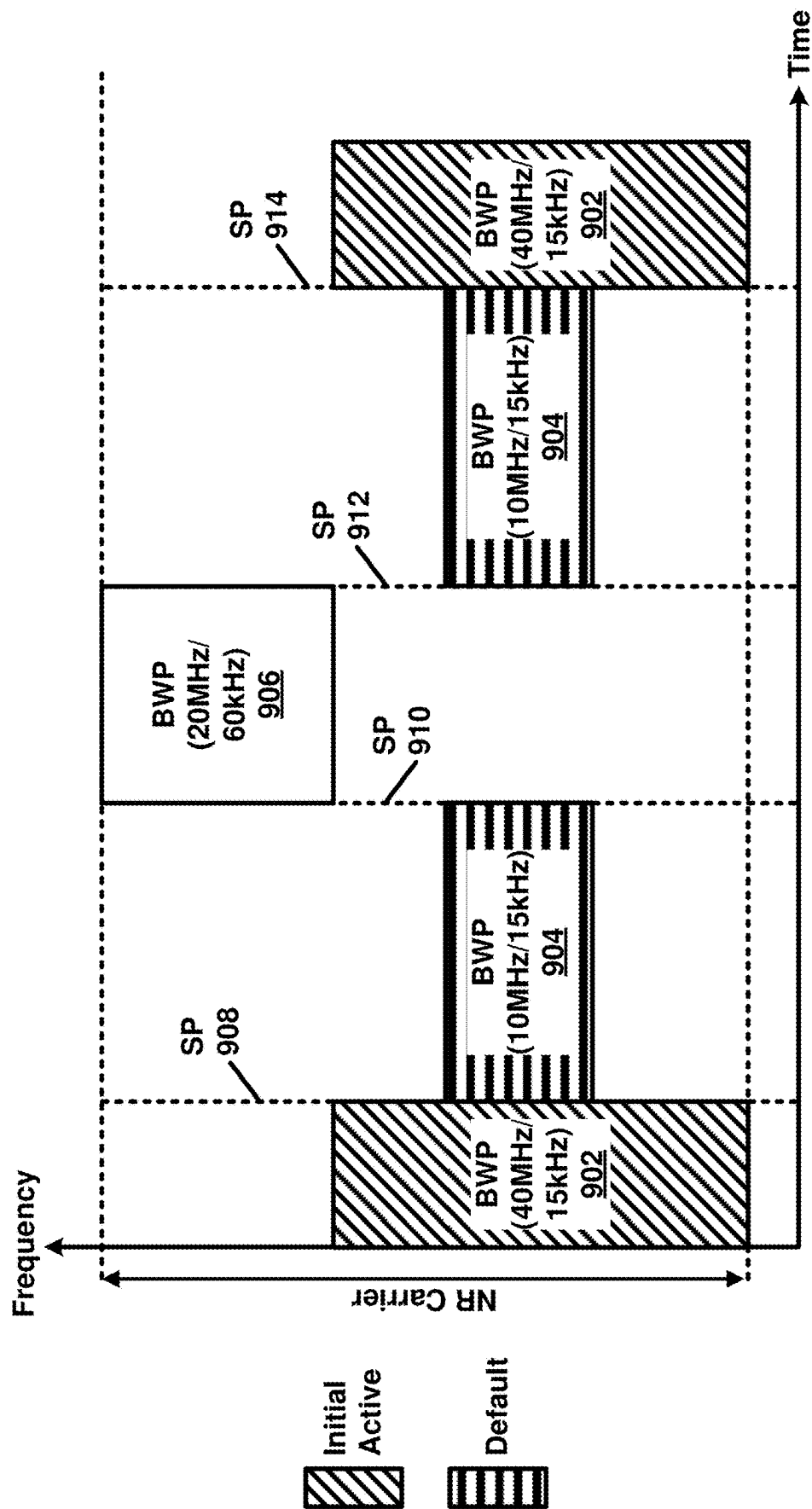


FIG. 9

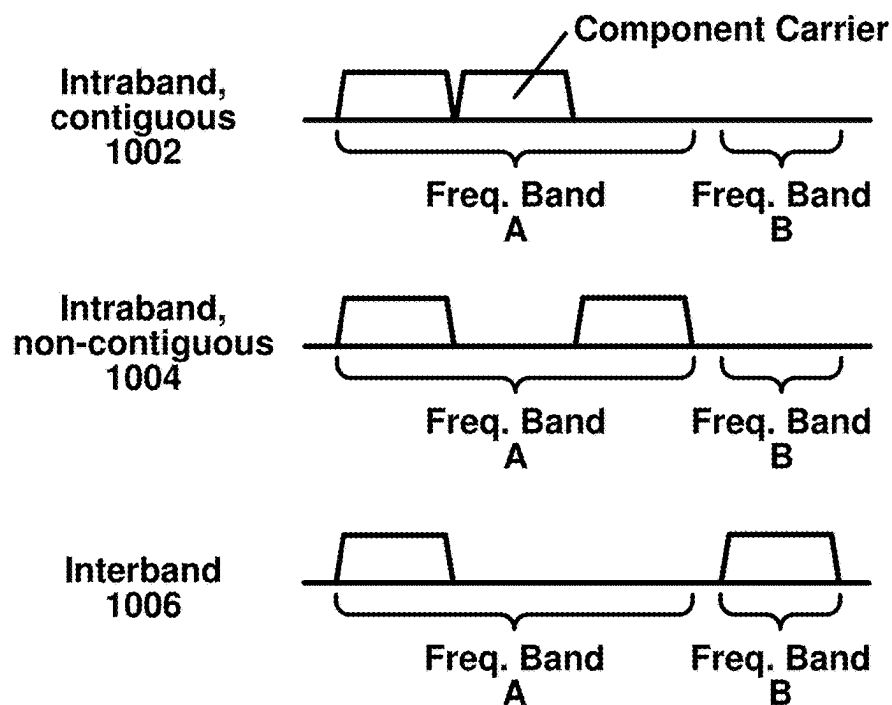


FIG. 10A

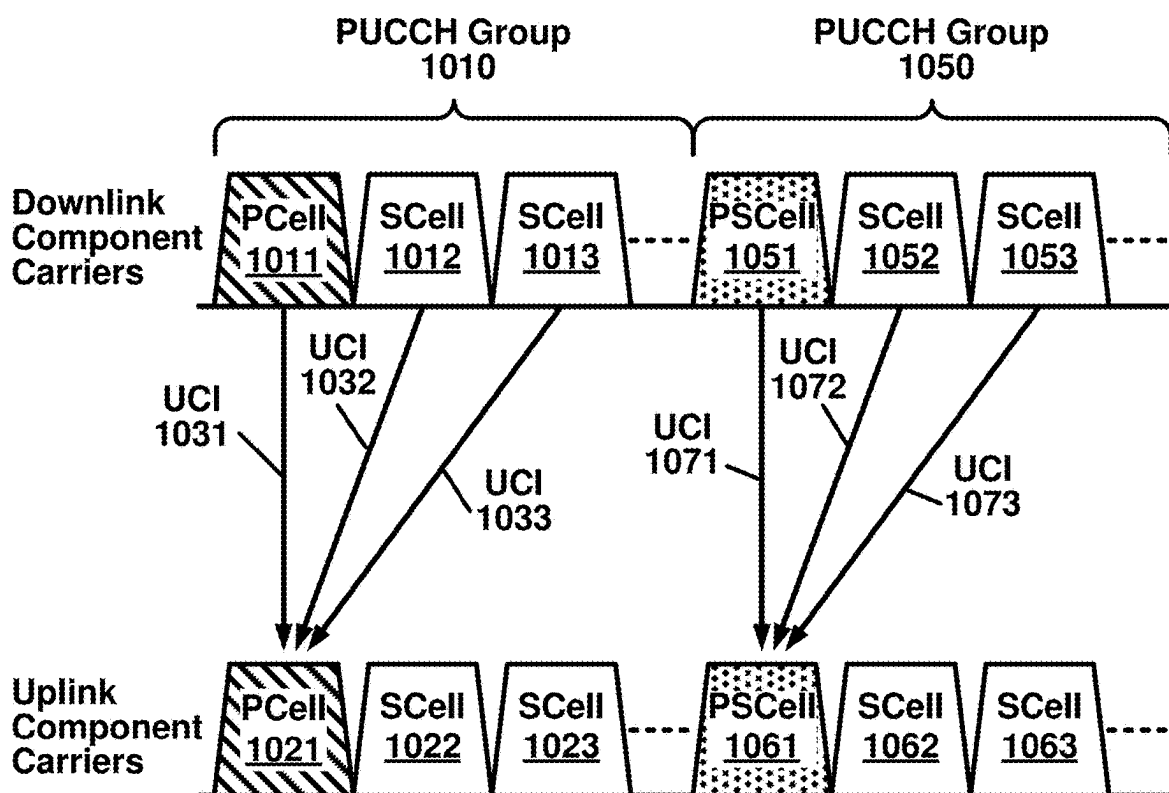


FIG. 10B

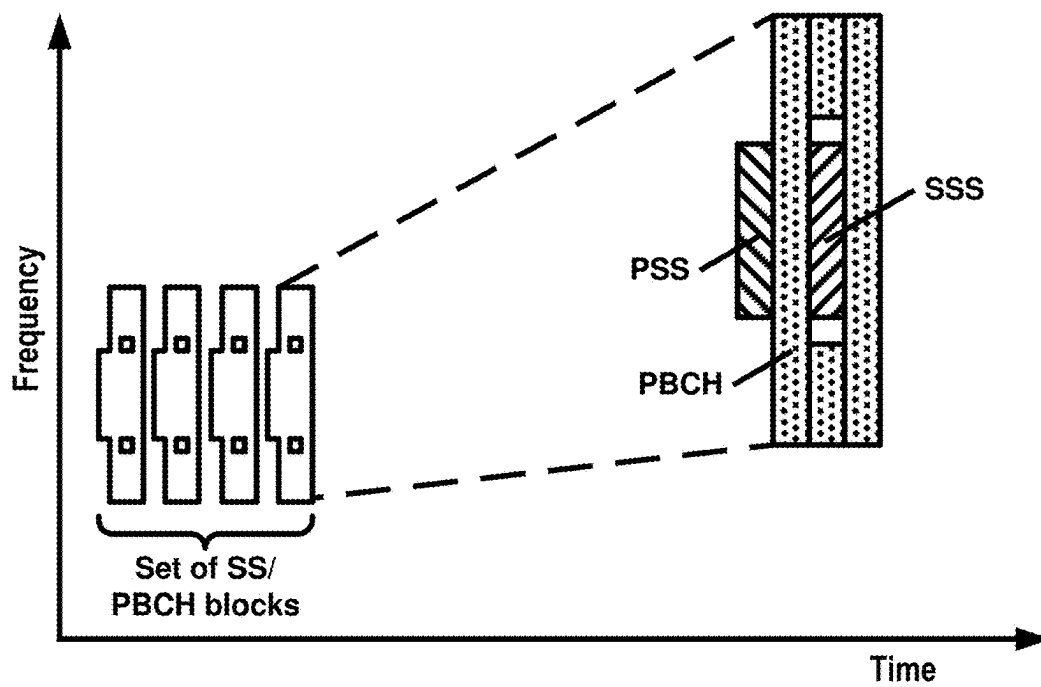


FIG. 11A

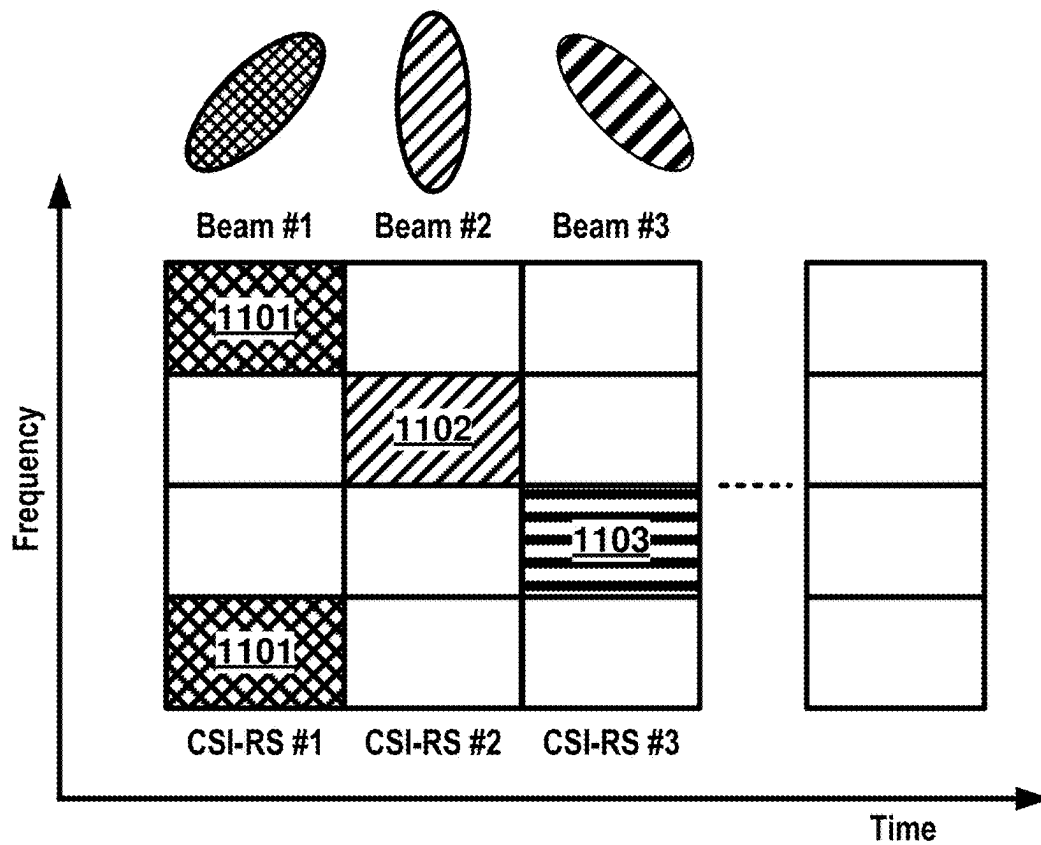


FIG. 11B

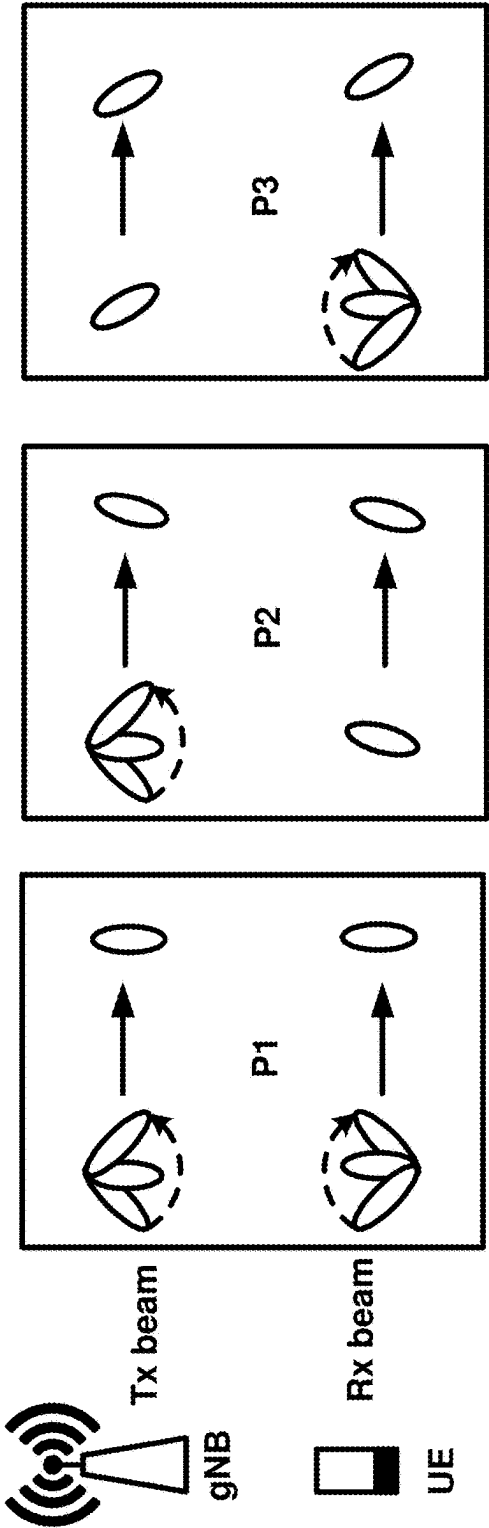


FIG. 12A

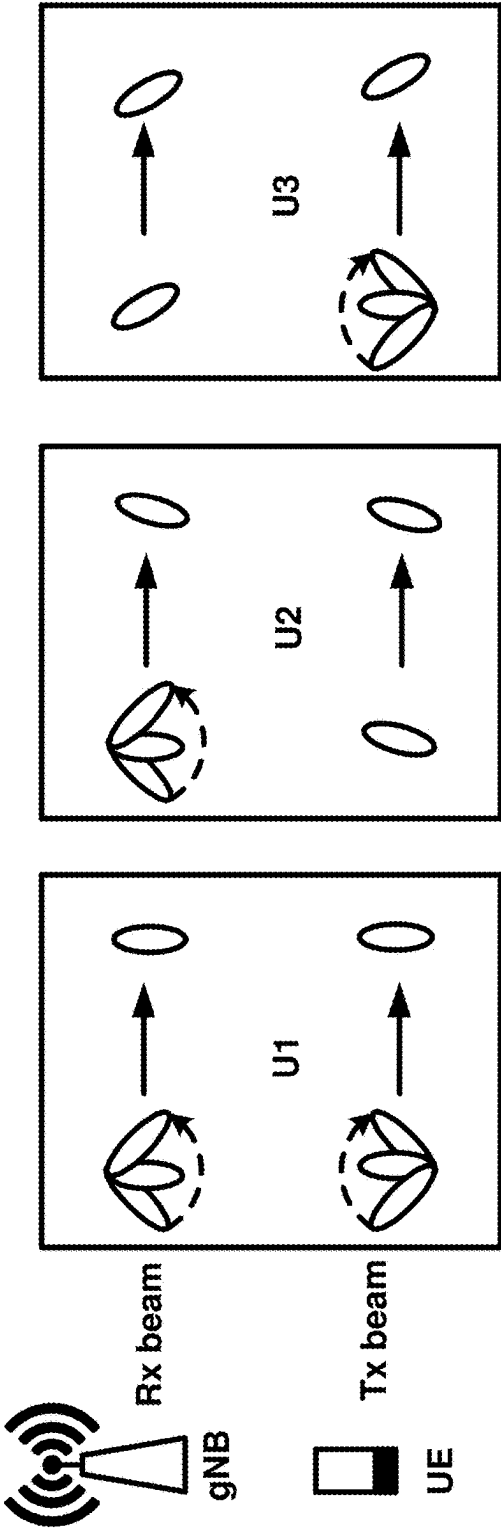


FIG. 12B

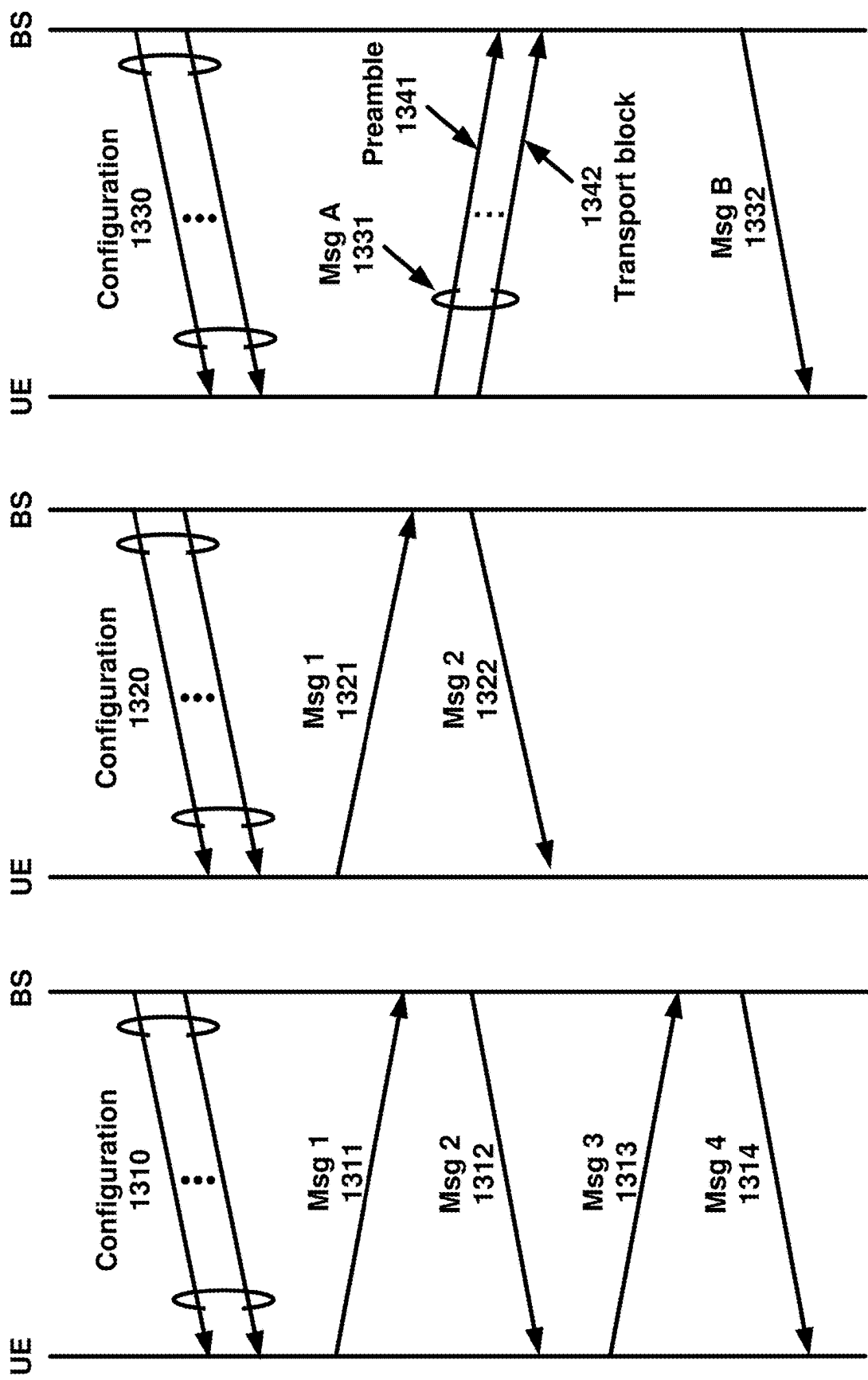


FIG. 13A

FIG. 13B

FIG. 13C

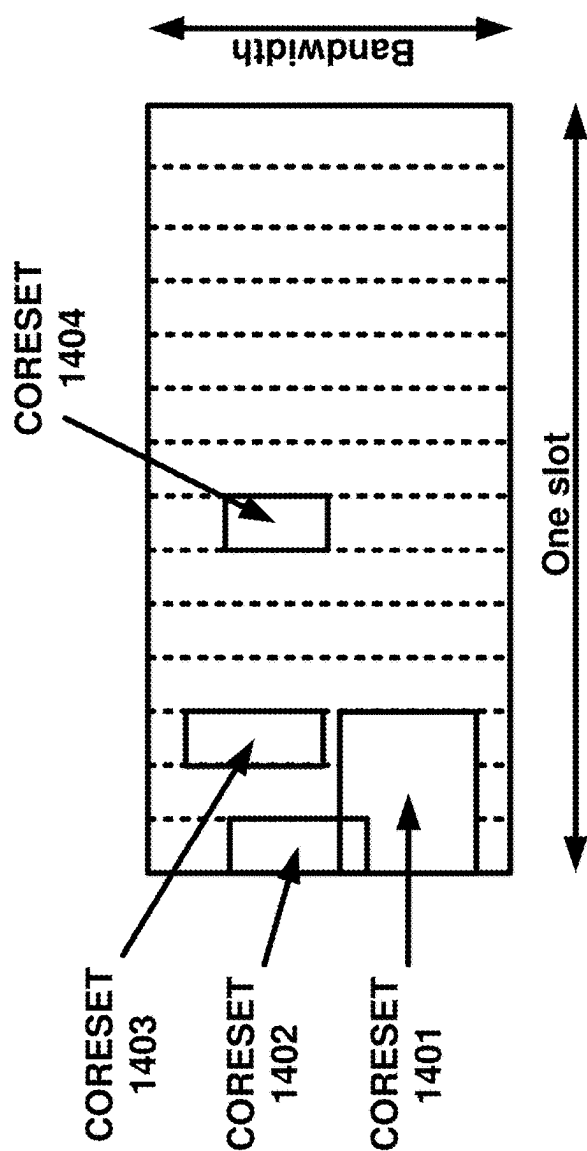


FIG. 14A

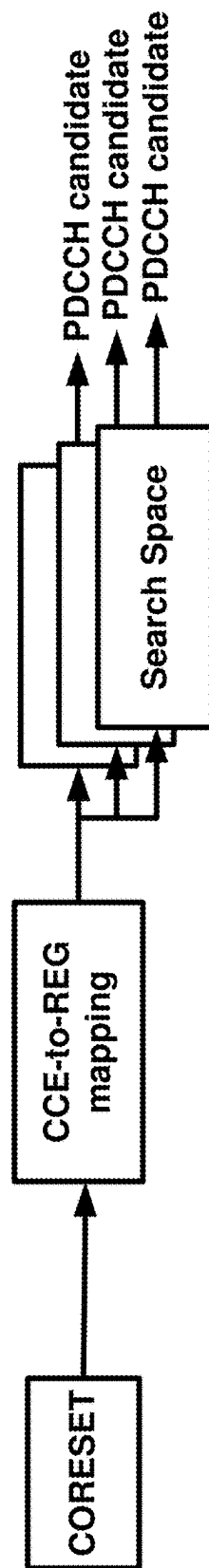


FIG. 14B

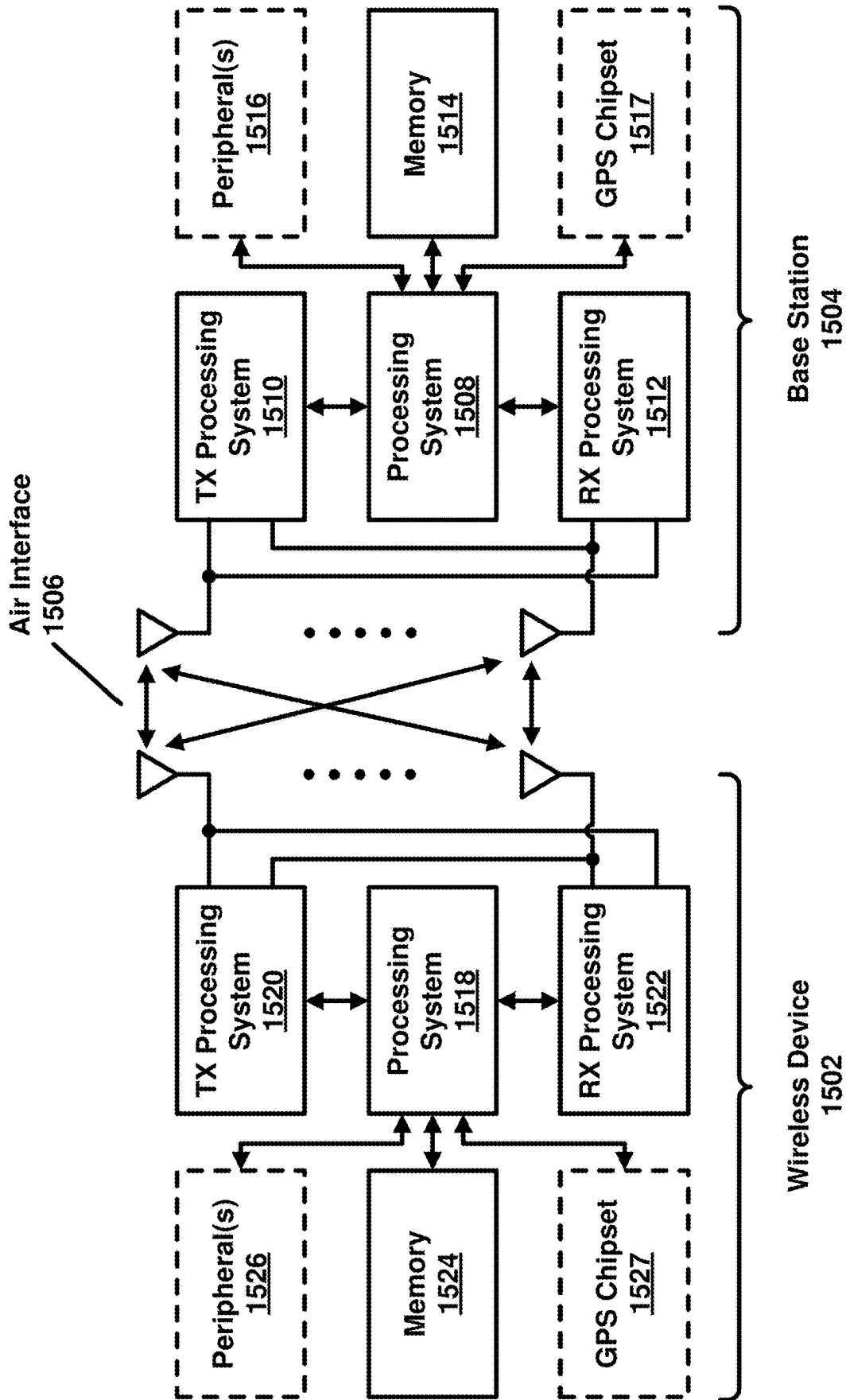


FIG. 15

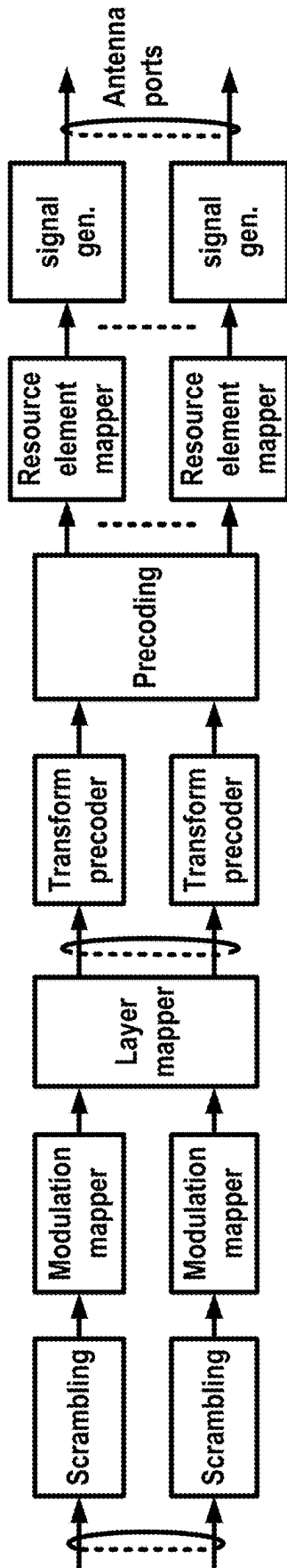


FIG. 16A

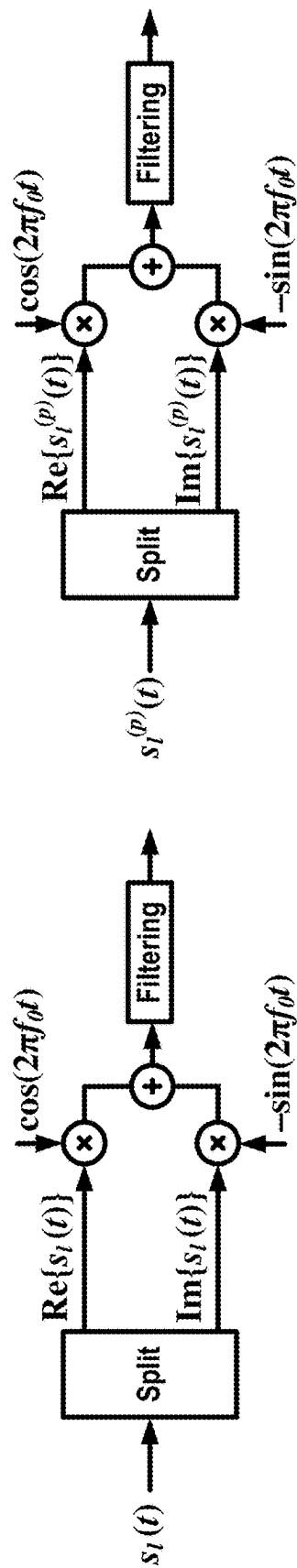


FIG. 16B

FIG. 16D

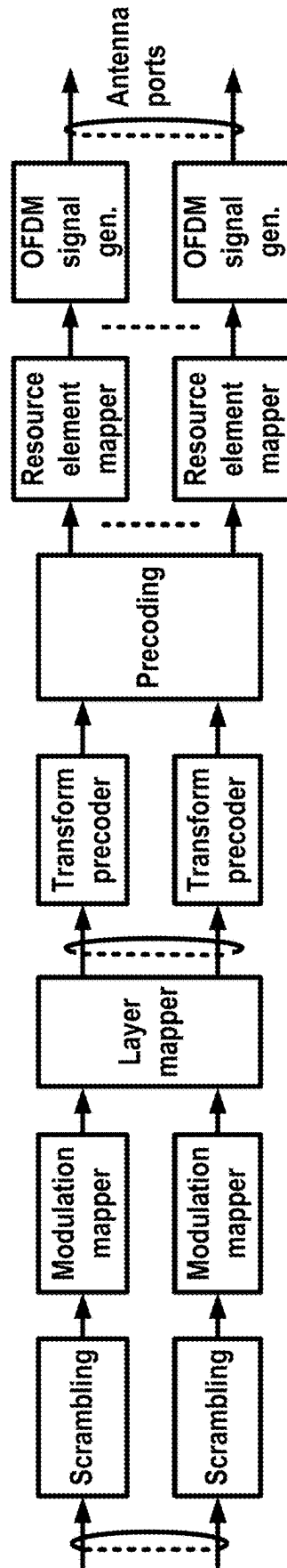


FIG. 16C

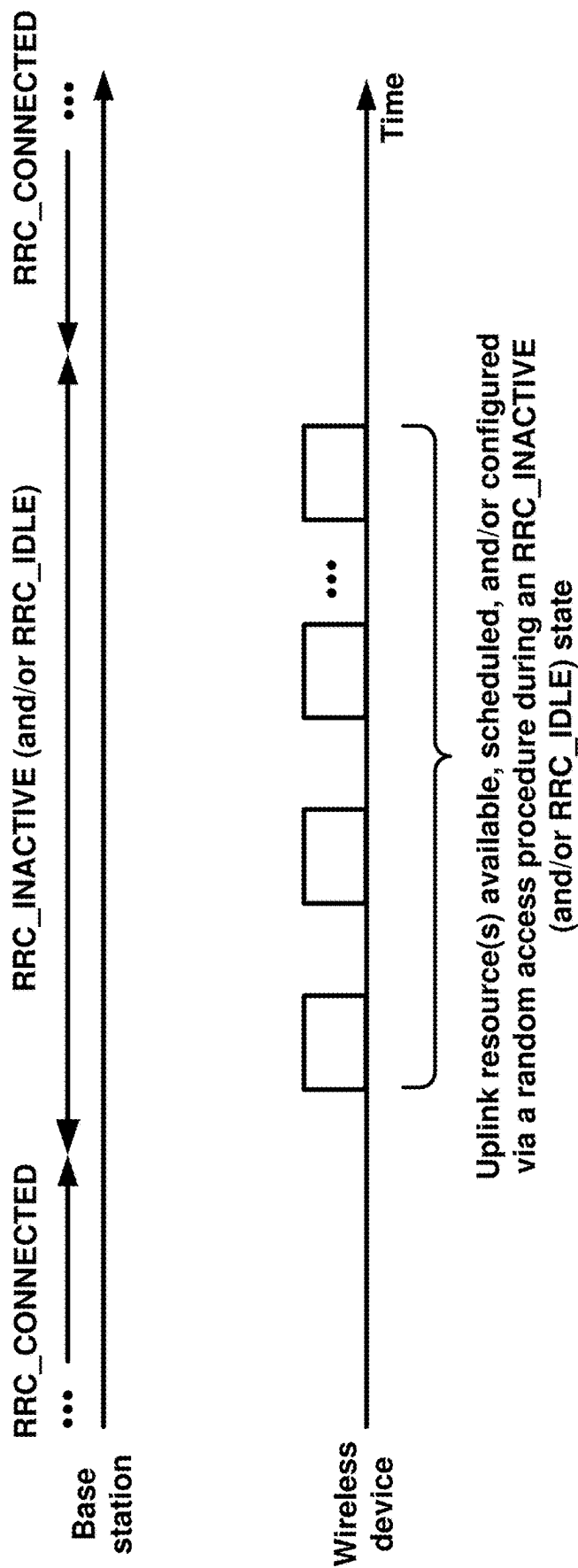


FIG. 17

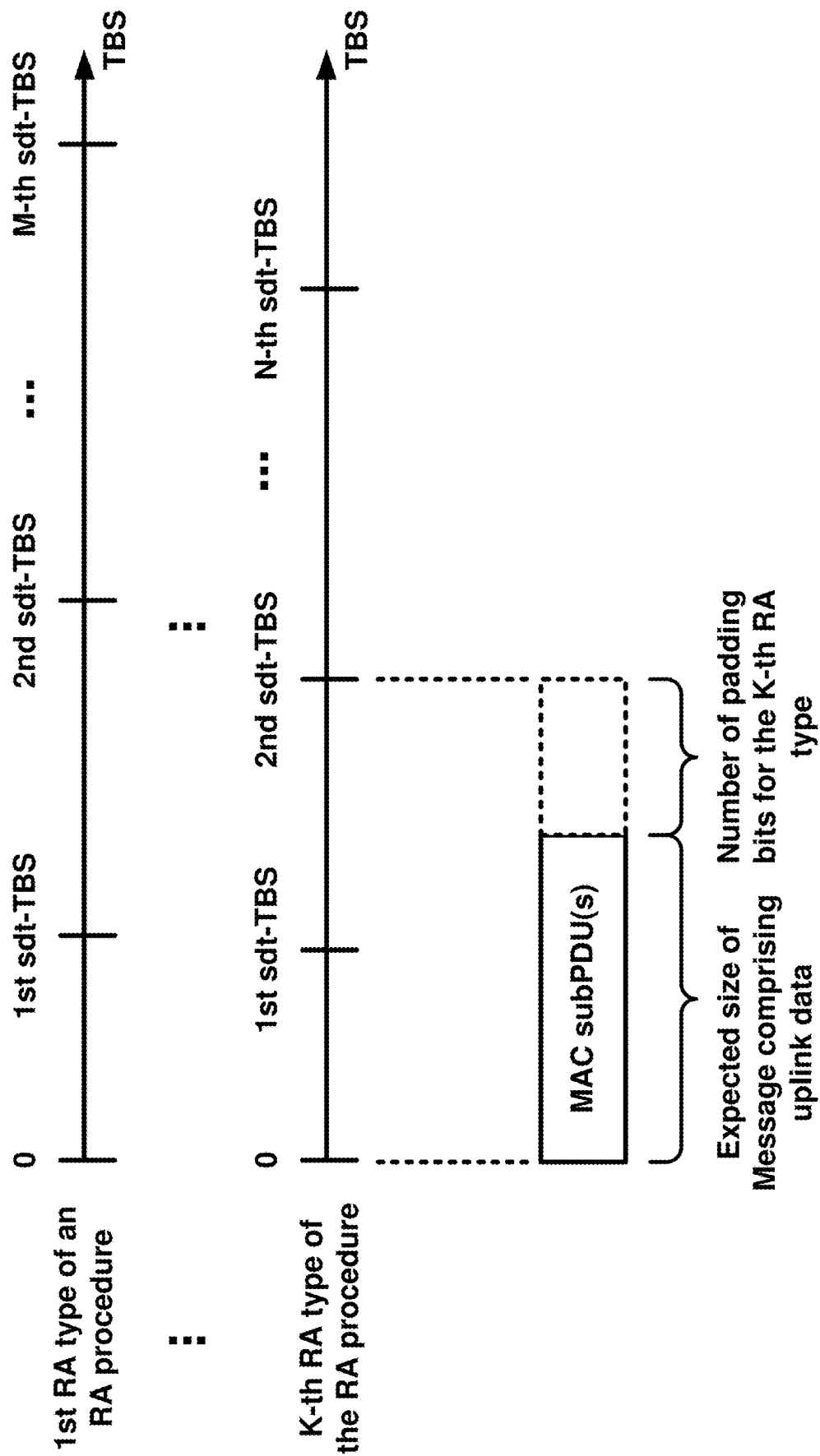


FIG. 18

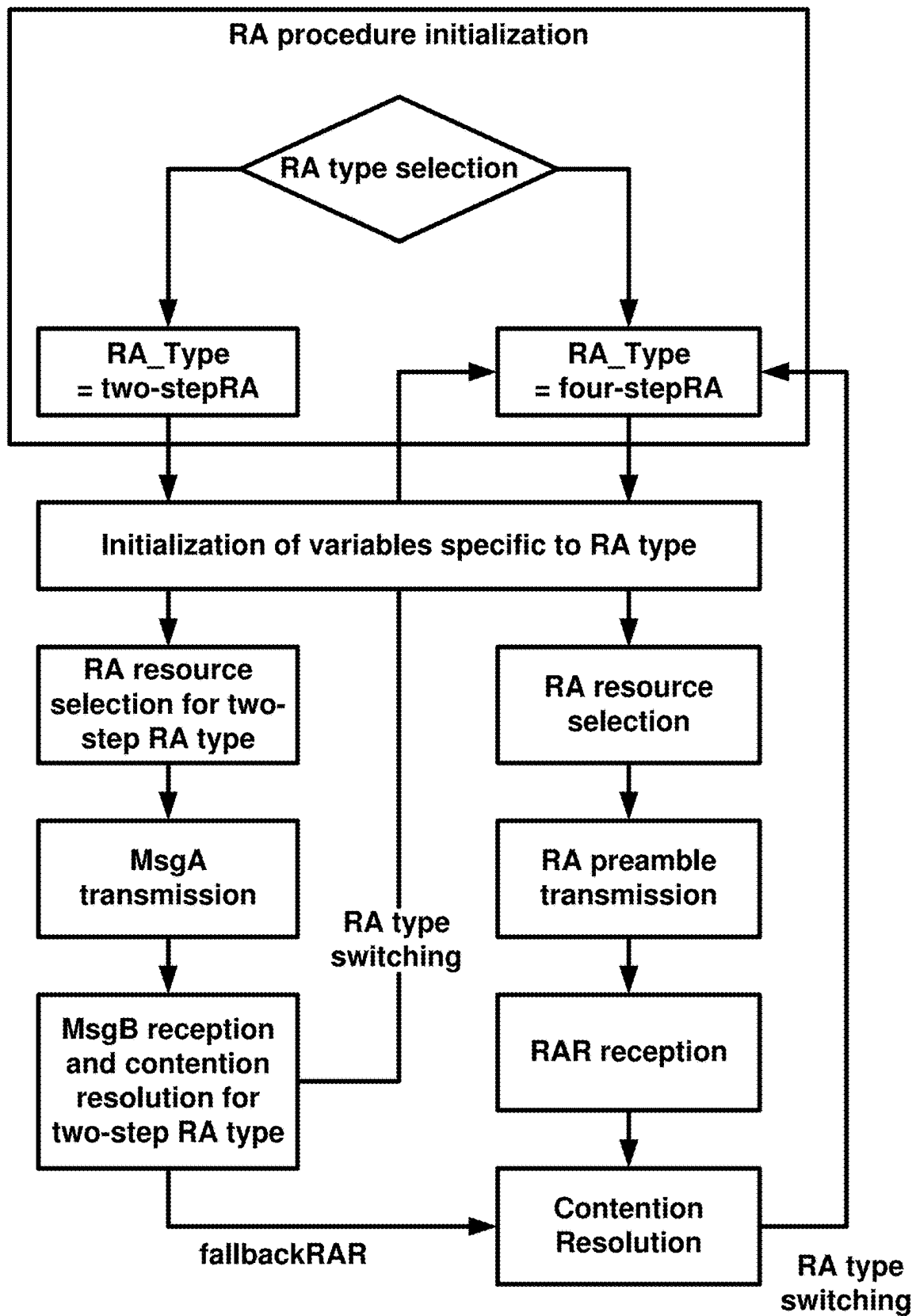


FIG. 19

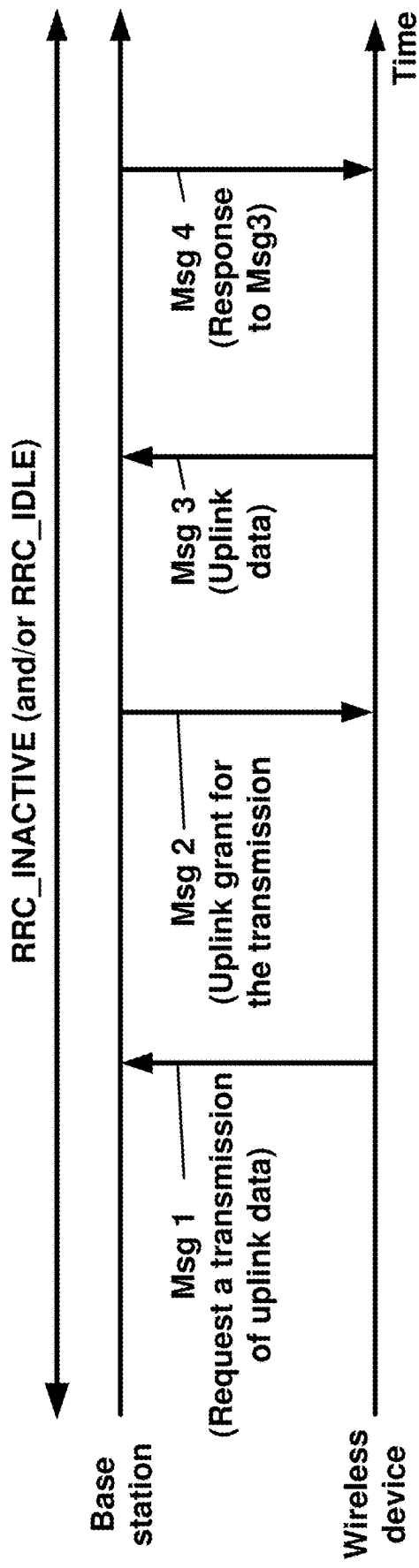


FIG. 20A

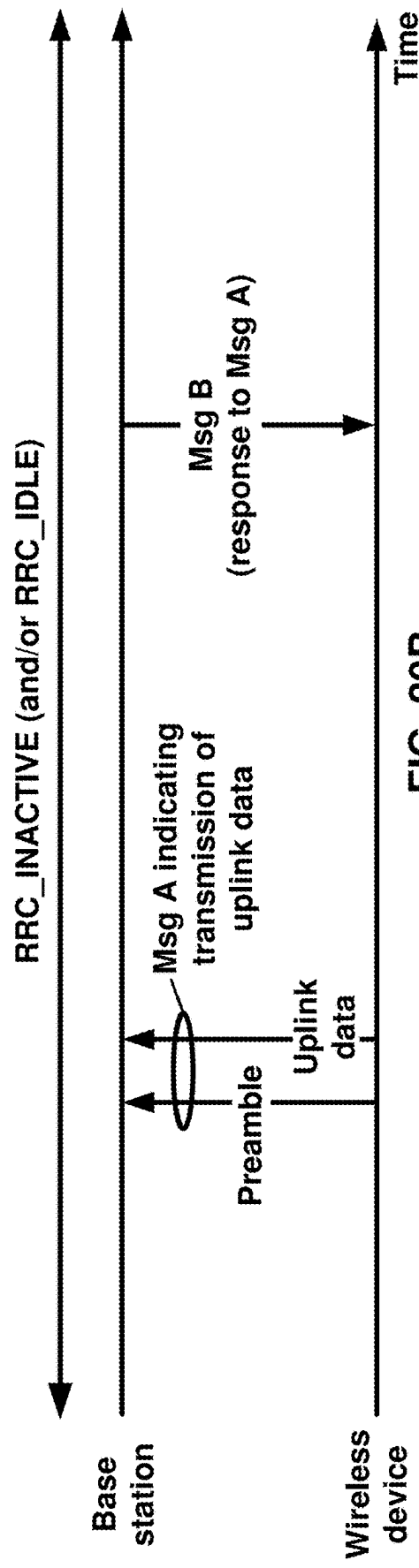


FIG. 20B

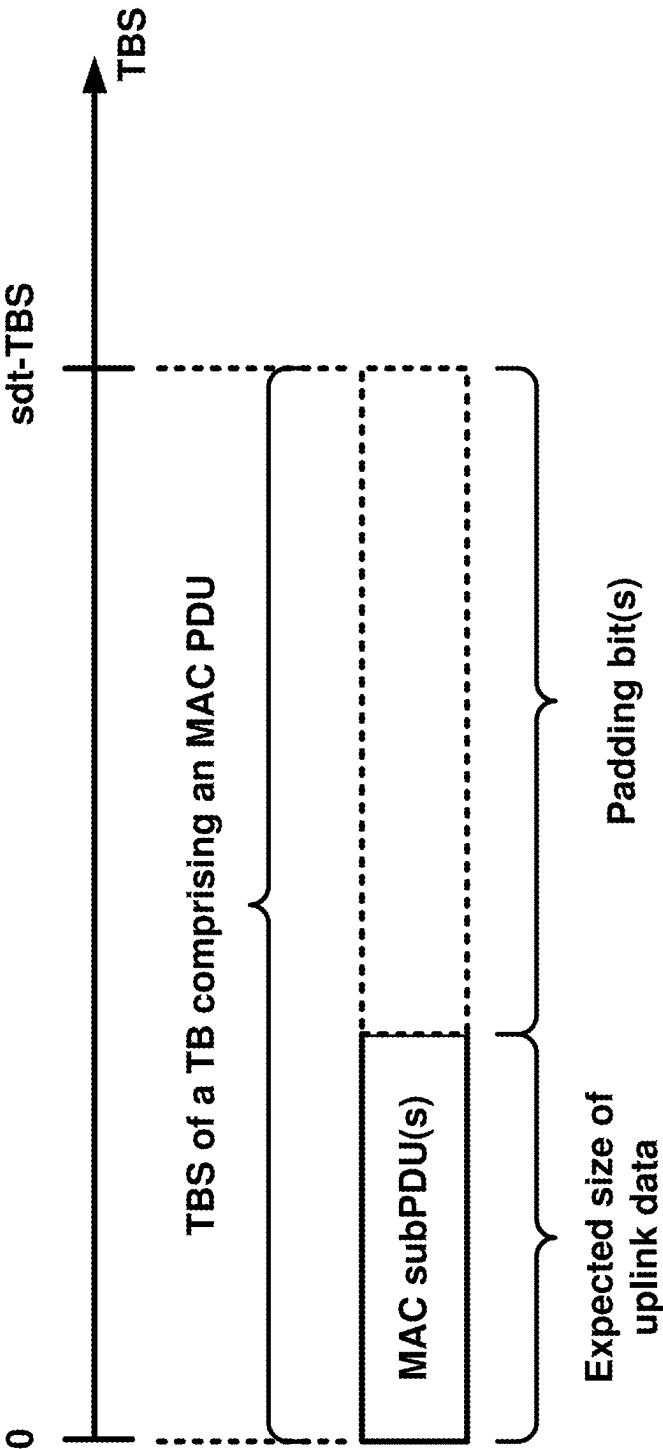


FIG. 21

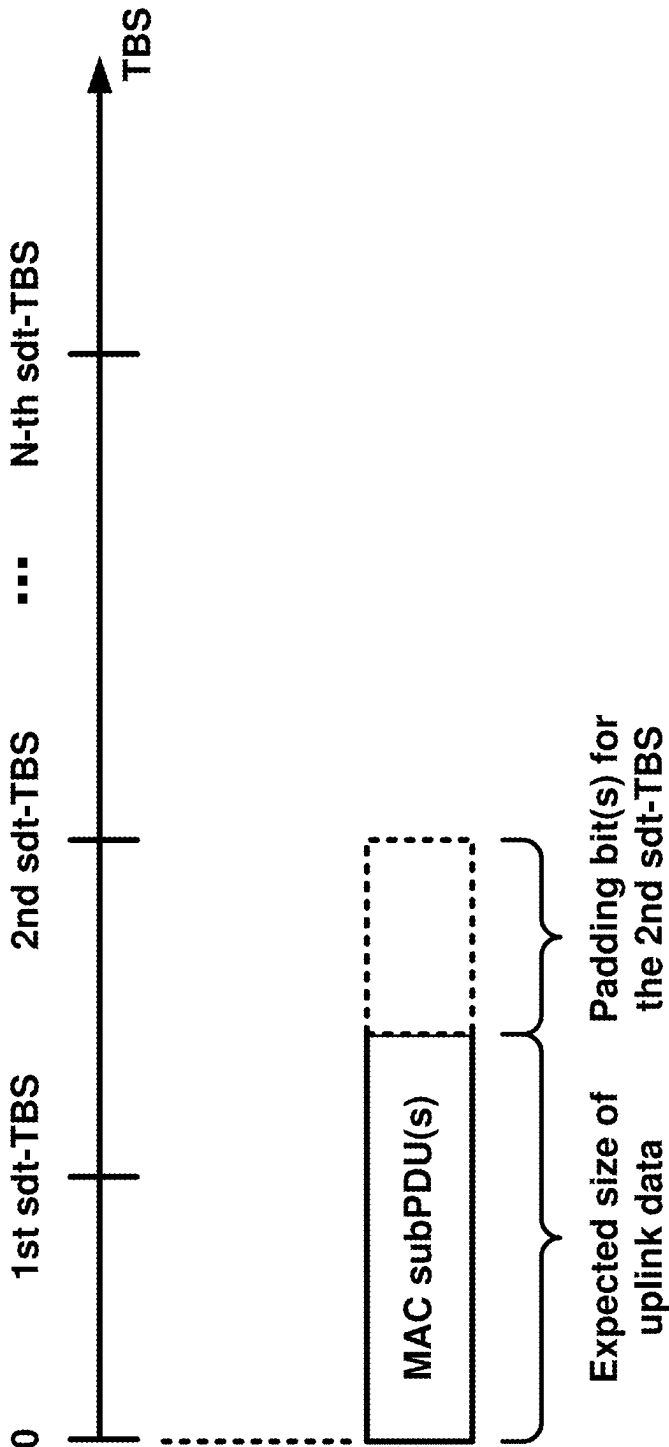


FIG. 22

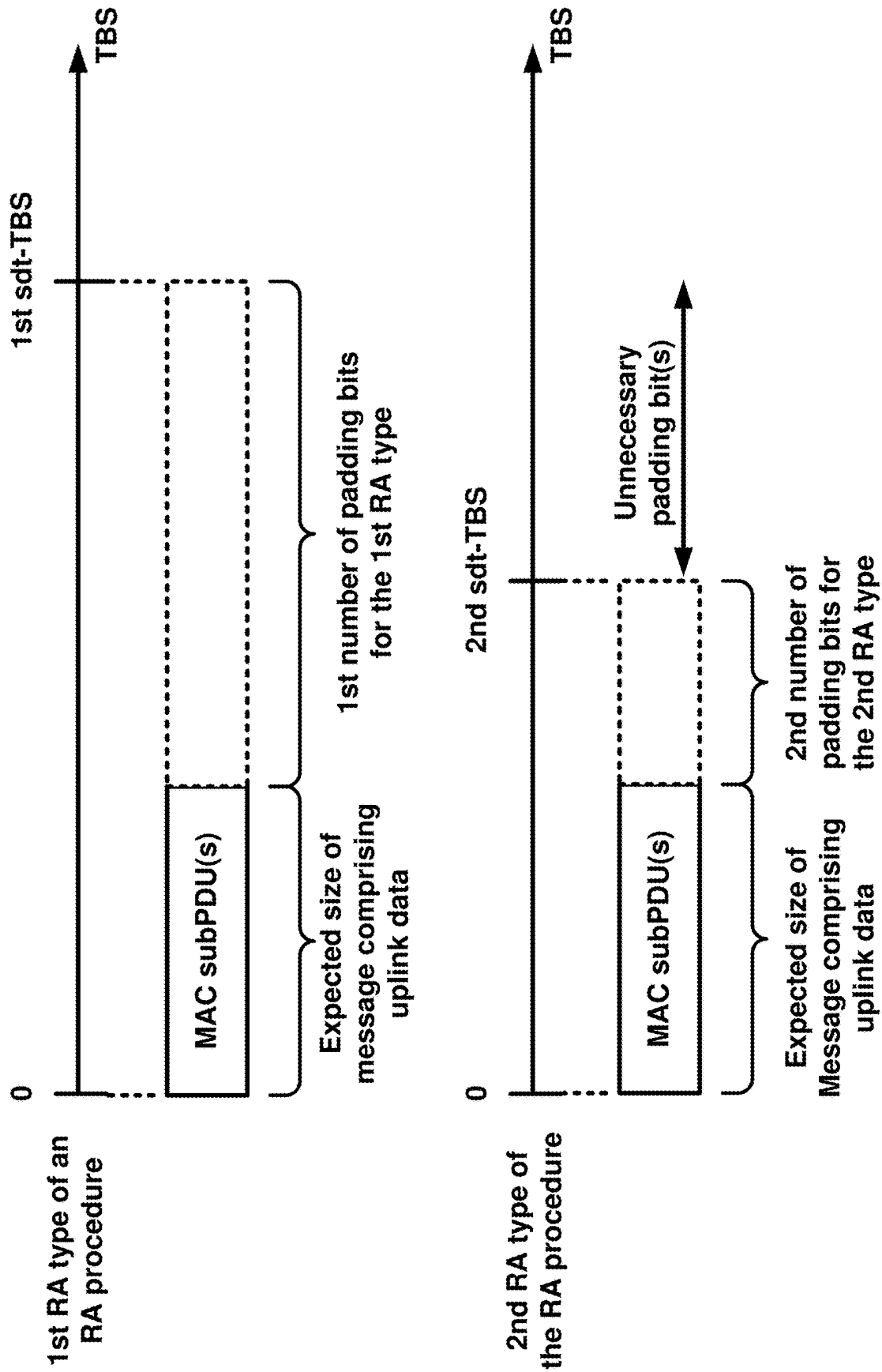


FIG. 23

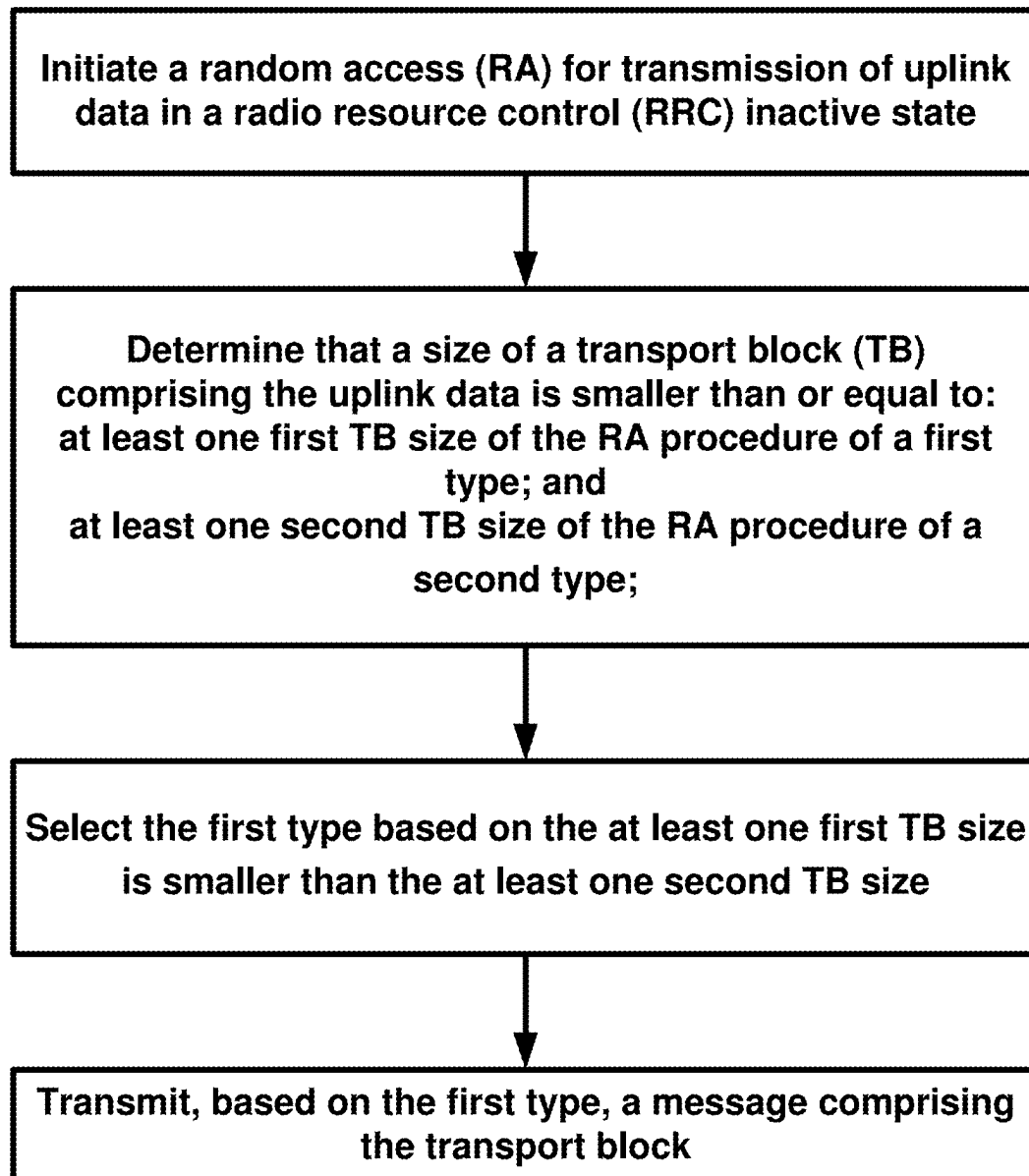


FIG. 24

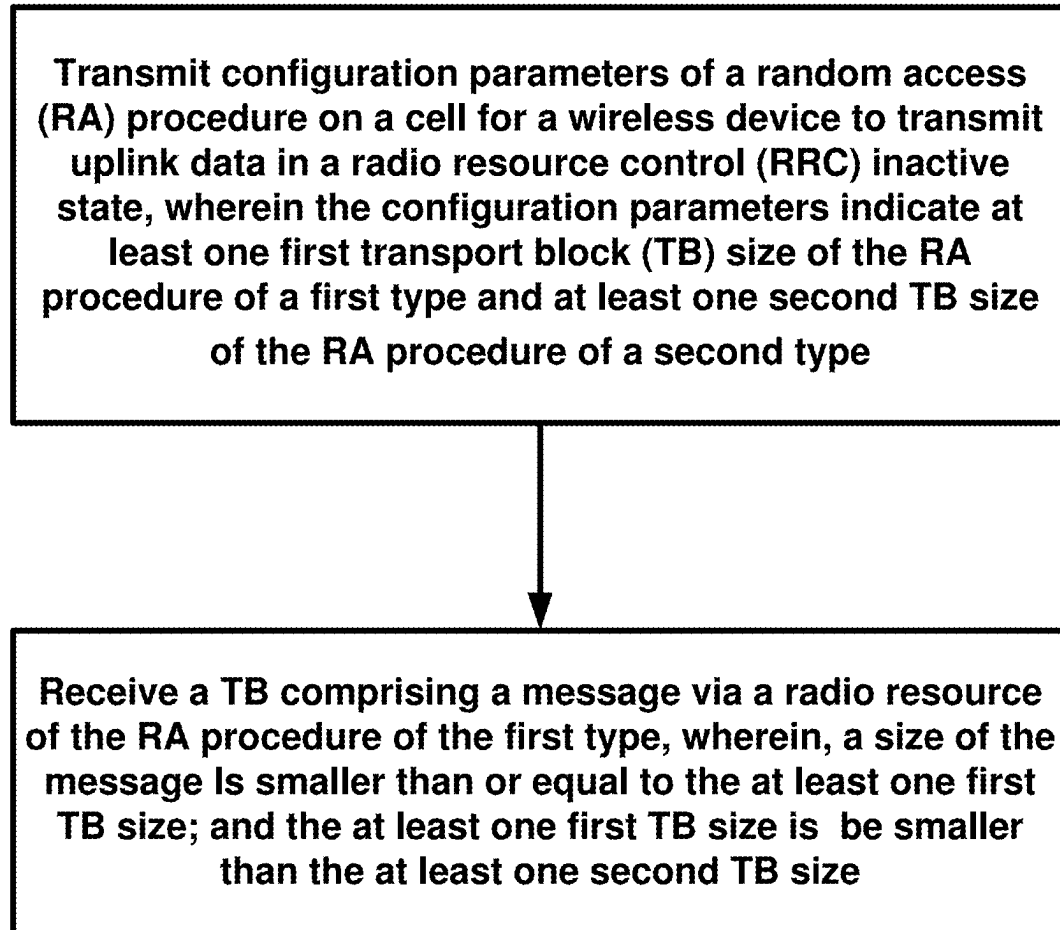


FIG. 25

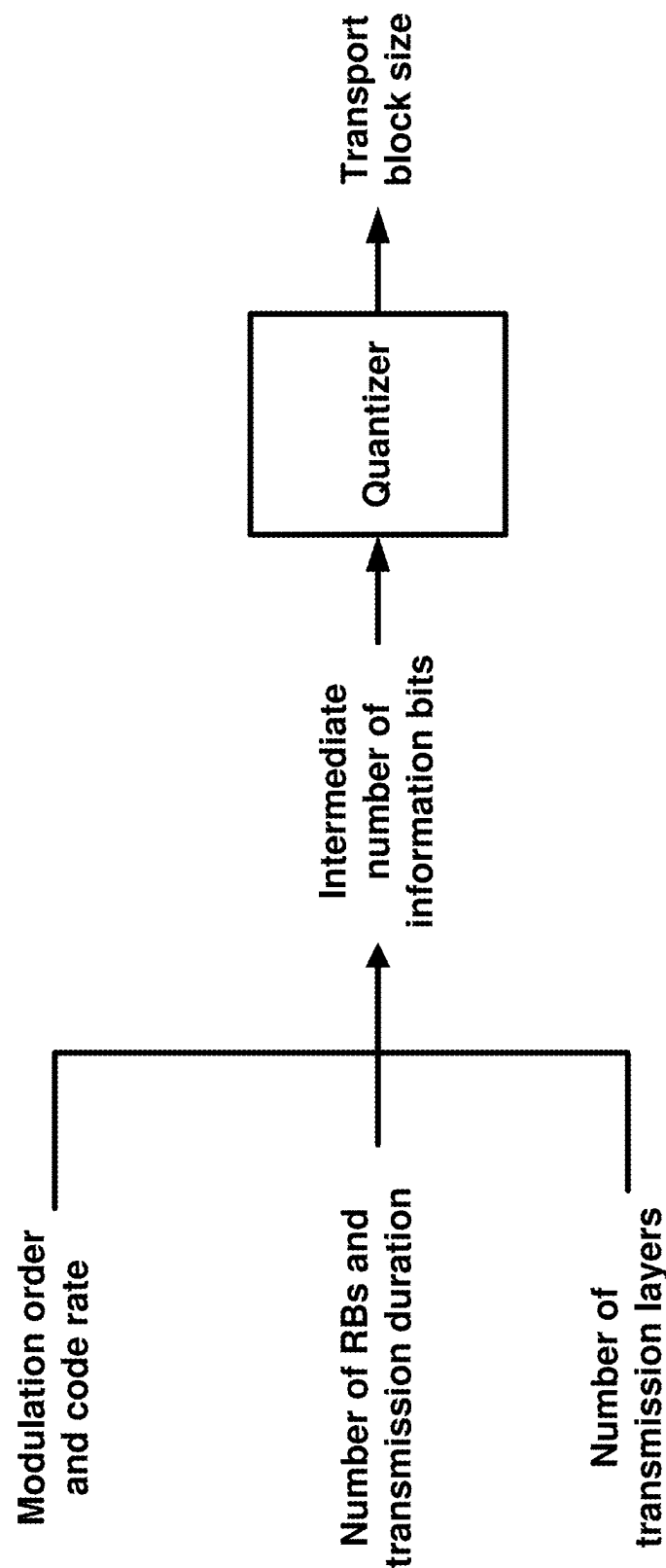


FIG. 26

RANDOM ACCESS TYPE SELECTION FOR SMALL DATA TRANSMISSION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/505,778, filed Oct. 20, 2021, which claims the benefit of U.S. Provisional Application No. 63/104,218, filed Oct. 22, 2020, the contents of which are hereby incorporated by reference in their entireties.

BRIEF DESCRIPTION OF THE DRAWINGS

Examples of several of the various embodiments of the present disclosure are described herein with reference to the drawings.

FIG. 1A and FIG. 1B illustrate example mobile communication networks in which embodiments of the present disclosure may be implemented.

FIG. 2A and FIG. 2B respectively illustrate a New Radio (NR) user plane and control plane protocol stack.

FIG. 3 illustrates an example of services provided between protocol layers of the NR user plane protocol stack of FIG. 2A.

FIG. 4A illustrates an example downlink data flow through the NR user plane protocol stack of FIG. 2A.

FIG. 4B illustrates an example format of a MAC sub-header in a MAC PDU.

FIG. 5A and FIG. 5B respectively illustrate a mapping between logical channels, transport channels, and physical channels for the downlink and uplink.

FIG. 6 is an example diagram showing RRC state transitions of a UE.

FIG. 7 illustrates an example configuration of an NR frame into which OFDM symbols are grouped.

FIG. 8 illustrates an example configuration of a slot in the time and frequency domain for an NR carrier.

FIG. 9 illustrates an example of bandwidth adaptation using three configured BWPs for an NR carrier.

FIG. 10A illustrates three carrier aggregation configurations with two component carriers.

FIG. 10B illustrates an example of how aggregated cells may be configured into one or more PUCCH groups.

FIG. 11A illustrates an example of an SS/PBCH block structure and location.

FIG. 11B illustrates an example of CSI-RSs that are mapped in the time and frequency domains.

FIG. 12A and FIG. 12B respectively illustrate examples of three downlink and uplink beam management procedures.

FIG. 13A, FIG. 13B, and FIG. 13C respectively illustrate a four-step contention-based random access procedure, a two-step contention-free random access procedure, and another two-step random access procedure.

FIG. 14A illustrates an example of CORESET configurations for a bandwidth part.

FIG. 14B illustrates an example of a CCE-to-REG mapping for DCI transmission on a CORESET and PDCCH processing.

FIG. 15 illustrates an example of a wireless device in communication with a base station.

FIG. 16A, FIG. 16B, FIG. 16C, and FIG. 16D illustrate example structures for uplink and downlink transmission.

FIG. 17 illustrates uplink data transmission in a Non-RRC_CONNECTED state, according to some embodiments.

FIG. 18 illustrates one or more sdt-TBS values configured for an RA procedure, according to some embodiments.

FIG. 19 illustrates a flow diagram of a method for an RA procedure, according to some embodiments.

FIG. 20A illustrates an RA-based SDT with a four-step RA procedure, according to some embodiments.

FIG. 20B illustrates an RA-based SDT with a two-step RA procedure, according to some embodiments.

FIG. 21 illustrates a MAC PDU comprising uplink data and/or padding bit(s), according to some embodiments.

FIG. 22 illustrates a MAC PDU comprising uplink data and/or padding bit(s) for one or more sdt-TBS values, according to some embodiments.

FIG. 23 illustrates an RA type selection for an RA-based SDT, according to some embodiments.

FIG. 24 illustrates a flow diagram of a method for RA type selection for an RA-based SDT, according to some embodiments.

FIG. 25 illustrates a flow diagram of a method for an RA-based SDT, according to some embodiments.

FIG. 26 illustrates a determination of TBS, according to some embodiments.

DETAILED DESCRIPTION

In the present disclosure, various embodiments are presented as examples of how the disclosed techniques may be implemented and/or how the disclosed techniques may be practiced in environments and scenarios. It will be apparent to persons skilled in the relevant art that various changes in form and detail can be made therein without departing from the scope. In fact, after reading the description, it will be apparent to one skilled in the relevant art how to implement alternative embodiments. The present embodiments should not be limited by any of the described exemplary embodiments. The embodiments of the present disclosure will be described with reference to the accompanying drawings. Limitations, features, and/or elements from the disclosed example embodiments may be combined to create further embodiments within the scope of the disclosure. Any figures which highlight the functionality and advantages, are presented for example purposes only. The disclosed architecture is sufficiently flexible and configurable, such that it may be utilized in ways other than that shown. For example, the actions listed in any flowchart may be re-ordered or only optionally used in some embodiments.

Embodiments may be configured to operate as needed. The disclosed mechanism may be performed when certain criteria are met, for example, in a wireless device, a base station, a radio environment, a network, a combination of the above, and/or the like. Example criteria may be based, at least in part, on for example, wireless device or network node configurations, traffic load, initial system set up, packet sizes, traffic characteristics, a combination of the above, and/or the like. When the one or more criteria are met, various example embodiments may be applied. Therefore, it may be possible to implement example embodiments that selectively implement disclosed protocols.

A base station may communicate with a mix of wireless devices. Wireless devices and/or base stations may support multiple technologies, and/or multiple releases of the same technology. Wireless devices may have some specific capability(ies) depending on wireless device category and/or capability(ies). When this disclosure refers to a base station communicating with a plurality of wireless devices, this disclosure may refer to a subset of the total wireless devices in a coverage area. This disclosure may refer to, for example,

a plurality of wireless devices of a given LTE or 5G release with a given capability and in a given sector of the base station. The plurality of wireless devices in this disclosure may refer to a selected plurality of wireless devices, and/or a subset of total wireless devices in a coverage area which perform according to disclosed methods, and/or the like. There may be a plurality of base stations or a plurality of wireless devices in a coverage area that may not comply with the disclosed methods, for example, those wireless devices or base stations may perform based on older releases of LTE or 5G technology.

In this disclosure, “a” and “an” and similar phrases are to be interpreted as “at least one” and “one or more.” Similarly, any term that ends with the suffix “(s)” is to be interpreted as “at least one” and “one or more.” In this disclosure, the term “may” is to be interpreted as “may, for example.” In other words, the term “may” is indicative that the phrase following the term “may” is an example of one of a multitude of suitable possibilities that may, or may not, be employed by one or more of the various embodiments. The terms “comprises” and “consists of”, as used herein, enumerate one or more components of the element being described. The term “comprises” is interchangeable with “includes” and does not exclude unenumerated components from being included in the element being described. By contrast, “consists of” provides a complete enumeration of the one or more components of the element being described. The term “based on”, as used herein, should be interpreted as “based at least in part on” rather than, for example, “based solely on”. The term “and/or” as used herein represents any possible combination of enumerated elements. For example, “A, B, and/or C” may represent A; B; C; A and B; A and C; B and C; or A, B, and C.

If A and B are sets and every element of A is an element of B, A is called a subset of B. In this specification, only non-empty sets and subsets are considered. For example, possible subsets of B={cell1, cell2} are: {cell1}, {cell2}, and {cell1, cell2}. The phrase “based on” (or equally “based at least on”) is indicative that the phrase following the term “based on” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments. The phrase “in response to” (or equally “in response at least to”) is indicative that the phrase following the phrase “in response to” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments. The phrase “depending on” (or equally “depending at least to”) is indicative that the phrase following the phrase “depending on” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments. The phrase “employing/using” (or equally “employing/using at least”) is indicative that the phrase following the phrase “employing/using” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments.

The term configured may relate to the capacity of a device whether the device is in an operational or non-operational state. Configured may refer to specific settings in a device that effect the operational characteristics of the device whether the device is in an operational or non-operational state. In other words, the hardware, software, firmware, registers, memory values, and/or the like may be “configured” within a device, whether the device is in an operational or nonoperational state, to provide the device with specific characteristics. Terms such as “a control message to cause in a device” may mean that a control message has

parameters that may be used to configure specific characteristics or may be used to implement certain actions in the device, whether the device is in an operational or non-operational state.

In this disclosure, parameters (or equally called, fields, or Information elements: IEs) may comprise one or more information objects, and an information object may comprise one or more other objects. For example, if parameter (IE) N comprises parameter (IE) M, and parameter (IE) M comprises parameter (IE) K, and parameter (IE) K comprises parameter (information element) J. Then, for example, N comprises K, and N comprises J. In an example embodiment, when one or more messages comprise a plurality of parameters, it implies that a parameter in the plurality of parameters is in at least one of the one or more messages, but does not have to be in each of the one or more messages.

Many features presented are described as being optional through the use of “may” or the use of parentheses. For the sake of brevity and legibility, the present disclosure does not explicitly recite each and every permutation that may be obtained by choosing from the set of optional features. The present disclosure is to be interpreted as explicitly disclosing all such permutations. For example, a system described as having three optional features may be embodied in seven ways, namely with just one of the three possible features, with any two of the three possible features or with three of the three possible features.

Many of the elements described in the disclosed embodiments may be implemented as modules. A module is defined here as an element that performs a defined function and has a defined interface to other elements. The modules described in this disclosure may be implemented in hardware, software in combination with hardware, firmware, wetware (e.g. hardware with a biological element) or a combination thereof, which may be behaviorally equivalent. For example, modules may be implemented as a software routine written in a computer language configured to be executed by a hardware machine (such as C, C++, Fortran, Java, Basic, MATLAB or the like) or a modeling/simulation program such as Simulink, Stateflow, GNU Octave, or LabVIEWMathScript. It may be possible to implement modules using physical hardware that incorporates discrete or programmable analog, digital and/or quantum hardware. Examples of programmable hardware comprise: computers, microcontrollers, microprocessors, application-specific integrated circuits (ASICs); field programmable gate arrays (FPGAs); and complex programmable logic devices (CPLDs). Computers, microcontrollers and microprocessors are programmed using languages such as assembly, C, C++ or the like. FPGAs, ASICs and CPLDs are often programmed using hardware description languages (HDL) such as VHSIC hardware description language (VHDL) or Verilog that configure connections between internal hardware modules with lesser functionality on a programmable device. The mentioned technologies are often used in combination to achieve the result of a functional module.

FIG. 1A illustrates an example of a mobile communication network **100** in which embodiments of the present disclosure may be implemented. The mobile communication network **100** may be, for example, a public land mobile network (PLMN) run by a network operator. As illustrated in FIG. 1A, the mobile communication network **100** includes a core network (CN) **102**, a radio access network (RAN) **104**, and a wireless device **106**.

The CN **102** may provide the wireless device **106** with an interface to one or more data networks (DNs), such as public

DNS (e.g., the Internet), private DNSs, and/or intra-operator DNSs. As part of the interface functionality, the CN **102** may set up end-to-end connections between the wireless device **106** and the one or more DNSs, authenticate the wireless device **106**, and provide charging functionality.

The RAN **104** may connect the CN **102** to the wireless device **106** through radio communications over an air interface. As part of the radio communications, the RAN **104** may provide scheduling, radio resource management, and retransmission protocols. The communication direction from the RAN **104** to the wireless device **106** over the air interface is known as the downlink and the communication direction from the wireless device **106** to the RAN **104** over the air interface is known as the uplink. Downlink transmissions may be separated from uplink transmissions using frequency division duplexing (FDD), time-division duplexing (TDD), and/or some combination of the two duplexing techniques.

The term wireless device may be used throughout this disclosure to refer to and encompass any mobile device or fixed (non-mobile) device for which wireless communication is needed or usable. For example, a wireless device may be a telephone, smart phone, tablet, computer, laptop, sensor, meter, wearable device, Internet of Things (IoT) device, vehicle road side unit (RSU), relay node, automobile, and/or any combination thereof. The term wireless device encompasses other terminology, including user equipment (UE), user terminal (UT), access terminal (AT), mobile station, handset, wireless transmit and receive unit (WTRU), and/or wireless communication device.

The RAN **104** may include one or more base stations (not shown). The term base station may be used throughout this disclosure to refer to and encompass a Node B (associated with UMTS and/or 3G standards), an Evolved Node B (CNB, associated with E-UTRA and/or 4G standards), a remote radio head (RRH), a baseband processing unit coupled to one or more RRHs, a repeater node or relay node used to extend the coverage area of a donor node, a Next Generation Evolved Node B (ng-eNB), a Generation Node B (gNB, associated with NR and/or 5G standards), an access point (AP, associated with, for example, WiFi or any other suitable wireless communication standard), and/or any combination thereof. A base station may comprise at least one gNB Central Unit (gNB-CU) and at least one a gNB Distributed Unit (gNB-DU).

A base station included in the RAN **104** may include one or more sets of antennas for communicating with the wireless device **106** over the air interface. For example, one or more of the base stations may include three sets of antennas to respectively control three cells (or sectors). The size of a cell may be determined by a range at which a receiver (e.g., a base station receiver) can successfully receive the transmissions from a transmitter (e.g., a wireless device transmitter) operating in the cell. Together, the cells of the base stations may provide radio coverage to the wireless device **106** over a wide geographic area to support wireless device mobility.

In addition to three-sector sites, other implementations of base stations are possible. For example, one or more of the base stations in the RAN **104** may be implemented as a sectorized site with more or less than three sectors. One or more of the base stations in the RAN **104** may be implemented as an access point, as a baseband processing unit coupled to several remote radio heads (RRHs), and/or as a repeater or relay node used to extend the coverage area of a donor node. A baseband processing unit coupled to RRHs may be part of a centralized or cloud RAN architecture,

where the baseband processing unit may be either centralized in a pool of baseband processing units or virtualized. A repeater node may amplify and rebroadcast a radio signal received from a donor node. A relay node may perform the same/similar functions as a repeater node but may decode the radio signal received from the donor node to remove noise before amplifying and rebroadcasting the radio signal.

The RAN **104** may be deployed as a homogenous network of macrocell base stations that have similar antenna patterns and similar high-level transmit powers. The RAN **104** may be deployed as a heterogeneous network. In heterogeneous networks, small cell base stations may be used to provide small coverage areas, for example, coverage areas that overlap with the comparatively larger coverage areas provided by macrocell base stations. The small coverage areas may be provided in areas with high data traffic (or so-called "hotspots") or in areas with weak macrocell coverage. Examples of small cell base stations include, in order of decreasing coverage area, microcell base stations, picocell base stations, and femtocell base stations or home base stations.

The Third-Generation Partnership Project (3GPP) was formed in 1998 to provide global standardization of specifications for mobile communication networks similar to the mobile communication network **100** in FIG. **1A**. To date, 3GPP has produced specifications for three generations of mobile networks: a third generation (3G) network known as Universal Mobile Telecommunications System (UMTS), a fourth generation (4G) network known as Long-Term Evolution (LTE), and a fifth generation (5G) network known as 5G System (5GS). Embodiments of the present disclosure are described with reference to the RAN of a 3GPP 5G network, referred to as next-generation RAN (NG-RAN). Embodiments may be applicable to RANs of other mobile communication networks, such as the RAN **104** in FIG. **1A**, the RANs of earlier 3G and 4G networks, and those of future networks yet to be specified (e.g., a 3GPP 6G network). NG-RAN implements 5G radio access technology known as New Radio (NR) and may be provisioned to implement 4G radio access technology or other radio access technologies, including non-3GPP radio access technologies.

FIG. **1B** illustrates another example mobile communication network **150** in which embodiments of the present disclosure may be implemented. Mobile communication network **150** may be, for example, a PLMN run by a network operator. As illustrated in FIG. **1B**, mobile communication network **150** includes a 5G core network (5G-CN) **152**, an NG-RAN **154**, and UEs **156A** and **156B** (collectively UEs **156**). These components may be implemented and operate in the same or similar manner as corresponding components described with respect to FIG. **1A**.

The 5G-CN **152** provides the UEs **156** with an interface to one or more DNSs, such as public DNS (e.g., the Internet), private DNSs, and/or intra-operator DNSs. As part of the interface functionality, the 5G-CN **152** may set up end-to-end connections between the UEs **156** and the one or more DNSs, authenticate the UEs **156**, and provide charging functionality. Compared to the CN of a 3GPP 4G network, the basis of the 5G-CN **152** may be a service-based architecture. This means that the architecture of the nodes making up the 5G-CN **152** may be defined as network functions that offer services via interfaces to other network functions. The network functions of the 5G-CN **152** may be implemented in several ways, including as network elements on dedicated or shared hardware, as software instances running on dedicated or shared hardware, or as virtualized functions instantiated on a platform (e.g., a cloud-based platform).

As illustrated in FIG. 1B, the 5G-CN 152 includes an Access and Mobility Management Function (AMF) 158A and a User Plane Function (UPF) 158B, which are shown as one component AMF/UPF 158 in FIG. 1B for case of illustration. The UPF 158B may serve as a gateway between the NG-RAN 154 and the one or more DNs. The UPF 158B may perform functions such as packet routing and forwarding, packet inspection and user plane policy rule enforcement, traffic usage reporting, uplink classification to support routing of traffic flows to the one or more DNs, quality of service (QoS) handling for the user plane (e.g., packet filtering, gating, uplink/downlink rate enforcement, and uplink traffic verification), downlink packet buffering, and downlink data notification triggering. The UPF 158B may serve as an anchor point for intra-/inter-Radio Access Technology (RAT) mobility, an external protocol (or packet) data unit (PDU) session point of interconnect to the one or more DNs, and/or a branching point to support a multi-homed PDU session. The UEs 156 may be configured to receive services through a PDU session, which is a logical connection between a UE and a DN.

The AMF 158A may perform functions such as Non-Access Stratum (NAS) signaling termination, NAS signaling security, Access Stratum (AS) security control, inter-CN node signaling for mobility between 3GPP access networks, idle mode UE reachability (e.g., control and execution of paging retransmission), registration area management, intra-system and inter-system mobility support, access authentication, access authorization including checking of roaming rights, mobility management control (subscription and policies), network slicing support, and/or session management function (SMF) selection. NAS may refer to the functionality operating between a CN and a UE, and AS may refer to the functionality operating between the UE and a RAN.

The 5G-CN 152 may include one or more additional network functions that are not shown in FIG. 1B for the sake of clarity. For example, the 5G-CN 152 may include one or more of a Session Management Function (SMF), an NR Repository Function (NRF), a Policy Control Function (PCF), a Network Exposure Function (NEF), a Unified Data Management (UDM), an Application Function (AF), and/or an Authentication Server Function (AUSF).

The NG-RAN 154 may connect the 5G-CN 152 to the UEs 156 through radio communications over the air interface. The NG-RAN 154 may include one or more gNBs, illustrated as gNB 160A and gNB 160B (collectively gNBs 160) and/or one or more ng-eNBs, illustrated as ng-eNB 162A and ng-eNB 162B (collectively ng-eNBs 162). The gNBs 160 and ng-eNBs 162 may be more generically referred to as base stations. The gNBs 160 and ng-eNBs 162 may include one or more sets of antennas for communicating with the UEs 156 over an air interface. For example, one or more of the gNBs 160 and/or one or more of the ng-eNBs 162 may include three sets of antennas to respectively control three cells (or sectors). Together, the cells of the gNBs 160 and the ng-eNBs 162 may provide radio coverage to the UEs 156 over a wide geographic area to support UE mobility.

As shown in FIG. 1B, the gNBs 160 and/or the ng-eNBs 162 may be connected to the 5G-CN 152 by means of an NG interface and to other base stations by an Xn interface. The NG and Xn interfaces may be established using direct physical connections and/or indirect connections over an underlying transport network, such as an internet protocol (IP) transport network. The gNBs 160 and/or the ng-eNBs 162 may be connected to the UEs 156 by means of a Uu interface. For example, as illustrated in FIG. 1B, gNB 160A

may be connected to the UE 156A by means of a Uu interface. The NG, Xn, and Uu interfaces are associated with a protocol stack. The protocol stacks associated with the interfaces may be used by the network elements in FIG. 1B to exchange data and signaling messages and may include two planes: a user plane and a control plane. The user plane may handle data of interest to a user. The control plane may handle signaling messages of interest to the network elements.

The gNBs 160 and/or the ng-eNBs 162 may be connected to one or more AMF/UPF functions of the 5G-CN 152, such as the AMF/UPF 158, by means of one or more NG interfaces. For example, the gNB 160A may be connected to the UPF 158B of the AMF/UPF 158 by means of an NG-User plane (NG-U) interface. The NG-U interface may provide delivery (e.g., non-guaranteed delivery) of user plane PDUs between the gNB 160A and the UPF 158B. The gNB 160A may be connected to the AMF 158A by means of an NG-Control plane (NG-C) interface. The NG-C interface may provide, for example, NG interface management, UE context management, UE mobility management, transport of NAS messages, paging, PDU session management, and configuration transfer and/or warning message transmission.

The gNBs 160 may provide NR user plane and control plane protocol terminations towards the UEs 156 over the Uu interface. For example, the gNB 160A may provide NR user plane and control plane protocol terminations toward the UE 156A over a Uu interface associated with a first protocol stack. The ng-eNBs 162 may provide Evolved UMTS Terrestrial Radio Access (E-UTRA) user plane and control plane protocol terminations towards the UEs 156 over a Uu interface, where E-UTRA refers to the 3GPP 4G radio-access technology. For example, the ng-eNB 162B may provide E-UTRA user plane and control plane protocol terminations towards the UE 156B over a Uu interface associated with a second protocol stack.

The 5G-CN 152 was described as being configured to handle NR and 4G radio accesses. It will be appreciated by one of ordinary skill in the art that it may be possible for NR to connect to a 4G core network in a mode known as “non-standalone operation.” In non-standalone operation, a 4G core network is used to provide (or at least support) control-plane functionality (e.g., initial access, mobility, and paging). Although only one AMF/UPF 158 is shown in FIG. 1B, one gNB or ng-eNB may be connected to multiple AMF/UPF nodes to provide redundancy and/or to load share across the multiple AMF/UPF nodes.

As discussed, an interface (e.g., Uu, Xn, and NG interfaces) between the network elements in FIG. 1B may be associated with a protocol stack that the network elements use to exchange data and signaling messages. A protocol stack may include two planes: a user plane and a control plane. The user plane may handle data of interest to a user, and the control plane may handle signaling messages of interest to the network elements.

FIG. 2A and FIG. 2B respectively illustrate examples of NR user plane and NR control plane protocol stacks for the Uu interface that lies between a UE 210 and a gNB 220. The protocol stacks illustrated in FIG. 2A and FIG. 2B may be the same or similar to those used for the Uu interface between, for example, the UE 156A and the gNB 160A shown in FIG. 1B.

FIG. 2A illustrates a NR user plane protocol stack comprising five layers implemented in the UE 210 and the gNB 220. At the bottom of the protocol stack, physical layers (PHYs) 211 and 221 may provide transport services to the higher layers of the protocol stack and may correspond to

layer 1 of the Open Systems Interconnection (OSI) model. The next four protocols above PHYs **211** and **221** comprise media access control layers (MACs) **212** and **222**, radio link control layers (RLCs) **213** and **223**, packet data convergence protocol layers (PDCPs) **214** and **224**, and service data application protocol layers (SDAPs) **215** and **225**. Together, these four protocols may make up layer 2, or the data link layer, of the OSI model.

FIG. 3 illustrates an example of services provided between protocol layers of the NR user plane protocol stack. Starting from the top of FIG. 2A and FIG. 3, the SDAPs **215** and **225** may perform QoS flow handling. The UE **210** may receive services through a PDU session, which may be a logical connection between the UE **210** and a DN. The PDU session may have one or more QoS flows. A UPF of a CN (e.g., the UPF **158B**) may map IP packets to the one or more QoS flows of the PDU session based on QoS requirements (e.g., in terms of delay, data rate, and/or error rate). The SDAPs **215** and **225** may perform mapping/de-mapping between the one or more QoS flows and one or more data radio bearers. The mapping/de-mapping between the QoS flows and the data radio bearers may be determined by the SDAP **225** at the gNB **220**. The SDAP **215** at the UE **210** may be informed of the mapping between the QoS flows and the data radio bearers through reflective mapping or control signaling received from the gNB **220**. For reflective mapping, the SDAP **225** at the gNB **220** may mark the downlink packets with a QoS flow indicator (QFI), which may be observed by the SDAP **215** at the UE **210** to determine the mapping/de-mapping between the QoS flows and the data radio bearers.

The PDCPs **214** and **224** may perform header compression/decompression to reduce the amount of data that needs to be transmitted over the air interface, ciphering/deciphering to prevent unauthorized decoding of data transmitted over the air interface, and integrity protection (to ensure control messages originate from intended sources). The PDCPs **214** and **224** may perform retransmissions of undelivered packets, in-sequence delivery and reordering of packets, and removal of packets received in duplicate due to, for example, an intra-gNB handover. The PDCPs **214** and **224** may perform packet duplication to improve the likelihood of the packet being received and, at the receiver, remove any duplicate packets. Packet duplication may be useful for services that require high reliability.

Although not shown in FIG. 3, PDCPs **214** and **224** may perform mapping/de-mapping between a split radio bearer and RLC channels in a dual connectivity scenario. Dual connectivity is a technique that allows a UE to connect to two cells or, more generally, two cell groups: a master cell group (MCG) and a secondary cell group (SCG). A split bearer is when a single radio bearer, such as one of the radio bearers provided by the PDCPs **214** and **224** as a service to the SDAPs **215** and **225**, is handled by cell groups in dual connectivity. The PDCPs **214** and **224** may map/de-map the split radio bearer between RLC channels belonging to cell groups.

The RLCs **213** and **223** may perform segmentation, retransmission through Automatic Repeat Request (ARQ), and removal of duplicate data units received from MACs **212** and **222**, respectively. The RLCs **213** and **223** may support three transmission modes: transparent mode (TM); unacknowledged mode (UM); and acknowledged mode (AM). Based on the transmission mode an RLC is operating, the RLC may perform one or more of the noted functions. The RLC configuration may be per logical channel with no dependency on numerologies and/or Transmission Time

Interval (TTI) durations. As shown in FIG. 3, the RLCs **213** and **223** may provide RLC channels as a service to PDCPs **214** and **224**, respectively.

The MACs **212** and **222** may perform multiplexing/demultiplexing of logical channels and/or mapping between logical channels and transport channels. The multiplexing/demultiplexing may include multiplexing/demultiplexing of data units, belonging to the one or more logical channels, into/from Transport Blocks (TBs) delivered to/from the PHYs **211** and **221**. The MAC **222** may be configured to perform scheduling, scheduling information reporting, and priority handling between UEs by means of dynamic scheduling. Scheduling may be performed in the gNB **220** (at the MAC **222**) for downlink and uplink. The MACs **212** and **222** may be configured to perform error correction through Hybrid Automatic Repeat Request (HARQ) (e.g., one HARQ entity per carrier in case of Carrier Aggregation (CA)), priority handling between logical channels of the UE **210** by means of logical channel prioritization, and/or padding. The MACs **212** and **222** may support one or more numerologies and/or transmission timings. In an example, mapping restrictions in a logical channel prioritization may control which numerology and/or transmission timing a logical channel may use. As shown in FIG. 3, the MACs **212** and **222** may provide logical channels as a service to the RLCs **213** and **223**.

The PHYs **211** and **221** may perform mapping of transport channels to physical channels and digital and analog signal processing functions for sending and receiving information over the air interface. These digital and analog signal processing functions may include, for example, coding/decoding and modulation/demodulation. The PHYs **211** and **221** may perform multi-antenna mapping. As shown in FIG. 3, the PHYs **211** and **221** may provide one or more transport channels as a service to the MACs **212** and **222**.

FIG. 4A illustrates an example downlink data flow through the NR user plane protocol stack. FIG. 4A illustrates a downlink data flow of three IP packets (n, n+1, and m) through the NR user plane protocol stack to generate two TBs at the gNB **220**. An uplink data flow through the NR user plane protocol stack may be similar to the downlink data flow depicted in FIG. 4A.

The downlink data flow of FIG. 4A begins when SDAP **225** receives the three IP packets from one or more QoS flows and maps the three packets to radio bearers. In FIG. 4A, the SDAP **225** maps IP packets n and n+1 to a first radio bearer **402** and maps IP packet m to a second radio bearer **404**. An SDAP header (labeled with an "H" in FIG. 4A) is added to an IP packet. The data unit from/to a higher protocol layer is referred to as a service data unit (SDU) of the lower protocol layer and the data unit to/from a lower protocol layer is referred to as a protocol data unit (PDU) of the higher protocol layer. As shown in FIG. 4A, the data unit from the SDAP **225** is an SDU of lower protocol layer PDCP **224** and is a PDU of the SDAP **225**.

The remaining protocol layers in FIG. 4A may perform their associated functionality (e.g., with respect to FIG. 3), add corresponding headers, and forward their respective outputs to the next lower layer. For example, the PDCP **224** may perform IP-header compression and ciphering and forward its output to the RLC **223**. The RLC **223** may optionally perform segmentation (e.g., as shown for IP packet m in FIG. 4A) and forward its output to the MAC **222**. The MAC **222** may multiplex a number of RLC PDUs and may attach a MAC subheader to an RLC PDU to form a transport block. In NR, the MAC subheaders may be distributed across the MAC PDU, as illustrated in FIG. 4A.

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In LTE, the MAC subheaders may be entirely located at the beginning of the MAC PDU. The NR MAC PDU structure may reduce processing time and associated latency because the MAC PDU subheaders may be computed before the full MAC PDU is assembled.

FIG. 4B illustrates an example format of a MAC subheader in a MAC PDU. The MAC subheader includes: an SDU length field for indicating the length (e.g., in bytes) of the MAC SDU to which the MAC subheader corresponds; a logical channel identifier (LCID) field for identifying the logical channel from which the MAC SDU originated to aid in the demultiplexing process; a flag (F) for indicating the size of the SDU length field; and a reserved bit (R) field for future use.

FIG. 4B further illustrates MAC control elements (CEs) inserted into the MAC PDU by a MAC, such as MAC 223 or MAC 222. For example, FIG. 4B illustrates two MAC CEs inserted into the MAC PDU. MAC CEs may be inserted at the beginning of a MAC PDU for downlink transmissions (as shown in FIG. 4B) and at the end of a MAC PDU for uplink transmissions. MAC CEs may be used for in-band control signaling. Example MAC CEs include: scheduling-related MAC CEs, such as buffer status reports and power headroom reports; activation/deactivation MAC CEs, such as those for activation/deactivation of PDCP duplication detection, channel state information (CSI) reporting, sounding reference signal (SRS) transmission, and prior configured components; discontinuous reception (DRX) related MAC CEs; timing advance MAC CEs; and random access related MAC CEs. A MAC CE may be preceded by a MAC subheader with a similar format as described for MAC SDUs and may be identified with a reserved value in the LCID field that indicates the type of control information included in the MAC CE.

Before describing the NR control plane protocol stack, logical channels, transport channels, and physical channels are first described as well as a mapping between the channel types. One or more of the channels may be used to carry out functions associated with the NR control plane protocol stack described later below.

FIG. 5A and FIG. 5B illustrate, for downlink and uplink respectively, a mapping between logical channels, transport channels, and physical channels. Information is passed through channels between the RLC, the MAC, and the PHY of the NR protocol stack. A logical channel may be used between the RLC and the MAC and may be classified as a control channel that carries control and configuration information in the NR control plane or as a traffic channel that carries data in the NR user plane. A logical channel may be classified as a dedicated logical channel that is dedicated to a specific UE or as a common logical channel that may be used by more than one UE. A logical channel may also be defined by the type of information it carries. The set of logical channels defined by NR include, for example:

- a paging control channel (PCCH) for carrying paging messages used to page a UE whose location is not known to the network on a cell level;
- a broadcast control channel (BCCH) for carrying system information messages in the form of a master information block (MIB) and several system information blocks (SIBs), wherein the system information messages may be used by the UEs to obtain information about how a cell is configured and how to operate within the cell;
- a common control channel (CCCH) for carrying control messages together with random access;

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a dedicated control channel (DCCH) for carrying control messages to/from a specific the UE to configure the UE; and

a dedicated traffic channel (DTCH) for carrying user data to/from a specific the UE.

Transport channels are used between the MAC and PHY layers and may be defined by how the information they carry is transmitted over the air interface. The set of transport channels defined by NR include, for example:

a paging channel (PCH) for carrying paging messages that originated from the PCCH;

a broadcast channel (BCH) for carrying the MIB from the BCCH;

a downlink shared channel (DL-SCH) for carrying downlink data and signaling messages, including the SIBs from the BCCH;

an uplink shared channel (UL-SCH) for carrying uplink data and signaling messages; and

a random access channel (RACH) for allowing a UE to contact the network without any prior scheduling.

The PHY may use physical channels to pass information between processing levels of the PHY. A physical channel may have an associated set of time-frequency resources for carrying the information of one or more transport channels. The PHY may generate control information to support the low-level operation of the PHY and provide the control information to the lower levels of the PHY via physical control channels, known as L1/L2 control channels. The set of physical channels and physical control channels defined by NR include, for example:

a physical broadcast channel (PBCH) for carrying the MIB from the BCH;

a physical downlink shared channel (PDSCH) for carrying downlink data and signaling messages from the DL-SCH, as well as paging messages from the PCH;

a physical downlink control channel (PDCCH) for carrying downlink control information (DCI), which may include downlink scheduling commands, uplink scheduling grants, and uplink power control commands;

a physical uplink shared channel (PUSCH) for carrying uplink data and signaling messages from the UL-SCH and in some instances uplink control information (UCI) as described below;

a physical uplink control channel (PUCCH) for carrying UCI, which may include HARQ acknowledgments, channel quality indicators (CQI), pre-coding matrix indicators (PMI), rank indicators (RI), and scheduling requests (SR); and

a physical random access channel (PRACH) for random access.

Similar to the physical control channels, the physical layer generates physical signals to support the low-level operation of the physical layer. As shown in FIG. 5A and FIG. 5B, the physical layer signals defined by NR include: primary synchronization signals (PSS), secondary synchronization signals (SSS), channel state information reference signals (CSI-RS), demodulation reference signals (DMRS), sounding reference signals (SRS), and phase-tracking reference signals (PT-RS). These physical layer signals will be described in greater detail below.

FIG. 2B illustrates an example NR control plane protocol stack. As shown in FIG. 2B, the NR control plane protocol stack may use the same/similar first four protocol layers as the example NR user plane protocol stack. These four protocol layers include the PHYs 211 and 221, the MACs 212 and 222, the RLCs 213 and 223, and the PDCPs 214 and 224. Instead of having the SDAPs 215 and 225 at the top of

the stack as in the NR user plane protocol stack, the NR control plane stack has radio resource controls (RRCs) **216** and **226** and NAS protocols **217** and **237** at the top of the NR control plane protocol stack.

The NAS protocols **217** and **237** may provide control plane functionality between the UE **210** and the AMF **230** (e.g., the AMF **158A**) or, more generally, between the UE **210** and the CN. The NAS protocols **217** and **237** may provide control plane functionality between the UE **210** and the AMF **230** via signaling messages, referred to as NAS messages. There is no direct path between the UE **210** and the AMF **230** through which the NAS messages can be transported. The NAS messages may be transported using the AS of the Uu and NG interfaces. NAS protocols **217** and **237** may provide control plane functionality such as authentication, security, connection setup, mobility management, and session management.

The RRCs **216** and **226** may provide control plane functionality between the UE **210** and the gNB **220** or, more generally, between the UE **210** and the RAN. The RRCs **216** and **226** may provide control plane functionality between the UE **210** and the gNB **220** via signaling messages, referred to as RRC messages. RRC messages may be transmitted between the UE **210** and the RAN using signaling radio bearers and the same/similar PDCP, RLC, MAC, and PHY protocol layers. The MAC may multiplex control-plane and user-plane data into the same transport block (TB). The RRCs **216** and **226** may provide control plane functionality such as: broadcast of system information related to AS and NAS; paging initiated by the CN or the RAN; establishment, maintenance and release of an RRC connection between the UE **210** and the RAN; security functions including key management; establishment, configuration, maintenance and release of signaling radio bearers and data radio bearers; mobility functions; QoS management functions; the UE measurement reporting and control of the reporting; detection of and recovery from radio link failure (RLF); and/or NAS message transfer. As part of establishing an RRC connection, RRCs **216** and **226** may establish an RRC context, which may involve configuring parameters for communication between the UE **210** and the RAN.

FIG. 6 is an example diagram showing RRC state transitions of a UE. The UE may be the same or similar to the wireless device **106** depicted in FIG. 1A, the UE **210** depicted in FIG. 2A and FIG. 2B, or any other wireless device described in the present disclosure. As illustrated in FIG. 6, a UE may be in at least one of three RRC states: RRC connected **602** (e.g., RRC_CONNECTED), RRC idle **604** (e.g., RRC_IDLE), and RRC inactive **606** (e.g., RRC_INACTIVE).

In RRC connected **602**, the UE has an established RRC context and may have at least one RRC connection with a base station. The base station may be similar to one of the one or more base stations included in the RAN **104** depicted in FIG. 1A, one of the gNBs **160** or ng-eNBs **162** depicted in FIG. 1B, the gNB **220** depicted in FIG. 2A and FIG. 2B, or any other base station described in the present disclosure. The base station with which the UE is connected may have the RRC context for the UE. The RRC context, referred to as the UE context, may comprise parameters for communication between the UE and the base station. These parameters may include, for example: one or more AS contexts; one or more radio link configuration parameters; bearer configuration information (e.g., relating to a data radio bearer, signaling radio bearer, logical channel, QoS flow, and/or PDU session); security information; and/or PHY, MAC, RLC, PDCP, and/or SDAP layer configuration infor-

mation. While in RRC connected **602**, mobility of the UE may be managed by the RAN (e.g., the RAN **104** or the NG-RAN **154**). The UE may measure the signal levels (e.g., reference signal levels) from a serving cell and neighboring cells and report these measurements to the base station currently serving the UE. The UE's serving base station may request a handover to a cell of one of the neighboring base stations based on the reported measurements. The RRC state may transition from RRC connected **602** to RRC idle **604** through a connection release procedure **608** or to RRC inactive **606** through a connection inactivation procedure **610**.

In RRC idle **604**, an RRC context may not be established for the UE. In RRC idle **604**, the UE may not have an RRC connection with the base station. While in RRC idle **604**, the UE may be in a sleep state for the majority of the time (e.g., to conserve battery power). The UE may wake up periodically (e.g., once in every discontinuous reception cycle) to monitor for paging messages from the RAN. Mobility of the UE may be managed by the UE through a procedure known as cell reselection. The RRC state may transition from RRC idle **604** to RRC connected **602** through a connection establishment procedure **612**, which may involve a random access procedure as discussed in greater detail below.

In RRC inactive **606**, the RRC context previously established is maintained in the UE and the base station. This allows for a fast transition to RRC connected **602** with reduced signaling overhead as compared to the transition from RRC idle **604** to RRC connected **602**. While in RRC inactive **606**, the UE may be in a sleep state and mobility of the UE may be managed by the UE through cell reselection. The RRC state may transition from RRC inactive **606** to RRC connected **602** through a connection resume procedure **614** or to RRC idle **604** through a connection release procedure **616** that may be the same as or similar to connection release procedure **608**.

An RRC state may be associated with a mobility management mechanism. In RRC idle **604** and RRC inactive **606**, mobility is managed by the UE through cell reselection. The purpose of mobility management in RRC idle **604** and RRC inactive **606** is to allow the network to be able to notify the UE of an event via a paging message without having to broadcast the paging message over the entire mobile communications network. The mobility management mechanism used in RRC idle **604** and RRC inactive **606** may allow the network to track the UE on a cell-group level so that the paging message may be broadcast over the cells of the cell group that the UE currently resides within instead of the entire mobile communication network. The mobility management mechanisms for RRC idle **604** and RRC inactive **606** track the UE on a cell-group level. They may do so using different granularities of grouping. For example, there may be three levels of cell-grouping granularity: individual cells; cells within a RAN area identified by a RAN area identifier (RAI); and cells within a group of RAN areas, referred to as a tracking area and identified by a tracking area identifier (TAI).

Tracking areas may be used to track the UE at the CN level. The CN (e.g., the CN **102** or the 5G-CN **152**) may provide the UE with a list of TAIs associated with a UE registration area. If the UE moves, through cell reselection, to a cell associated with a TAI not included in the list of TAIs associated with the UE registration area, the UE may perform a registration update with the CN to allow the CN to update the UE's location and provide the UE with a new the UE registration area.

RAN areas may be used to track the UE at the RAN level. For a UE in RRC inactive **606** state, the UE may be assigned a RAN notification area. A RAN notification area may comprise one or more cell identities, a list of RANs, or a list of TAIs. In an example, a base station may belong to one or more RAN notification areas. In an example, a cell may belong to one or more RAN notification areas. If the UE moves, through cell reselection, to a cell not included in the RAN notification area assigned to the UE, the UE may perform a notification area update with the RAN to update the UE's RAN notification area.

A base station storing an RRC context for a UE or a last serving base station of the UE may be referred to as an anchor base station. An anchor base station may maintain an RRC context for the UE at least during a period of time that the UE stays in a RAN notification area of the anchor base station and/or during a period of time that the UE stays in RRC inactive **606**.

A gNB, such as gNBs **160** in FIG. 1B, may be split in two parts: a central unit (gNB-CU), and one or more distributed units (gNB-DU). A gNB-CU may be coupled to one or more gNB-DUs using an F1 interface. The gNB-CU may comprise the RRC, the PDCP, and the SDAP. A gNB-DU may comprise the RLC, the MAC, and the PHY.

In NR, the physical signals and physical channels (discussed with respect to FIG. 5A and FIG. 5B) may be mapped onto orthogonal frequency divisional multiplexing (OFDM) symbols. OFDM is a multicarrier communication scheme that transmits data over F orthogonal subcarriers (or tones). Before transmission, the data may be mapped to a series of complex symbols (e.g., M-quadrature amplitude modulation (M-QAM) or M-phase shift keying (M-PSK) symbols), referred to as source symbols, and divided into F parallel symbol streams. The F parallel symbol streams may be treated as though they are in the frequency domain and used as inputs to an Inverse Fast Fourier Transform (IFFT) block that transforms them into the time domain. The IFFT block may take in F source symbols at a time, one from each of the F parallel symbol streams, and use each source symbol to modulate the amplitude and phase of one of F sinusoidal basis functions that correspond to the F orthogonal subcarriers. The output of the IFFT block may be F time-domain samples that represent the summation of the F orthogonal subcarriers. The F time-domain samples may form a single OFDM symbol. After some processing (e.g., addition of a cyclic prefix) and up-conversion, an OFDM symbol provided by the IFFT block may be transmitted over the air interface on a carrier frequency. The F parallel symbol streams may be mixed using an FFT block before being processed by the IFFT block. This operation produces Discrete Fourier Transform (DFT)-precoded OFDM symbols and may be used by UEs in the uplink to reduce the peak to average power ratio (PAPR). Inverse processing may be performed on the OFDM symbol at a receiver using an FFT block to recover the data mapped to the source symbols.

FIG. 7 illustrates an example configuration of an NR frame into which OFDM symbols are grouped. An NR frame may be identified by a system frame number (SFN). The SFN may repeat with a period of 1024 frames. As illustrated, one NR frame may be 10 milliseconds (ms) in duration and may include 10 subframes that are 1 ms in duration. A subframe may be divided into slots that include, for example, 14 OFDM symbols per slot.

The duration of a slot may depend on the numerology used for the OFDM symbols of the slot. In NR, a flexible numerology is supported to accommodate different cell deployments (e.g., cells with carrier frequencies below 1

GHz up to cells with carrier frequencies in the mm-wave range). A numerology may be defined in terms of subcarrier spacing and cyclic prefix duration. For a numerology in NR, subcarrier spacings may be scaled up by powers of two from a baseline subcarrier spacing of 15 kHz, and cyclic prefix durations may be scaled down by powers of two from a baseline cyclic prefix duration of 4.7 μ s. For example, NR defines numerologies with the following subcarrier spacing/cyclic prefix duration combinations: 15 kHz/4.7 μ s; 30 kHz/2.3 μ s; 60 kHz/1.2 μ s; 120 kHz/0.59 μ s; and 240 kHz/0.29 μ s.

A slot may have a fixed number of OFDM symbols (e.g., 14 OFDM symbols). A numerology with a higher subcarrier spacing has a shorter slot duration and, correspondingly, more slots per subframe. FIG. 7 illustrates this numerology-dependent slot duration and slots-per-subframe transmission structure (the numerology with a subcarrier spacing of 240 kHz is not shown in FIG. 7 for case of illustration). A subframe in NR may be used as a numerology-independent time reference, while a slot may be used as the unit upon which uplink and downlink transmissions are scheduled. To support low latency, scheduling in NR may be decoupled from the slot duration and start at any OFDM symbol and last for as many symbols as needed for a transmission. These partial slot transmissions may be referred to as mini-slot or subslot transmissions.

FIG. 8 illustrates an example configuration of a slot in the time and frequency domain for an NR carrier. The slot includes resource elements (REs) and resource blocks (RBs). An RE is the smallest physical resource in NR. An RE spans one OFDM symbol in the time domain by one subcarrier in the frequency domain as shown in FIG. 8. An RB spans twelve consecutive REs in the frequency domain as shown in FIG. 8. An NR carrier may be limited to a width of 275 RBs or $275 \times 12 = 3300$ subcarriers. Such a limitation, if used, may limit the NR carrier to 50, 100, 200, and 400 MHz for subcarrier spacings of 15, 30, 60, and 120 kHz, respectively, where the 400 MHz bandwidth may be set based on a 400 MHz per carrier bandwidth limit.

FIG. 8 illustrates a single numerology being used across the entire bandwidth of the NR carrier. In other example configurations, multiple numerologies may be supported on the same carrier.

NR may support wide carrier bandwidths (e.g., up to 400 MHz for a subcarrier spacing of 120 kHz). Not all UEs may be able to receive the full carrier bandwidth (e.g., due to hardware limitations). Also, receiving the full carrier bandwidth may be prohibitive in terms of UE power consumption. In an example, to reduce power consumption and/or for other purposes, a UE may adapt the size of the UE's receive bandwidth based on the amount of traffic the UE is scheduled to receive. This is referred to as bandwidth adaptation.

NR defines bandwidth parts (BWPs) to support UEs not capable of receiving the full carrier bandwidth and to support bandwidth adaptation. In an example, a BWP may be defined by a subset of contiguous RBs on a carrier. A UE may be configured (e.g., via RRC layer) with one or more downlink BWPs and one or more uplink BWPs per serving cell (e.g., up to four downlink BWPs and up to four uplink BWPs per serving cell). At a given time, one or more of the configured BWPs for a serving cell may be active. These one or more BWPs may be referred to as active BWPs of the serving cell. When a serving cell is configured with a secondary uplink carrier, the serving cell may have one or more first active BWPs in the uplink carrier and one or more second active BWPs in the secondary uplink carrier.

For unpaired spectra, a downlink BWP from a set of configured downlink BWPs may be linked with an uplink BWP from a set of configured uplink BWPs if a downlink BWP index of the downlink BWP and an uplink BWP index of the uplink BWP are the same. For unpaired spectra, a UE may expect that a center frequency for a downlink BWP is the same as a center frequency for an uplink BWP.

For a downlink BWP in a set of configured downlink BWPs on a primary cell (PCell), a base station may configure a UE with one or more control resource sets (CORE-SETs) for at least one search space. A search space is a set of locations in the time and frequency domains where the UE may find control information. The search space may be a UE-specific search space or a common search space (potentially usable by a plurality of UEs). For example, a base station may configure a UE with a common search space, on a PCell or on a primary secondary cell (PSCell), in an active downlink BWP.

For an uplink BWP in a set of configured uplink BWPs, a BS may configure a UE with one or more resource sets for one or more PUCCH transmissions. A UE may receive downlink receptions (e.g., PDCCH or PDSCH) in a downlink BWP according to a configured numerology (e.g., subcarrier spacing and cyclic prefix duration) for the downlink BWP. The UE may transmit uplink transmissions (e.g., PUCCH or PUSCH) in an uplink BWP according to a configured numerology (e.g., subcarrier spacing and cyclic prefix length for the uplink BWP).

One or more BWP indicator fields may be provided in Downlink Control Information (DCI). A value of a BWP indicator field may indicate which BWP in a set of configured BWPs is an active downlink BWP for one or more downlink receptions. The value of the one or more BWP indicator fields may indicate an active uplink BWP for one or more uplink transmissions.

A base station may semi-statically configure a UE with a default downlink BWP within a set of configured downlink BWPs associated with a PCell. If the base station does not provide the default downlink BWP to the UE, the default downlink BWP may be an initial active downlink BWP. The UE may determine which BWP is the initial active downlink BWP based on a CORESET configuration obtained using the PBCH.

A base station may configure a UE with a BWP inactivity timer value for a PCell. The UE may start or restart a BWP inactivity timer at any appropriate time. For example, the UE may start or restart the BWP inactivity timer (a) when the UE detects a DCI indicating an active downlink BWP other than a default downlink BWP for a paired spectra operation; or (b) when a UE detects a DCI indicating an active downlink BWP or active uplink BWP other than a default downlink BWP or uplink BWP for an unpaired spectra operation. If the UE does not detect DCI during an interval of time (e.g., 1 ms or 0.5 ms), the UE may run the BWP inactivity timer toward expiration (for example, increment from zero to the BWP inactivity timer value, or decrement from the BWP inactivity timer value to zero). When the BWP inactivity timer expires, the UE may switch from the active downlink BWP to the default downlink BWP.

In an example, a base station may semi-statically configure a UE with one or more BWPs. A UE may switch an active BWP from a first BWP to a second BWP in response to receiving a DCI indicating the second BWP as an active BWP and/or in response to an expiry of the BWP inactivity timer (e.g., if the second BWP is the default BWP).

Downlink and uplink BWP switching (where BWP switching refers to switching from a currently active BWP to a not currently active BWP) may be performed independently in paired spectra. In unpaired spectra, downlink and uplink BWP switching may be performed simultaneously. Switching between configured BWPs may occur based on RRC signaling, DCI, expiration of a BWP inactivity timer, and/or an initiation of random access.

FIG. 9 illustrates an example of bandwidth adaptation using three configured BWPs for an NR carrier. A UE configured with the three BWPs may switch from one BWP to another BWP at a switching point. In the example illustrated in FIG. 9, the BWPs include: a BWP 902 with a bandwidth of 40 MHz and a subcarrier spacing of 15 kHz; a BWP 904 with a bandwidth of 10 MHz and a subcarrier spacing of 15 kHz; and a BWP 906 with a bandwidth of 20 MHz and a subcarrier spacing of 60 kHz. The BWP 902 may be an initial active BWP, and the BWP 904 may be a default BWP. The UE may switch between BWPs at switching points. In the example of FIG. 9, the UE may switch from the BWP 902 to the BWP 904 at a switching point 908. The switching at the switching point 908 may occur for any suitable reason, for example, in response to an expiry of a BWP inactivity timer (indicating switching to the default BWP) and/or in response to receiving a DCI indicating BWP 904 as the active BWP. The UE may switch at a switching point 910 from active BWP 904 to BWP 906 in response receiving a DCI indicating BWP 906 as the active BWP. The UE may switch at a switching point 912 from active BWP 906 to BWP 904 in response to an expiry of a BWP inactivity timer and/or in response receiving a DCI indicating BWP 904 as the active BWP. The UE may switch at a switching point 914 from active BWP 904 to BWP 902 in response receiving a DCI indicating BWP 902 as the active BWP.

If a UE is configured for a secondary cell with a default downlink BWP in a set of configured downlink BWPs and a timer value, UE procedures for switching BWPs on a secondary cell may be the same/similar as those on a primary cell. For example, the UE may use the timer value and the default downlink BWP for the secondary cell in the same/similar manner as the UE would use these values for a primary cell.

To provide for greater data rates, two or more carriers can be aggregated and simultaneously transmitted to/from the same UE using carrier aggregation (CA). The aggregated carriers in CA may be referred to as component carriers (CCs). When CA is used, there are a number of serving cells for the UE, one for a CC. The CCs may have three configurations in the frequency domain.

FIG. 10A illustrates the three CA configurations with two CCs. In the intraband, contiguous configuration 1002, the two CCs are aggregated in the same frequency band (frequency band A) and are located directly adjacent to each other within the frequency band. In the intraband, non-contiguous configuration 1004, the two CCs are aggregated in the same frequency band (frequency band A) and are separated in the frequency band by a gap. In the interband configuration 1006, the two CCs are located in frequency bands (frequency band A and frequency band B).

In an example, up to 32 CCs may be aggregated. The aggregated CCs may have the same or different bandwidths, subcarrier spacing, and/or duplexing schemes (TDD or FDD). A serving cell for a UE using CA may have a downlink CC. For FDD, one or more uplink CCs may be optionally configured for a serving cell. The ability to aggregate more downlink carriers than uplink carriers may

be useful, for example, when the UE has more data traffic in the downlink than in the uplink.

When CA is used, one of the aggregated cells for a UE may be referred to as a primary cell (PCell). The PCell may be the serving cell that the UE initially connects to at RRC connection establishment, reestablishment, and/or handover. The PCell may provide the UE with NAS mobility information and the security input. UEs may have different PCells. In the downlink, the carrier corresponding to the PCell may be referred to as the downlink primary CC (DL PCC). In the uplink, the carrier corresponding to the PCell may be referred to as the uplink primary CC (UL PCC). The other aggregated cells for the UE may be referred to as secondary cells (SCells). In an example, the SCells may be configured after the PCell is configured for the UE. For example, an SCell may be configured through an RRC Connection Reconfiguration procedure. In the downlink, the carrier corresponding to an SCell may be referred to as a downlink secondary CC (DL SCC). In the uplink, the carrier corresponding to the SCell may be referred to as the uplink secondary CC (UL SCC).

Configured SCells for a UE may be activated and deactivated based on, for example, traffic and channel conditions. Deactivation of an SCell may mean that PDCCH and PDSCH reception on the SCell is stopped and PUSCH, SRS, and CQI transmissions on the SCell are stopped. Configured SCells may be activated and deactivated using a MAC CE with respect to FIG. 4B. For example, a MAC CE may use a bitmap (e.g., one bit per SCell) to indicate which SCells (e.g., in a subset of configured SCells) for the UE are activated or deactivated. Configured SCells may be deactivated in response to an expiration of an SCell deactivation timer (e.g., one SCell deactivation timer per SCell).

Downlink control information, such as scheduling assignments and scheduling grants, for a cell may be transmitted on the cell corresponding to the assignments and grants, which is known as self-scheduling. The DCI for the cell may be transmitted on another cell, which is known as cross-carrier scheduling. Uplink control information (e.g., HARQ acknowledgments and channel state feedback, such as CQI, PMI, and/or RI) for aggregated cells may be transmitted on the PUCCH of the PCell. For a larger number of aggregated downlink CCs, the PUCCH of the PCell may become overloaded. Cells may be divided into multiple PUCCH groups.

FIG. 10B illustrates an example of how aggregated cells may be configured into one or more PUCCH groups. A PUCCH group 1010 and a PUCCH group 1050 may include one or more downlink CCs, respectively. In the example of FIG. 10B, the PUCCH group 1010 includes three downlink CCs: a PCell 1011, an SCell 1012, and an SCell 1013. The PUCCH group 1050 includes three downlink CCs in the present example: a PCell 1051, an SCell 1052, and an SCell 1053. One or more uplink CCs may be configured as a PCell 1021, an SCell 1022, and an SCell 1023. One or more other uplink CCs may be configured as a primary Scell (PSCell) 1061, an SCell 1062, and an SCell 1063. Uplink control information (UCI) related to the downlink CCs of the PUCCH group 1010, shown as UCI 1031, UCI 1032, and UCI 1033, may be transmitted in the uplink of the PCell 1021. Uplink control information (UCI) related to the downlink CCs of the PUCCH group 1050, shown as UCI 1071, UCI 1072, and UCI 1073, may be transmitted in the uplink of the PSCell 1061. In an example, if the aggregated cells depicted in FIG. 10B were not divided into the PUCCH group 1010 and the PUCCH group 1050, a single uplink PCell to transmit UCI relating to the downlink CCs, and the

PCell may become overloaded. By dividing transmissions of UCI between the PCell 1021 and the PSCell 1061, overloading may be prevented.

A cell, comprising a downlink carrier and optionally an uplink carrier, may be assigned with a physical cell ID and a cell index. The physical cell ID or the cell index may identify a downlink carrier and/or an uplink carrier of the cell, for example, depending on the context in which the physical cell ID is used. A physical cell ID may be determined using a synchronization signal transmitted on a downlink component carrier. A cell index may be determined using RRC messages. In the disclosure, a physical cell ID may be referred to as a carrier ID, and a cell index may be referred to as a carrier index. For example, when the disclosure refers to a first physical cell ID for a first downlink carrier, the disclosure may mean the first physical cell ID is for a cell comprising the first downlink carrier. The same/similar concept may apply to, for example, a carrier activation. When the disclosure indicates that a first carrier is activated, the specification may mean that a cell comprising the first carrier is activated.

In CA, a multi-carrier nature of a PHY may be exposed to a MAC. In an example, a HARQ entity may operate on a serving cell. A transport block may be generated per assignment/grant per serving cell. A transport block and potential HARQ retransmissions of the transport block may be mapped to a serving cell.

In the downlink, a base station may transmit (e.g., unicast, multicast, and/or broadcast) one or more Reference Signals (RSs) to a UE (e.g., PSS, SSS, CSI-RS, DMRS, and/or PT-RS, as shown in FIG. 5A). In the uplink, the UE may transmit one or more RSs to the base station (e.g., DMRS, PT-RS, and/or SRS, as shown in FIG. 5B). The PSS and the SSS may be transmitted by the base station and used by the UE to synchronize the UE to the base station. The PSS and the SSS may be provided in a synchronization signal (SS)/physical broadcast channel (PBCH) block that includes the PSS, the SSS, and the PBCH. The base station may periodically transmit a burst of SS/PBCH blocks.

FIG. 11A illustrates an example of an SS/PBCH block's structure and location. A burst of SS/PBCH blocks may include one or more SS/PBCH blocks (e.g., 4 SS/PBCH blocks, as shown in FIG. 11A). Bursts may be transmitted periodically (e.g., every 2 frames or 20 ms). A burst may be restricted to a half-frame (e.g., a first half-frame having a duration of 5 ms). It will be understood that FIG. 11A is an example, and that these parameters (number of SS/PBCH blocks per burst, periodicity of bursts, position of burst within the frame) may be configured based on, for example: a carrier frequency of a cell in which the SS/PBCH block is transmitted; a numerology or subcarrier spacing of the cell; a configuration by the network (e.g., using RRC signaling); or any other suitable factor. In an example, the UE may assume a subcarrier spacing for the SS/PBCH block based on the carrier frequency being monitored, unless the radio network configured the UE to assume a different subcarrier spacing.

The SS/PBCH block may span one or more OFDM symbols in the time domain (e.g., 4 OFDM symbols, as shown in the example of FIG. 11A) and may span one or more subcarriers in the frequency domain (e.g., 240 contiguous subcarriers). The PSS, the SSS, and the PBCH may have a common center frequency. The PSS may be transmitted first and may span, for example, 1 OFDM symbol and 127 subcarriers. The SSS may be transmitted after the PSS (e.g., two symbols later) and may span 1 OFDM symbol and

127 subcarriers. The PBCH may be transmitted after the PSS (e.g., across the next 3 OFDM symbols) and may span 240 subcarriers.

The location of the SS/PBCH block in the time and frequency domains may not be known to the UE (e.g., if the UE is searching for the cell). To find and select the cell, the UE may monitor a carrier for the PSS. For example, the UE may monitor a frequency location within the carrier. If the PSS is not found after a certain duration (e.g., 20 ms), the UE may search for the PSS at a different frequency location within the carrier, as indicated by a synchronization raster. If the PSS is found at a location in the time and frequency domains, the UE may determine, based on a known structure of the SS/PBCH block, the locations of the SSS and the PBCH, respectively. The SS/PBCH block may be a cell-defining SS block (CD-SSB). In an example, a primary cell may be associated with a CD-SSB. The CD-SSB may be located on a synchronization raster. In an example, a cell selection/search and/or reselection may be based on the CD-SSB.

The SS/PBCH block may be used by the UE to determine one or more parameters of the cell. For example, the UE may determine a physical cell identifier (PCI) of the cell based on the sequences of the PSS and the SSS, respectively. The UE may determine a location of a frame boundary of the cell based on the location of the SS/PBCH block. For example, the SS/PBCH block may indicate that it has been transmitted in accordance with a transmission pattern, wherein a SS/PBCH block in the transmission pattern is a known distance from the frame boundary.

The PBCH may use a QPSK modulation and may use forward error correction (FEC). The FEC may use polar coding. One or more symbols spanned by the PBCH may carry one or more DMRSs for demodulation of the PBCH. The PBCH may include an indication of a current system frame number (SFN) of the cell and/or a SS/PBCH block timing index. These parameters may facilitate time synchronization of the UE to the base station. The PBCH may include a master information block (MIB) used to provide the UE with one or more parameters. The MIB may be used by the UE to locate remaining minimum system information (RMSI) associated with the cell. The RMSI may include a System Information Block Type 1 (SIB1). The SIB1 may contain information needed by the UE to access the cell. The UE may use one or more parameters of the MIB to monitor PDCCH, which may be used to schedule PDSCH. The PDSCH may include the SIB1. The SIB1 may be decoded using parameters provided in the MIB. The PBCH may indicate an absence of SIB1. Based on the PBCH indicating the absence of SIB1, the UE may be pointed to a frequency. The UE may search for an SS/PBCH block at the frequency to which the UE is pointed.

The UE may assume that one or more SS/PBCH blocks transmitted with a same SS/PBCH block index are quasi co-located (QCLed) (e.g., having the same/similar Doppler spread, Doppler shift, average gain, average delay, and/or spatial Rx parameters). The UE may not assume QCL for SS/PBCH block transmissions having different SS/PBCH block indices.

SS/PBCH blocks (e.g., those within a half-frame) may be transmitted in spatial directions (e.g., using different beams that span a coverage area of the cell). In an example, a first SS/PBCH block may be transmitted in a first spatial direction using a first beam, and a second SS/PBCH block may be transmitted in a second spatial direction using a second beam.

In an example, within a frequency span of a carrier, a base station may transmit a plurality of SS/PBCH blocks. In an example, a first PCI of a first SS/PBCH block of the plurality of SS/PBCH blocks may be different from a second PCI of a second SS/PBCH block of the plurality of SS/PBCH blocks. The PCIs of SS/PBCH blocks transmitted in different frequency locations may be different or the same.

The CSI-RS may be transmitted by the base station and used by the UE to acquire channel state information (CSI). The base station may configure the UE with one or more CSI-RSs for channel estimation or any other suitable purpose. The base station may configure a UE with one or more of the same/similar CSI-RSs. The UE may measure the one or more CSI-RSs. The UE may estimate a downlink channel state and/or generate a CSI report based on the measuring of the one or more downlink CSI-RSs. The UE may provide the CSI report to the base station. The base station may use feedback provided by the UE (e.g., the estimated downlink channel state) to perform link adaptation.

The base station may semi-statically configure the UE with one or more CSI-RS resource sets. A CSI-RS resource may be associated with a location in the time and frequency domains and a periodicity. The base station may selectively activate and/or deactivate a CSI-RS resource. The base station may indicate to the UE that a CSI-RS resource in the CSI-RS resource set is activated and/or deactivated.

The base station may configure the UE to report CSI measurements. The base station may configure the UE to provide CSI reports periodically, aperiodically, or semi-persistently. For periodic CSI reporting, the UE may be configured with a timing and/or periodicity of a plurality of CSI reports. For aperiodic CSI reporting, the base station may request a CSI report. For example, the base station may command the UE to measure a configured CSI-RS resource and provide a CSI report relating to the measurements. For semi-persistent CSI reporting, the base station may configure the UE to transmit periodically, and selectively activate or deactivate the periodic reporting. The base station may configure the UE with a CSI-RS resource set and CSI reports using RRC signaling.

The CSI-RS configuration may comprise one or more parameters indicating, for example, up to 32 antenna ports. The UE may be configured to employ the same OFDM symbols for a downlink CSI-RS and a control resource set (CORESET) when the downlink CSI-RS and CORESET are spatially QCLed and resource elements associated with the downlink CSI-RS are outside of the physical resource blocks (PRBs) configured for the CORESET. The UE may be configured to employ the same OFDM symbols for downlink CSI-RS and SS/PBCH blocks when the downlink CSI-RS and SS/PBCH blocks are spatially QCLed and resource elements associated with the downlink CSI-RS are outside of PRBs configured for the SS/PBCH blocks.

Downlink DMRSs may be transmitted by a base station and used by a UE for channel estimation. For example, the downlink DMRS may be used for coherent demodulation of one or more downlink physical channels (e.g., PDSCH). An NR network may support one or more variable and/or configurable DMRS patterns for data demodulation. At least one downlink DMRS configuration may support a front-loaded DMRS pattern. A front-loaded DMRS may be mapped over one or more OFDM symbols (e.g., one or two adjacent OFDM symbols). A base station may semi-statically configure the UE with a number (e.g., a maximum number) of front-loaded DMRS symbols for PDSCH. A DMRS configuration may support one or more DMRS ports. For example, for single user-MIMO, a DMRS configuration

may support up to eight orthogonal downlink DMRS ports per UE. For multiuser-MIMO, a DMRS configuration may support up to 4 orthogonal downlink DMRS ports per UE. A radio network may support (e.g., at least for CP-OFDM) a common DMRS structure for downlink and uplink, wherein a DMRS location, a DMRS pattern, and/or a scrambling sequence may be the same or different. The base station may transmit a downlink DMRS and a corresponding PDSCH using the same precoding matrix. The UE may use the one or more downlink DMRSs for coherent demodulation/channel estimation of the PDSCH.

In an example, a transmitter (e.g., a base station) may use a precoder matrices for a part of a transmission bandwidth. For example, the transmitter may use a first precoder matrix for a first bandwidth and a second precoder matrix for a second bandwidth. The first precoder matrix and the second precoder matrix may be different based on the first bandwidth being different from the second bandwidth. The UE may assume that a same precoding matrix is used across a set of PRBs. The set of PRBs may be denoted as a precoding resource block group (PRG).

A PDSCH may comprise one or more layers. The UE may assume that at least one symbol with DMRS is present on a layer of the one or more layers of the PDSCH. A higher layer may configure up to 3 DMRSs for the PDSCH.

Downlink PT-RS may be transmitted by a base station and used by a UE for phase-noise compensation. Whether a downlink PT-RS is present or not may depend on an RRC configuration. The presence and/or pattern of the downlink PT-RS may be configured on a UE-specific basis using a combination of RRC signaling and/or an association with one or more parameters employed for other purposes (e.g., modulation and coding scheme (MCS)), which may be indicated by DCI. When configured, a dynamic presence of a downlink PT-RS may be associated with one or more DCI parameters comprising at least MCS. An NR network may support a plurality of PT-RS densities defined in the time and/or frequency domains. When present, a frequency domain density may be associated with at least one configuration of a scheduled bandwidth. The UE may assume a same precoding for a DMRS port and a PT-RS port. A number of PT-RS ports may be fewer than a number of DMRS ports in a scheduled resource. Downlink PT-RS may be confined in the scheduled time/frequency duration for the UE. Downlink PT-RS may be transmitted on symbols to facilitate phase tracking at the receiver.

The UE may transmit an uplink DMRS to a base station for channel estimation. For example, the base station may use the uplink DMRS for coherent demodulation of one or more uplink physical channels. For example, the UE may transmit an uplink DMRS with a PUSCH and/or a PUCCH. The uplink DM-RS may span a range of frequencies that is similar to a range of frequencies associated with the corresponding physical channel. The base station may configure the UE with one or more uplink DMRS configurations. At least one DMRS configuration may support a front-loaded DMRS pattern. The front-loaded DMRS may be mapped over one or more OFDM symbols (e.g., one or two adjacent OFDM symbols). One or more uplink DMRSs may be configured to transmit at one or more symbols of a PUSCH and/or a PUCCH. The base station may semi-statically configure the UE with a number (e.g. maximum number) of front-loaded DMRS symbols for the PUSCH and/or the PUCCH, which the UE may use to schedule a single-symbol DMRS and/or a double-symbol DMRS. An NR network may support (e.g., for cyclic prefix orthogonal frequency division multiplexing (CP-OFDM)) a common DMRS

structure for downlink and uplink, wherein a DMRS location, a DMRS pattern, and/or a scrambling sequence for the DMRS may be the same or different.

A PUSCH may comprise one or more layers, and the UE may transmit at least one symbol with DMRS present on a layer of the one or more layers of the PUSCH. In an example, a higher layer may configure up to three DMRSs for the PUSCH.

Uplink PT-RS (which may be used by a base station for phase tracking and/or phase-noise compensation) may or may not be present depending on an RRC configuration of the UE. The presence and/or pattern of uplink PT-RS may be configured on a UE-specific basis by a combination of RRC signaling and/or one or more parameters employed for other purposes (e.g., Modulation and Coding Scheme (MCS)), which may be indicated by DCI. When configured, a dynamic presence of uplink PT-RS may be associated with one or more DCI parameters comprising at least MCS. A radio network may support a plurality of uplink PT-RS densities defined in time/frequency domain. When present, a frequency domain density may be associated with at least one configuration of a scheduled bandwidth. The UE may assume a same precoding for a DMRS port and a PT-RS port. A number of PT-RS ports may be fewer than a number of DMRS ports in a scheduled resource. For example, uplink PT-RS may be confined in the scheduled time/frequency duration for the UE.

SRS may be transmitted by a UE to a base station for channel state estimation to support uplink channel dependent scheduling and/or link adaptation. SRS transmitted by the UE may allow a base station to estimate an uplink channel state at one or more frequencies. A scheduler at the base station may employ the estimated uplink channel state to assign one or more resource blocks for an uplink PUSCH transmission from the UE. The base station may semi-statically configure the UE with one or more SRS resource sets. For an SRS resource set, the base station may configure the UE with one or more SRS resources. An SRS resource set applicability may be configured by a higher layer (e.g., RRC) parameter. For example, when a higher layer parameter indicates beam management, an SRS resource in a SRS resource set of the one or more SRS resource sets (e.g., with the same/similar time domain behavior, periodic, aperiodic, and/or the like) may be transmitted at a time instant (e.g., simultaneously). The UE may transmit one or more SRS resources in SRS resource sets. An NR network may support aperiodic, periodic and/or semi-persistent SRS transmissions. The UE may transmit SRS resources based on one or more trigger types, wherein the one or more trigger types may comprise higher layer signaling (e.g., RRC) and/or one or more DCI formats. In an example, at least one DCI format may be employed for the UE to select at least one of one or more configured SRS resource sets. An SRS trigger type 0 may refer to an SRS triggered based on a higher layer signaling. An SRS trigger type 1 may refer to an SRS triggered based on one or more DCI formats. In an example, when PUSCH and SRS are transmitted in a same slot, the UE may be configured to transmit SRS after a transmission of a PUSCH and a corresponding uplink DMRS.

The base station may semi-statically configure the UE with one or more SRS configuration parameters indicating at least one of following: a SRS resource configuration identifier; a number of SRS ports; time domain behavior of an SRS resource configuration (e.g., an indication of periodic, semi-persistent, or aperiodic SRS); slot, mini-slot, and/or subframe level periodicity; offset for a periodic and/or an aperiodic SRS resource; a number of OFDM symbols in an

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SRS resource; a starting OFDM symbol of an SRS resource; an SRS bandwidth; a frequency hopping bandwidth; a cyclic shift; and/or an SRS sequence ID.

An antenna port is defined such that the channel over which a symbol on the antenna port is conveyed can be inferred from the channel over which another symbol on the same antenna port is conveyed. If a first symbol and a second symbol are transmitted on the same antenna port, the receiver may infer the channel (e.g., fading gain, multipath delay, and/or the like) for conveying the second symbol on the antenna port, from the channel for conveying the first symbol on the antenna port. A first antenna port and a second antenna port may be referred to as quasi co-located (QCLed) if one or more large-scale properties of the channel over which a first symbol on the first antenna port is conveyed may be inferred from the channel over which a second symbol on a second antenna port is conveyed. The one or more large-scale properties may comprise at least one of: a delay spread; a Doppler spread; a Doppler shift; an average gain; an average delay; and/or spatial Receiving (Rx) parameters.

Channels that use beamforming require beam management. Beam management may comprise beam measurement, beam selection, and beam indication. A beam may be associated with one or more reference signals. For example, a beam may be identified by one or more beamformed reference signals. The UE may perform downlink beam measurement based on downlink reference signals (e.g., a channel state information reference signal (CSI-RS)) and generate a beam measurement report. The UE may perform the downlink beam measurement procedure after an RRC connection is set up with a base station.

FIG. 11B illustrates an example of channel state information reference signals (CSI-RSs) that are mapped in the time and frequency domains. A square shown in FIG. 11B may span a resource block (RB) within a bandwidth of a cell. A base station may transmit one or more RRC messages comprising CSI-RS resource configuration parameters indicating one or more CSI-RSs. One or more of the following parameters may be configured by higher layer signaling (e.g., RRC and/or MAC signaling) for a CSI-RS resource configuration: a CSI-RS resource configuration identity, a number of CSI-RS ports, a CSI-RS configuration (e.g., symbol and resource element (RE) locations in a subframe), a CSI-RS subframe configuration (e.g., subframe location, offset, and periodicity in a radio frame), a CSI-RS power parameter, a CSI-RS sequence parameter, a code division multiplexing (CDM) type parameter, a frequency density, a transmission comb, quasi co-location (QCL) parameters (e.g., QCL-scrambling identity, crs-portscount, mbsfn-subframeconfiglist, csi-rs-configZPId, qcl-csi-rs-configNZPId), and/or other radio resource parameters.

The three beams illustrated in FIG. 11B may be configured for a UE in a UE-specific configuration. Three beams are illustrated in FIG. 11B (beam #1, beam #2, and beam #3), more or fewer beams may be configured. Beam #1 may be allocated with CSI-RS 1101 that may be transmitted in one or more subcarriers in an RB of a first symbol. Beam #2 may be allocated with CSI-RS 1102 that may be transmitted in one or more subcarriers in an RB of a second symbol. Beam #3 may be allocated with CSI-RS 1103 that may be transmitted in one or more subcarriers in an RB of a third symbol. By using frequency division multiplexing (FDM), a base station may use other subcarriers in a same RB (for example, those that are not used to transmit CSI-RS 1101) to transmit another CSI-RS associated with a beam for another UE. By using time domain multiplexing (TDM),

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beams used for the UE may be configured such that beams for the UE use symbols from beams of other UEs.

CSI-RSs such as those illustrated in FIG. 11B (e.g., CSI-RS 1101, 1102, 1103) may be transmitted by the base station and used by the UE for one or more measurements. For example, the UE may measure a reference signal received power (RSRP) of configured CSI-RS resources. The base station may configure the UE with a reporting configuration and the UE may report the RSRP measurements to a network (for example, via one or more base stations) based on the reporting configuration. In an example, the base station may determine, based on the reported measurement results, one or more transmission configuration indication (TCI) states comprising a number of reference signals. In an example, the base station may indicate one or more TCI states to the UE (e.g., via RRC signaling, a MAC CE, and/or a DCI). The UE may receive a downlink transmission with a receive (Rx) beam determined based on the one or more TCI states. In an example, the UE may or may not have a capability of beam correspondence. If the UE has the capability of beam correspondence, the UE may determine a spatial domain filter of a transmit (Tx) beam based on a spatial domain filter of the corresponding Rx beam. If the UE does not have the capability of beam correspondence, the UE may perform an uplink beam selection procedure to determine the spatial domain filter of the Tx beam. The UE may perform the uplink beam selection procedure based on one or more sounding reference signal (SRS) resources configured to the UE by the base station. The base station may select and indicate uplink beams for the UE based on measurements of the one or more SRS resources transmitted by the UE.

In a beam management procedure, a UE may assess (e.g., measure) a channel quality of one or more beam pair links, a beam pair link comprising a transmitting beam transmitted by a base station and a receiving beam received by the UE. Based on the assessment, the UE may transmit a beam measurement report indicating one or more beam pair quality parameters comprising, e.g., one or more beam identifications (e.g., a beam index, a reference signal index, or the like), RSRP, a precoding matrix indicator (PMI), a channel quality indicator (CQI), and/or a rank indicator (RI).

FIG. 12A illustrates examples of three downlink beam management procedures: P1, P2, and P3. Procedure P1 may enable a UE measurement on transmit (Tx) beams of a transmission reception point (TRP) (or multiple TRPs), e.g., to support a selection of one or more base station Tx beams and/or UE Rx beams (shown as ovals in the top row and bottom row, respectively, of P1). Beamforming at a TRP may comprise a Tx beam sweep for a set of beams (shown, in the top rows of P1 and P2, as ovals rotated in a counter-clockwise direction indicated by the dashed arrow). Beamforming at a UE may comprise an Rx beam sweep for a set of beams (shown, in the bottom rows of P1 and P3, as ovals rotated in a clockwise direction indicated by the dashed arrow). Procedure P2 may be used to enable a UE measurement on Tx beams of a TRP (shown, in the top row of P2, as ovals rotated in a counter-clockwise direction indicated by the dashed arrow). The UE and/or the base station may perform procedure P2 using a smaller set of beams than is used in procedure P1, or using narrower beams than the beams used in procedure P1. This may be referred to as beam refinement. The UE may perform procedure P3 for Rx beam determination by using the same Tx beam at the base station and sweeping an Rx beam at the UE.

FIG. 12B illustrates examples of three uplink beam management procedures: U1, U2, and U3. Procedure U1 may be

used to enable a base station to perform a measurement on Tx beams of a UE, e.g., to support a selection of one or more UE Tx beams and/or base station Rx beams (shown as ovals in the top row and bottom row, respectively, of U1). Beamforming at the UE may include, e.g., a Tx beam sweep from a set of beams (shown in the bottom rows of U1 and U3 as ovals rotated in a clockwise direction indicated by the dashed arrow). Beamforming at the base station may include, e.g., an Rx beam sweep from a set of beams (shown, in the top rows of U1 and U2, as ovals rotated in a counter-clockwise direction indicated by the dashed arrow). Procedure U2 may be used to enable the base station to adjust its Rx beam when the UE uses a fixed Tx beam. The UE and/or the base station may perform procedure U2 using a smaller set of beams than is used in procedure P1, or using narrower beams than the beams used in procedure P1. This may be referred to as beam refinement. The UE may perform procedure U3 to adjust its Tx beam when the base station uses a fixed Rx beam.

A UE may initiate a beam failure recovery (BFR) procedure based on detecting a beam failure. The UE may transmit a BFR request (e.g., a preamble, a UCI, an SR, a MAC CE, and/or the like) based on the initiating of the BFR procedure. The UE may detect the beam failure based on a determination that a quality of beam pair link(s) of an associated control channel is unsatisfactory (e.g., having an error rate higher than an error rate threshold, a received signal power lower than a received signal power threshold, an expiration of a timer, and/or the like).

The UE may measure a quality of a beam pair link using one or more reference signals (RSs) comprising one or more SS/PBCH blocks, one or more CSI-RS resources, and/or one or more demodulation reference signals (DMRSs). A quality of the beam pair link may be based on one or more of a block error rate (BLER), an RSRP value, a signal to interference plus noise ratio (SINR) value, a reference signal received quality (RSRQ) value, and/or a CSI value measured on RS resources. The base station may indicate that an RS resource is quasi co-located (QCLed) with one or more DM-RSs of a channel (e.g., a control channel, a shared data channel, and/or the like). The RS resource and the one or more DMRSs of the channel may be QCLed when the channel characteristics (e.g., Doppler shift, Doppler spread, average delay, delay spread, spatial Rx parameter, fading, and/or the like) from a transmission via the RS resource to the UE are similar or the same as the channel characteristics from a transmission via the channel to the UE.

A network (e.g., a gNB and/or an ng-eNB of a network) and/or the UE may initiate a random access procedure. A UE in an RRC_IDLE state and/or an RRC_INACTIVE state may initiate the random access procedure to request a connection setup to a network. The UE may initiate the random access procedure from an RRC_CONNECTED state. The UE may initiate the random access procedure to request uplink resources (e.g., for uplink transmission of an SR when there is no PUCCH resource available) and/or acquire uplink timing (e.g., when uplink synchronization status is non-synchronized). The UE may initiate the random access procedure to request one or more system information blocks (SIBs) (e.g., other system information such as SIB2, SIB3, and/or the like). The UE may initiate the random access procedure for a beam failure recovery request. A network may initiate a random access procedure for a handover and/or for establishing time alignment for an SCell addition.

FIG. 13A illustrates a four-step contention-based random access procedure. Prior to initiation of the procedure, a base

station may transmit a configuration message 1310 to the UE. The procedure illustrated in FIG. 13A comprises transmission of four messages: a Msg 1 1311, a Msg 2 1312, a Msg 3 1313, and a Msg 4 1314. The Msg 1 1311 may include and/or be referred to as a preamble (or a random access preamble). The Msg 2 1312 may include and/or be referred to as a random access response (RAR).

The configuration message 1310 may be transmitted, for example, using one or more RRC messages. The one or more RRC messages may indicate one or more random access channel (RACH) parameters to the UE. The one or more RACH parameters may comprise at least one of the following: general parameters for one or more random access procedures (e.g., RACH-configGeneral); cell-specific parameters (e.g., RACH-ConfigCommon); and/or dedicated parameters (e.g., RACH-configDedicated). The base station may broadcast or multicast the one or more RRC messages to one or more UEs. The one or more RRC messages may be UE-specific (e.g., dedicated RRC messages transmitted to a UE in an RRC_CONNECTED state and/or in an RRC_INACTIVE state). The UE may determine, based on the one or more RACH parameters, a time-frequency resource and/or an uplink transmit power for transmission of the Msg 1 1311 and/or the Msg 3 1313. Based on the one or more RACH parameters, the UE may determine a reception timing and a downlink channel for receiving the Msg 2 1312 and the Msg 4 1314.

The one or more RACH parameters provided in the configuration message 1310 may indicate one or more Physical RACH (PRACH) occasions available for transmission of the Msg 1 1311. The one or more PRACH occasions may be predefined. The one or more RACH parameters may indicate one or more available sets of one or more PRACH occasions (e.g., prach-ConfigIndex). The one or more RACH parameters may indicate an association between (a) one or more PRACH occasions and (b) one or more reference signals. The one or more RACH parameters may indicate an association between (a) one or more preambles and (b) one or more reference signals. The one or more reference signals may be SS/PBCH blocks and/or CSI-RSs. For example, the one or more RACH parameters may indicate a number of SS/PBCH blocks mapped to a PRACH occasion and/or a number of preambles mapped to a SS/PBCH blocks.

The one or more RACH parameters provided in the configuration message 1310 may be used to determine an uplink transmit power of Msg 1 1311 and/or Msg 3 1313. For example, the one or more RACH parameters may indicate a reference power for a preamble transmission (e.g., a received target power and/or an initial power of the preamble transmission). There may be one or more power offsets indicated by the one or more RACH parameters. For example, the one or more RACH parameters may indicate: a power ramping step; a power offset between SSB and CSI-RS; a power offset between transmissions of the Msg 1 1311 and the Msg 3 1313; and/or a power offset value between preamble groups. The one or more RACH parameters may indicate one or more thresholds based on which the UE may determine at least one reference signal (e.g., an SSB and/or CSI-RS) and/or an uplink carrier (e.g., a normal uplink (NUL) carrier and/or a supplemental uplink (SUL) carrier).

The Msg 1 1311 may include one or more preamble transmissions (e.g., a preamble transmission and one or more preamble retransmissions). An RRC message may be used to configure one or more preamble groups (e.g., group A and/or group B). A preamble group may comprise one or

more preambles. The UE may determine the preamble group based on a pathloss measurement and/or a size of the Msg 3 1313. The UE may measure an RSRP of one or more reference signals (e.g., SSBs and/or CSI-RSs) and determine at least one reference signal having an RSRP above an RSRP threshold (e.g., rsrp-ThresholdSSB and/or $\text{rsrp-Threshold-CSI-RS}$). The UE may select at least one preamble associated with the one or more reference signals and/or a selected preamble group, for example, if the association between the one or more preambles and the at least one reference signal is configured by an RRC message.

The UE may determine the preamble based on the one or more RACH parameters provided in the configuration message 1310. For example, the UE may determine the preamble based on a pathloss measurement, an RSRP measurement, and/or a size of the Msg 3 1313. As another example, the one or more RACH parameters may indicate: a preamble format; a maximum number of preamble transmissions; and/or one or more thresholds for determining one or more preamble groups (e.g., group A and group B). A base station may use the one or more RACH parameters to configure the UE with an association between one or more preambles and one or more reference signals (e.g., SSBs and/or CSI-RSs). If the association is configured, the UE may determine the preamble to include in Msg 1 1311 based on the association. The Msg 1 1311 may be transmitted to the base station via one or more PRACH occasions. The UE may use one or more reference signals (e.g., SSBs and/or CSI-RSs) for selection of the preamble and for determining of the PRACH occasion. One or more RACH parameters (e.g., $\text{ra-ssb-OccasionMskIndex}$ and/or ra-OccasionList) may indicate an association between the PRACH occasions and the one or more reference signals.

The UE may perform a preamble retransmission if no response is received following a preamble transmission. The UE may increase an uplink transmit power for the preamble retransmission. The UE may select an initial preamble transmit power based on a pathloss measurement and/or a target received preamble power configured by the network. The UE may determine to retransmit a preamble and may ramp up the uplink transmit power. The UE may receive one or more RACH parameters (e.g., $\text{PREAMBLE_POWER_RAMPING_STEP}$) indicating a ramping step for the preamble retransmission. The ramping step may be an amount of incremental increase in uplink transmit power for a retransmission. The UE may ramp up the uplink transmit power if the UE determines a reference signal (e.g., SSB and/or CSI-RS) that is the same as a previous preamble transmission. The UE may count a number of preamble transmissions and/or retransmissions (e.g., $\text{PREAMBLE_TRANSMISSION_COUNTER}$). The UE may determine that a random access procedure completed unsuccessfully, for example, if the number of preamble transmissions exceeds a threshold configured by the one or more RACH parameters (e.g., preambleTransMax).

The Msg 2 1312 received by the UE may include an RAR. In some scenarios, the Msg 2 1312 may include multiple RARs corresponding to multiple UEs. The Msg 2 1312 may be received after or in response to the transmitting of the Msg 1 1311. The Msg 2 1312 may be scheduled on the DL-SCH and indicated on a PDCCH using a random access RNTI (RA-RNTI). The Msg 2 1312 may indicate that the Msg 1 1311 was received by the base station. The Msg 2 1312 may include a time-alignment command that may be used by the UE to adjust the UE's transmission timing, a scheduling grant for transmission of the Msg 3 1313, and/or a Temporary Cell RNTI (TC-RNTI). After transmitting a

preamble, the UE may start a time window (e.g., $\text{ra-Response Window}$) to monitor a PDCCH for the Msg 2 1312. The UE may determine when to start the time window based on a PRACH occasion that the UE uses to transmit the preamble. For example, the UE may start the time window one or more symbols after a last symbol of the preamble (e.g., at a first PDCCH occasion from an end of a preamble transmission). The one or more symbols may be determined based on a numerology. The PDCCH may be in a common search space (e.g., a Type1-PDCCH common search space) configured by an RRC message. The UE may identify the RAR based on a Radio Network Temporary Identifier (RNTI). RNTIs may be used depending on one or more events initiating the random access procedure. The UE may use random access RNTI (RA-RNTI). The RA-RNTI may be associated with PRACH occasions in which the UE transmits a preamble. For example, the UE may determine the RA-RNTI based on: an OFDM symbol index; a slot index; a frequency domain index; and/or a UL carrier indicator of the PRACH occasions. An example of RA-RNTI may be as follows:

$$\text{RA-RNTI} = 1 + s_id + 14 \times t_id + 14 \times 80 \times f_id + 14 \times 80 \times 8 \times ul_carrier_id$$

where s_id may be an index of a first OFDM symbol of the PRACH occasion (e.g., $0 \leq s_id < 14$), t_id may be an index of a first slot of the PRACH occasion in a system frame (e.g., $0 \leq t_id < 80$), f_id may be an index of the PRACH occasion in the frequency domain (e.g., $0 \leq f_id < 8$), and $ul_carrier_id$ may be a UL carrier used for a preamble transmission (e.g., 0 for an NUL carrier, and 1 for an SUL carrier).

The UE may transmit the Msg 3 1313 in response to a successful reception of the Msg 2 1312 (e.g., using resources identified in the Msg 2 1312). The Msg 3 1313 may be used for contention resolution in, for example, the contention-based random access procedure illustrated in FIG. 13A. In some scenarios, a plurality of UEs may transmit a same preamble to a base station and the base station may provide an RAR that corresponds to a UE. Collisions may occur if the plurality of UEs interpret the RAR as corresponding to themselves. Contention resolution (e.g., using the Msg 3 1313 and the Msg 4 1314) may be used to increase the likelihood that the UE does not incorrectly use an identity of another the UE. To perform contention resolution, the UE may include a device identifier in the Msg 3 1313 (e.g., a C-RNTI if assigned, a TC RNTI included in the Msg 2 1312, and/or any other suitable identifier).

The Msg 4 1314 may be received after or in response to the transmitting of the Msg 3 1313. If a C-RNTI was included in the Msg 3 1313, the base station will address the UE on the PDCCH using the C-RNTI. If the UE's unique C-RNTI is detected on the PDCCH, the random access procedure is determined to be successfully completed. If a TC-RNTI is included in the Msg 3 1313 (e.g., if the UE is in an RRC_IDLE state or not otherwise connected to the base station), Msg 4 1314 will be received using a DL-SCH associated with the TC-RNTI. If a MAC PDU is successfully decoded and a MAC PDU comprises the UE contention resolution identity MAC CE that matches or otherwise corresponds with the CCCH SDU sent (e.g., transmitted) in Msg 3 1313, the UE may determine that the contention resolution is successful and/or the UE may determine that the random access procedure is successfully completed.

The UE may be configured with a supplementary uplink (SUL) carrier and a normal uplink (NUL) carrier. An initial access (e.g., random access procedure) may be supported in an uplink carrier. For example, a base station may configure

the UE with two separate RACH configurations: one for an SUL carrier and the other for an NUL carrier. For random access in a cell configured with an SUL carrier, the network may indicate which carrier to use (NUL or SUL). The UE may determine the SUL carrier, for example, if a measured quality of one or more reference signals is lower than a broadcast threshold. Uplink transmissions of the random access procedure (e.g., the Msg 1 1311 and/or the Msg 3 1313) may remain on the selected carrier. The UE may switch an uplink carrier during the random access procedure (e.g., between the Msg 1 1311 and the Msg 3 1313) in one or more cases. For example, the UE may determine and/or switch an uplink carrier for the Msg 1 1311 and/or the Msg 3 1313 based on a channel clear assessment (e.g., a listen-before-talk).

FIG. 13B illustrates a two-step contention-free random access procedure. Similar to the four-step contention-based random access procedure illustrated in FIG. 13A, a base station may, prior to initiation of the procedure, transmit a configuration message 1320 to the UE. The configuration message 1320 may be analogous in some respects to the configuration message 1310. The procedure illustrated in FIG. 13B comprises transmission of two messages: a Msg 1 1321 and a Msg 2 1322. The Msg 1 1321 and the Msg 2 1322 may be analogous in some respects to the Msg 1 1311 and a Msg 2 1312 illustrated in FIG. 13A, respectively. As will be understood from FIGS. 13A and 13B, the contention-free random access procedure may not include messages analogous to the Msg 3 1313 and/or the Msg 4 1314.

The contention-free random access procedure illustrated in FIG. 13B may be initiated for a beam failure recovery, other SI request, SCell addition, and/or handover. For example, a base station may indicate or assign to the UE the preamble to be used for the Msg 1 1321. The UE may receive, from the base station via PDCCH and/or RRC, an indication of a preamble (e.g., ra-PreambleIndex).

After transmitting a preamble, the UE may start a time window (e.g., ra-Response Window) to monitor a PDCCH for the RAR. In the event of a beam failure recovery request, the base station may configure the UE with a separate time window and/or a separate PDCCH in a search space indicated by an RRC message (e.g., recoverySearchSpaceId). The UE may monitor for a PDCCH transmission addressed to a Cell RNTI (C-RNTI) on the search space. In the contention-free random access procedure illustrated in FIG. 13B, the UE may determine that a random access procedure successfully completes after or in response to transmission of Msg 1 1321 and reception of a corresponding Msg 2 1322. The UE may determine that a random access procedure successfully completes, for example, if a PDCCH transmission is addressed to a C-RNTI. The UE may determine that a random access procedure successfully completes, for example, if the UE receives an RAR comprising a preamble identifier corresponding to a preamble transmitted by the UE and/or the RAR comprises a MAC sub-PDU with the preamble identifier. The UE may determine the response as an indication of an acknowledgement for an SI request.

FIG. 13C illustrates another two-step random access procedure. Similar to the random access procedures illustrated in FIGS. 13A and 13B, a base station may, prior to initiation of the procedure, transmit a configuration message 1330 to the UE. The configuration message 1330 may be analogous in some respects to the configuration message 1310 and/or the configuration message 1320. The procedure illustrated in FIG. 13C comprises transmission of two messages: a Msg A 1331 and a Msg B 1332.

Msg A 1331 may be transmitted in an uplink transmission by the UE. Msg A 1331 may comprise one or more transmissions of a preamble 1341 and/or one or more transmissions of a transport block 1342. The transport block 1342 may comprise contents that are similar and/or equivalent to the contents of the Msg 3 1313 illustrated in FIG. 13A. The transport block 1342 may comprise UCI (e.g., an SR, a HARQ ACK/NACK, and/or the like). The UE may receive the Msg B 1332 after or in response to transmitting the Msg A 1331. The Msg B 1332 may comprise contents that are similar and/or equivalent to the contents of the Msg 2 1312 (e.g., an RAR) illustrated in FIGS. 13A and 13B and/or the Msg 4 1314 illustrated in FIG. 13A.

The UE may initiate the two-step random access procedure in FIG. 13C for licensed spectrum and/or unlicensed spectrum. The UE may determine, based on one or more factors, whether to initiate the two-step random access procedure. The one or more factors may be: a radio access technology in use (e.g., LTE, NR, and/or the like); whether the UE has valid TA or not; a cell size; the UE's RRC state; a type of spectrum (e.g., licensed vs. unlicensed); and/or any other suitable factors.

The UE may determine, based on two-step RACH parameters included in the configuration message 1330, a radio resource and/or an uplink transmit power for the preamble 1341 and/or the transport block 1342 included in the Msg A 1331. The RACH parameters may indicate a modulation and coding schemes (MCS), a time-frequency resource, and/or a power control for the preamble 1341 and/or the transport block 1342. A time-frequency resource for transmission of the preamble 1341 (e.g., a PRACH) and a time-frequency resource for transmission of the transport block 1342 (e.g., a PUSCH) may be multiplexed using FDM, TDM, and/or CDM. The RACH parameters may enable the UE to determine a reception timing and a downlink channel for monitoring for and/or receiving Msg B 1332.

The transport block 1342 may comprise data (e.g., delay-sensitive data), an identifier of the UE, security information, and/or device information (e.g., an International Mobile Subscriber Identity (IMSI)). The base station may transmit the Msg B 1332 as a response to the Msg A 1331. The Msg B 1332 may comprise at least one of following: a preamble identifier; a timing advance command; a power control command; an uplink grant (e.g., a radio resource assignment and/or an MCS); a UE identifier for contention resolution; and/or an RNTI (e.g., a C-RNTI or a TC-RNTI). The UE may determine that the two-step random access procedure is successfully completed if: a preamble identifier in the Msg B 1332 is matched to a preamble transmitted by the UE; and/or the identifier of the UE in Msg B 1332 is matched to the identifier of the UE in the Msg A 1331 (e.g., the transport block 1342).

A UE and a base station may exchange control signaling. The control signaling may be referred to as L1/L2 control signaling and may originate from the PHY layer (e.g., layer 1) and/or the MAC layer (e.g., layer 2). The control signaling may comprise downlink control signaling transmitted from the base station to the UE and/or uplink control signaling transmitted from the UE to the base station.

The downlink control signaling may comprise: a downlink scheduling assignment; an uplink scheduling grant indicating uplink radio resources and/or a transport format; a slot format information; a preemption indication; a power control command; and/or any other suitable signaling. The UE may receive the downlink control signaling in a payload transmitted by the base station on a physical downlink control channel (PDCCH). The payload transmitted on the

PDCCH may be referred to as downlink control information (DCI). In some scenarios, the PDCCH may be a group common PDCCH (GC-PDCCH) that is common to a group of UEs.

A base station may attach one or more cyclic redundancy check (CRC) parity bits to a DCI in order to facilitate detection of transmission errors. When the DCI is intended for a UE (or a group of the UEs), the base station may scramble the CRC parity bits with an identifier of the UE (or an identifier of the group of the UEs). Scrambling the CRC parity bits with the identifier may comprise Modulo-2 addition (or an exclusive OR operation) of the identifier value and the CRC parity bits. The identifier may comprise a 16-bit value of a radio network temporary identifier (RNTI).

DCIs may be used for different purposes. A purpose may be indicated by the type of RNTI used to scramble the CRC parity bits. For example, a DCI having CRC parity bits scrambled with a paging RNTI (P-RNTI) may indicate paging information and/or a system information change notification. The P-RNTI may be predefined as "FFFE" in hexadecimal. A DCI having CRC parity bits scrambled with a system information RNTI (SI-RNTI) may indicate a broadcast transmission of the system information. The SI-RNTI may be predefined as "FFFF" in hexadecimal. A DCI having CRC parity bits scrambled with a random access RNTI (RA-RNTI) may indicate a random access response (RAR). A DCI having CRC parity bits scrambled with a cell RNTI (C-RNTI) may indicate a dynamically scheduled unicast transmission and/or a triggering of PDCCH-ordered random access. A DCI having CRC parity bits scrambled with a temporary cell RNTI (TC-RNTI) may indicate a contention resolution (e.g., a Msg 3 analogous to the Msg 3 1313 illustrated in FIG. 13A). Other RNTIs configured to the UE by a base station may comprise a Configured Scheduling RNTI (CS-RNTI), a Transmit Power Control-PUCCH RNTI (TPC-PUCCH-RNTI), a Transmit Power Control-PUSCH RNTI (TPC-PUSCH-RNTI), a Transmit Power Control-SRS RNTI (TPC-SRS-RNTI), an Interruption RNTI (INT-RNTI), a Slot Format Indication RNTI (SFI-RNTI), a Semi-Persistent CSI RNTI (SP-CSI-RNTI), a Modulation and Coding Scheme Cell RNTI (MCS-C-RNTI), and/or the like.

Depending on the purpose and/or content of a DCI, the base station may transmit the DCIs with one or more DCI formats. For example, DCI format 0_0 may be used for scheduling of PUSCH in a cell. DCI format 0_0 may be a fallback DCI format (e.g., with compact DCI payloads). DCI format 0_1 may be used for scheduling of PUSCH in a cell (e.g., with more DCI payloads than DCI format 0_0). DCI format 1_0 may be used for scheduling of PDSCH in a cell. DCI format 1_0 may be a fallback DCI format (e.g., with compact DCI payloads). DCI format 1_1 may be used for scheduling of PDSCH in a cell (e.g., with more DCI payloads than DCI format 1_0). DCI format 2_0 may be used for providing a slot format indication to a group of UEs. DCI format 2_1 may be used for notifying a group of UEs of a physical resource block and/or OFDM symbol where the UE may assume no transmission is intended to the UE. DCI format 2_2 may be used for transmission of a transmit power control (TPC) command for PUCCH or PUSCH. DCI format 2_3 may be used for transmission of a group of TPC commands for SRS transmissions by one or more UEs. DCI format(s) for new functions may be defined in future releases. DCI formats may have different DCI sizes, or may share the same DCI size.

After scrambling a DCI with a RNTI, the base station may process the DCI with channel coding (e.g., polar coding),

rate matching, scrambling and/or QPSK modulation. A base station may map the coded and modulated DCI on resource elements used and/or configured for a PDCCH. Based on a payload size of the DCI and/or a coverage of the base station, the base station may transmit the DCI via a PDCCH occupying a number of contiguous control channel elements (CCEs). The number of the contiguous CCEs (referred to as aggregation level) may be 1, 2, 4, 8, 16, and/or any other suitable number. A CCE may comprise a number (e.g., 6) of resource-element groups (REGs). A REG may comprise a resource block in an OFDM symbol. The mapping of the coded and modulated DCI on the resource elements may be based on mapping of CCEs and REGs (e.g., CCE-to-REG mapping).

FIG. 14A illustrates an example of CORESET configurations for a bandwidth part. The base station may transmit a DCI via a PDCCH on one or more control resource sets (CORESETs). A CORESET may comprise a time-frequency resource in which the UE tries to decode a DCI using one or more search spaces. The base station may configure a CORESET in the time-frequency domain. In the example of FIG. 14A, a first CORESET 1401 and a second CORESET 1402 occur at the first symbol in a slot. The first CORESET 1401 overlaps with the second CORESET 1402 in the frequency domain. A third CORESET 1403 occurs at a third symbol in the slot. A fourth CORESET 1404 occurs at the seventh symbol in the slot. CORESETs may have a different number of resource blocks in frequency domain.

FIG. 14B illustrates an example of a CCE-to-REG mapping for DCI transmission on a CORESET and PDCCH processing. The CCE-to-REG mapping may be an interleaved mapping (e.g., for the purpose of providing frequency diversity) or a non-interleaved mapping (e.g., for the purposes of facilitating interference coordination and/or frequency-selective transmission of control channels). The base station may perform different or same CCE-to-REG mapping on different CORESETs. A CORESET may be associated with a CCE-to-REG mapping by RRC configuration. A CORESET may be configured with an antenna port quasi co-location (QCL) parameter. The antenna port QCL parameter may indicate QCL information of a demodulation reference signal (DMRS) for PDCCH reception in the CORESET.

The base station may transmit, to the UE, RRC messages comprising configuration parameters of one or more CORESETs and one or more search space sets. The configuration parameters may indicate an association between a search space set and a CORESET. A search space set may comprise a set of PDCCH candidates formed by CCEs at a given aggregation level. The configuration parameters may indicate: a number of PDCCH candidates to be monitored per aggregation level; a PDCCH monitoring periodicity and a PDCCH monitoring pattern; one or more DCI formats to be monitored by the UE; and/or whether a search space set is a common search space set or a UE-specific search space set. A set of CCEs in the common search space set may be predefined and known to the UE. A set of CCEs in the UE-specific search space set may be configured based on the UE's identity (e.g., C-RNTI).

As shown in FIG. 14B, the UE may determine a time-frequency resource for a CORESET based on RRC messages. The UE may determine a CCE-to-REG mapping (e.g., interleaved or non-interleaved, and/or mapping parameters) for the CORESET based on configuration parameters of the CORESET. The UE may determine a number (e.g., at most 10) of search space sets configured on the CORESET based on the RRC messages. The UE may monitor a set of PDCCH

candidates according to configuration parameters of a search space set. The UE may monitor a set of PDCCH candidates in one or more CORESETs for detecting one or more DCIs. Monitoring may comprise decoding one or more PDCCH candidates of the set of the PDCCH candidates according to the monitored DCI formats. Monitoring may comprise decoding a DCI content of one or more PDCCH candidates with possible (or configured) PDCCH locations, possible (or configured) PDCCH formats (e.g., number of CCEs, number of PDCCH candidates in common search spaces, and/or number of PDCCH candidates in the UE-specific search spaces) and possible (or configured) DCI formats. The decoding may be referred to as blind decoding. The UE may determine a DCI as valid for the UE, in response to CRC checking (e.g., scrambled bits for CRC parity bits of the DCI matching a RNTI value). The UE may process information contained in the DCI (e.g., a scheduling assignment, an uplink grant, power control, a slot format indication, a downlink preemption, and/or the like).

The UE may transmit uplink control signaling (e.g., uplink control information (UCI)) to a base station. The uplink control signaling may comprise hybrid automatic repeat request (HARQ) acknowledgements for received DL-SCH transport blocks. The UE may transmit the HARQ acknowledgements after receiving a DL-SCH transport block. Uplink control signaling may comprise channel state information (CSI) indicating channel quality of a physical downlink channel. The UE may transmit the CSI to the base station. The base station, based on the received CSI, may determine transmission format parameters (e.g., comprising multi-antenna and beamforming schemes) for a downlink transmission. Uplink control signaling may comprise scheduling requests (SR). The UE may transmit an SR indicating that uplink data is available for transmission to the base station. The UE may transmit a UCI (e.g., HARQ acknowledgements (HARQ-ACK), CSI report, SR, and the like) via a physical uplink control channel (PUCCH) or a physical uplink shared channel (PUSCH). The UE may transmit the uplink control signaling via a PUCCH using one of several PUCCH formats.

There may be five PUCCH formats and the UE may determine a PUCCH format based on a size of the UCI (e.g., a number of uplink symbols of UCI transmission and a number of UCI bits). PUCCH format 0 may have a length of one or two OFDM symbols and may include two or fewer bits. The UE may transmit UCI in a PUCCH resource using PUCCH format 0 if the transmission is over one or two symbols and the number of HARQ-ACK information bits with positive or negative SR (HARQ-ACK/SR bits) is one or two. PUCCH format 1 may occupy a number between four and fourteen OFDM symbols and may include two or fewer bits. The UE may use PUCCH format 1 if the transmission is four or more symbols and the number of HARQ-ACK/SR bits is one or two. PUCCH format 2 may occupy one or two OFDM symbols and may include more than two bits. The UE may use PUCCH format 2 if the transmission is over one or two symbols and the number of UCI bits is two or more. PUCCH format 3 may occupy a number between four and fourteen OFDM symbols and may include more than two bits. The UE may use PUCCH format 3 if the transmission is four or more symbols, the number of UCI bits is two or more and PUCCH resource does not include an orthogonal cover code. PUCCH format 4 may occupy a number between four and fourteen OFDM symbols and may include more than two bits. The UE may use PUCCH format 4 if the transmission is four or more sym-

bols, the number of UCI bits is two or more and the PUCCH resource includes an orthogonal cover code.

The base station may transmit configuration parameters to the UE for a plurality of PUCCH resource sets using, for example, an RRC message. The plurality of PUCCH resource sets (e.g., up to four sets) may be configured on an uplink BWP of a cell. A PUCCH resource set may be configured with a PUCCH resource set index, a plurality of PUCCH resources with a PUCCH resource being identified by a PUCCH resource identifier (e.g., pucch-Resourceid), and/or a number (e.g. a maximum number) of UCI information bits the UE may transmit using one of the plurality of PUCCH resources in the PUCCH resource set. When configured with a plurality of PUCCH resource sets, the UE may select one of the plurality of PUCCH resource sets based on a total bit length of the UCI information bits (e.g., HARQ-ACK, SR, and/or CSI). If the total bit length of UCI information bits is two or fewer, the UE may select a first PUCCH resource set having a PUCCH resource set index equal to "0". If the total bit length of UCI information bits is greater than two and less than or equal to a first configured value, the UE may select a second PUCCH resource set having a PUCCH resource set index equal to "1". If the total bit length of UCI information bits is greater than the first configured value and less than or equal to a second configured value, the UE may select a third PUCCH resource set having a PUCCH resource set index equal to "2". If the total bit length of UCI information bits is greater than the second configured value and less than or equal to a third value (e.g., 1406), the UE may select a fourth PUCCH resource set having a PUCCH resource set index equal to "3".

After determining a PUCCH resource set from a plurality of PUCCH resource sets, the UE may determine a PUCCH resource from the PUCCH resource set for UCI (HARQ-ACK, CSI, and/or SR) transmission. The UE may determine the PUCCH resource based on a PUCCH resource indicator in a DCI (e.g., with a DCI format 1_0 or DCI for 1_1) received on a PDCCH. A three-bit PUCCH resource indicator in the DCI may indicate one of eight PUCCH resources in the PUCCH resource set. Based on the PUCCH resource indicator, the UE may transmit the UCI (HARQ-ACK, CSI and/or SR) using a PUCCH resource indicated by the PUCCH resource indicator in the DCI.

FIG. 15 illustrates an example of a wireless device 1502 in communication with a base station 1504 in accordance with embodiments of the present disclosure. The wireless device 1502 and base station 1504 may be part of a mobile communication network, such as the mobile communication network 100 illustrated in FIG. 1A, the mobile communication network 150 illustrated in FIG. 1B, or any other communication network. Only one wireless device 1502 and one base station 1504 are illustrated in FIG. 15, but it will be understood that a mobile communication network may include more than one UE and/or more than one base station, with the same or similar configuration as those shown in FIG. 15.

The base station 1504 may connect the wireless device 1502 to a core network (not shown) through radio communications over the air interface (or radio interface) 1506. The communication direction from the base station 1504 to the wireless device 1502 over the air interface 1506 is known as the downlink, and the communication direction from the wireless device 1502 to the base station 1504 over the air interface is known as the uplink. Downlink transmissions may be separated from uplink transmissions using FDD, TDD, and/or some combination of the two duplexing techniques.

In the downlink, data to be sent to the wireless device **1502** from the base station **1504** may be provided to the processing system **1508** of the base station **1504**. The data may be provided to the processing system **1508** by, for example, a core network. In the uplink, data to be sent to the base station **1504** from the wireless device **1502** may be provided to the processing system **1518** of the wireless device **1502**. The processing system **1508** and the processing system **1518** may implement layer 3 and layer 2 OSI functionality to process the data for transmission. Layer 2 may include an SDAP layer, a PDCP layer, an RLC layer, and a MAC layer, for example, with respect to FIG. 2A, FIG. 2B, FIG. 3, and FIG. 4A. Layer 3 may include an RRC layer as with respect to FIG. 2B.

After being processed by processing system **1508**, the data to be sent to the wireless device **1502** may be provided to a transmission processing system **1510** of base station **1504**. Similarly, after being processed by the processing system **1518**, the data to be sent to base station **1504** may be provided to a transmission processing system **1520** of the wireless device **1502**. The transmission processing system **1510** and the transmission processing system **1520** may implement layer 1 OSI functionality. Layer 1 may include a PHY layer with respect to FIG. 2A, FIG. 2B, FIG. 3, and FIG. 4A. For transmit processing, the PHY layer may perform, for example, forward error correction coding of transport channels, interleaving, rate matching, mapping of transport channels to physical channels, modulation of physical channel, multiple-input multiple-output (MIMO) or multi-antenna processing, and/or the like.

At the base station **1504**, a reception processing system **1512** may receive the uplink transmission from the wireless device **1502**. At the wireless device **1502**, a reception processing system **1522** may receive the downlink transmission from base station **1504**. The reception processing system **1512** and the reception processing system **1522** may implement layer 1 OSI functionality. Layer 1 may include a PHY layer with respect to FIG. 2A, FIG. 2B, FIG. 3, and FIG. 4A. For receive processing, the PHY layer may perform, for example, error detection, forward error correction decoding, deinterleaving, demapping of transport channels to physical channels, demodulation of physical channels, MIMO or multi-antenna processing, and/or the like.

As shown in FIG. 15, a wireless device **1502** and the base station **1504** may include multiple antennas. The multiple antennas may be used to perform one or more MIMO or multi-antenna techniques, such as spatial multiplexing (e.g., single-user MIMO or multi-user MIMO), transmit/receive diversity, and/or beamforming. In other examples, the wireless device **1502** and/or the base station **1504** may have a single antenna.

The processing system **1508** and the processing system **1518** may be associated with a memory **1514** and a memory **1524**, respectively. Memory **1514** and memory **1524** (e.g., one or more non-transitory computer readable mediums) may store computer program instructions or code that may be executed by the processing system **1508** and/or the processing system **1518** to carry out one or more of the functionalities discussed in the present application. Although not shown in FIG. 15, the transmission processing system **1510**, the transmission processing system **1520**, the reception processing system **1512**, and/or the reception processing system **1522** may be coupled to a memory (e.g., one or more non-transitory computer readable mediums) storing computer program instructions or code that may be executed to carry out one or more of their respective functionalities.

The processing system **1508** and/or the processing system **1518** may comprise one or more controllers and/or one or more processors. The one or more controllers and/or one or more processors may comprise, for example, a general-purpose processor, a digital signal processor (DSP), a micro-controller, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) and/or other programmable logic device, discrete gate and/or transistor logic, discrete hardware components, an on-board unit, or any combination thereof. The processing system **1508** and/or the processing system **1518** may perform at least one of signal coding/processing, data processing, power control, input/output processing, and/or any other functionality that may enable the wireless device **1502** and the base station **1504** to operate in a wireless environment.

The processing system **1508** and/or the processing system **1518** may be connected to one or more peripherals **1516** and one or more peripherals **1526**, respectively. The one or more peripherals **1516** and the one or more peripherals **1526** may include software and/or hardware that provide features and/or functionalities, for example, a speaker, a microphone, a keypad, a display, a touchpad, a power source, a satellite transceiver, a universal serial bus (USB) port, a hands-free headset, a frequency modulated (FM) radio unit, a media player, an Internet browser, an electronic control unit (e.g., for a motor vehicle), and/or one or more sensors (e.g., an accelerometer, a gyroscope, a temperature sensor, a radar sensor, a lidar sensor, an ultrasonic sensor, a light sensor, a camera, and/or the like). The processing system **1508** and/or the processing system **1518** may receive user input data from and/or provide user output data to the one or more peripherals **1516** and/or the one or more peripherals **1526**. The processing system **1518** in the wireless device **1502** may receive power from a power source and/or may be configured to distribute the power to the other components in the wireless device **1502**. The power source may comprise one or more sources of power, for example, a battery, a solar cell, a fuel cell, or any combination thereof. The processing system **1508** and/or the processing system **1518** may be connected to a GPS chipset **1517** and a GPS chipset **1527**, respectively. The GPS chipset **1517** and the GPS chipset **1527** may be configured to provide geographic location information of the wireless device **1502** and the base station **1504**, respectively.

FIG. 16A illustrates an example structure for uplink transmission. A baseband signal representing a physical uplink shared channel may perform one or more functions. The one or more functions may comprise at least one of: scrambling; modulation of scrambled bits to generate complex-valued symbols; mapping of the complex-valued modulation symbols onto one or several transmission layers; transform precoding to generate complex-valued symbols; precoding of the complex-valued symbols; mapping of precoded complex-valued symbols to resource elements; generation of complex-valued time-domain Single Carrier-Frequency Division Multiple Access (SC-FDMA) or CP-OFDM signal for an antenna port; and/or the like. In an example, when transform precoding is enabled, a SC-FDMA signal for uplink transmission may be generated. In an example, when transform precoding is not enabled, an CP-OFDM signal for uplink transmission may be generated by FIG. 16A. These functions are illustrated as examples and it is anticipated that other mechanisms may be implemented in various embodiments.

FIG. 16B illustrates an example structure for modulation and up-conversion of a baseband signal to a carrier frequency. The baseband signal may be a complex-valued

SC-FDMA or CP-OFDM baseband signal for an antenna port and/or a complex-valued Physical Random Access Channel (PRACH) baseband signal. Filtering may be employed prior to transmission.

FIG. 16C illustrates an example structure for downlink transmissions. A baseband signal representing a physical downlink channel may perform one or more functions. The one or more functions may comprise: scrambling of coded bits in a codeword to be transmitted on a physical channel; modulation of scrambled bits to generate complex-valued modulation symbols; mapping of the complex-valued modulation symbols onto one or several transmission layers; precoding of the complex-valued modulation symbols on a layer for transmission on the antenna ports; mapping of complex-valued modulation symbols for an antenna port to resource elements; generation of complex-valued time-domain OFDM signal for an antenna port; and/or the like. These functions are illustrated as examples and it is anticipated that other mechanisms may be implemented in various embodiments.

FIG. 16D illustrates another example structure for modulation and up-conversion of a baseband signal to a carrier frequency. The baseband signal may be a complex-valued OFDM baseband signal for an antenna port. Filtering may be employed prior to transmission.

A wireless device may receive from a base station one or more messages (e.g. RRC messages) comprising configuration parameters of a plurality of cells (e.g. primary cell, secondary cell). The wireless device may communicate with at least one base station (e.g. two or more base stations in dual-connectivity) via the plurality of cells. The one or more messages (e.g. as a part of the configuration parameters) may comprise parameters of physical, MAC, RLC, PCDP, SDAP, RRC layers for configuring the wireless device. For example, the configuration parameters may comprise parameters for configuring physical and MAC layer channels, bearers, etc. For example, the configuration parameters may comprise parameters indicating values of timers for physical, MAC, RLC, PCDP, SDAP, RRC layers, and/or communication channels.

A timer may begin running once it is started and continue running until it is stopped or until it expires. A timer may be started if it is not running or restarted if it is running. A timer may be associated with a value (e.g. the timer may be started or restarted from a value or may be started from zero and expire once it reaches the value). The duration of a timer may not be updated until the timer is stopped or expires (e.g., due to BWP switching). A timer may be used to measure a time period/window for a process. When the specification refers to an implementation and procedure related to one or more timers, it will be understood that there are multiple ways to implement the one or more timers. For example, it will be understood that one or more of the multiple ways to implement a timer may be used to measure a time period/window for the procedure. For example, a random access response window timer may be used for measuring a window of time for receiving a random access response. In an example, instead of starting and expiry of a random access response window timer, the time difference between two time stamps may be used. When a timer is restarted, a process for measurement of time window may be restarted. Other example implementations may be provided to restart a measurement of a time window.

A wireless device may an uplink data transmission (e.g., transmission of uplink data in DTCH) in an RRC_CONNECTED state. For example, the wireless device may not perform (e.g., may not be allowed to perform or may

prohibit) an uplink data transmission in an Non-RRC_CONNECTED (e.g., an RRC_INACTIVE state and/or an RRC_IDLE state). The wireless device may make (e.g., set up, (re-)establish, and/or resume) a connection to a network for transmission(s) of data. The data may be DL (e.g., mobile terminated (MT)) data and/or UL (e.g., mobile originated (MO)) data. For example, a wireless device may perform one or more procedures to make the connection to the network in the RRC_INACTIVE state (or the RRC_IDLE state). For example, the one or more procedures comprise a connection setup procedure, connection a (re-)establish procedure, and/or a connection resume procedure. For example, the wireless device may perform the one or more procedures, e.g., when DL (e.g., mobile terminated (MT)) and/or UL (e.g., mobile originated (MO)) data are available in a buffer. Based on the one or more procedures (e.g., in response to successfully completing the connection setup or resume procedure), the RRC state of the wireless device may transition to RRC_CONNECTED state from an RRC_INACTIVE state (or from an RRC_IDLE state). The wireless device may receive DL data and/or DL signal(s) via DL transmission(s) and/or may transmit UL data and/or UL signal(s) via UL transmission in the RRC_CONNECTED state. The wireless device may transition to the RRC_INACTIVE state (or to the RRC_IDLE state) from RRC_CONNECTED state, e.g., after or in response to no more DL data (e.g., to be received) and/or no more UL data (e.g., to be transmitted) in buffer(s). To transition to the RRC_INACTIVE state from the RRC_CONNECTED state, the wireless device may perform a connection release procedure. The connection release procedure (e.g., an RRC release procedure) may result in transitioning the RRC state to the Non-RRC_CONNECTED state (e.g., RRC_INACTIVE state or RRC_IDLE state).

A frequent RRC state transition between a Non-RRC_CONNECTED state and an RRC_CONNECTED state may require a wireless device to transmit and/or receive a plurality of control signals in one or more layers (e.g., RRC messages, MAC CEs, and/or DCIs). For example, for an RRC connection setup, the wireless device may transmit, to a base station, an RRC connection setup request and receive an RRC connection setup message as a respond to the RRC connection setup request. For example, for an RRC connection resume, the wireless device may transmit, to a base station, an RRC connection resume request and receive an RRC connection resume message as a respond to the RRC connection resume request. For example, for an RRC connection release, the wireless device may receive, from a base station, an RRC connection release request. For example, for DL and/or UL transmission of small data available (or arrival) in the Non-RRC_CONNECTED, it may be inefficient for a wireless device to make (or resume) an connection to a network (e.g., transition to RRC_CONNECTED from Non-RRC_CONNECTED) and release the connection (e.g., transition to RRC_INACTIVE or RRC_IDLE from RRC_CONNECTED) after or in response to perform the DL and/or UL transmission of small data in RRC_CONNECTED. This may result in increasing unnecessary power consumption and/or signaling overhead. For example, the signaling overhead (e.g., control signaling overhead) required to transmit a payload may be larger than the payload. For example, a frequent RRC state transition for the small and infrequent DL and/or UL data packet(s) may cause unnecessary power consumption and signaling overhead for the wireless device.

Examples of small and infrequent data packets may be such traffic generated from smartphone applications, Instant

Messaging (IM) services, heart-beat/keep-alive traffic from IM/email clients and other apps, push notifications from various applications, non-smartphone applications, wearables (e.g., positioning information), sensors (e.g., for transmitting temperature, pressure readings periodically or in an event triggered manner), and/or smart meters and smart meter networks sending meter readings.

A wireless device may perform uplink data transmission(s) in an RRC_INACTIVE state (or in an RRC_IDLE state). For example, a wireless device may transmit one or more data packets in an Non-RRC_CONNECTED state. For example, the wireless device may receive, from a base station, scheduling information (e.g., RRC message) indicating one or more uplink radio resources in the Non-RRC_CONNECTED state for the wireless device. The one or more uplink radio resources may be for infrequent data transmission. The one or more uplink radio resources may be for non-periodic data transmission. The one or more uplink radio resources may be for periodic data transmission. The wireless device may transmit the one or more data packets via the one or more radio resources while keeping its RRC state as the RRC_INACTIVE state (and/or RRC_IDLE state). For example, the wireless device may not transition its RRC state to the RRC_CONNECTED to transmit the one or more data packets.

The uplink transmission(s) via the one or more radio resources in an Non-RRC_CONNECTED state may be efficient and flexible (e.g., for low throughput short data bursts). The uplink transmission(s) via the one or more radio resources in an Non-RRC_CONNECTED state may provide efficient signaling mechanisms (e.g. signaling overhead is less than payload). The uplink transmission(s) via the one or more radio resources in an Non-RRC_CONNECTED state may reduce signaling overhead. The uplink transmission(s) via the one or more radio resources in an Non-RRC_CONNECTED state may improve the battery performance of the wireless device. For example, a wireless device that has intermittent small data packets in the Non-RRC_CONNECTED state may benefit from such uplink transmission(s) in the Non-RRC_CONNECTED state.

Uplink transmission(s) in an Non-RRC_CONNECTED state may be based on a random access (RA) procedure. For example, a wireless device may transmit uplink data (e.g., associated with DTCH) via MsgA PUSCH and/or Msg3 PUSCH. The wireless device may keep (or maintain) an RRC state as the Non-RRC_CONNECTED state during the RA procedure. After or in response to completing the transmission of the uplink data and/or completing the RA procedure, the wireless device may keep (or maintain) an RRC state as the Non-RRC_CONNECTED state.

Uplink transmission in an Non-RRC_CONNECTED state may be based on pre-configured PUSCH resource(s). For example, a wireless device may receive resource configuration parameters indicating UL grant(s) and/or the pre-configured PUSCH resource(s) of the UL grant(s). The wireless device may transmit uplink data (e.g., associated with DTCH) using the UL grant(s) and/or via the pre-configured PUSCH resource(s) of the UL grant(s) in the Non-RRC_CONNECTED state.

Uplink data transmission(s) in an Non-RRC_CONNECTED state may be referred to as small data transmission (SDT), early data transmission (EDT), and/or data transmission in an RRC_INACTIVE (or IDLE). For example, in the present disclosure, an SDT and/or an EDT may be interchangeable with uplink data transmission(s) in an Non-RRC_CONNECTED state. For example, an RA-based SDT and/or an RA-based EDT may be interchangeable with

uplink data transmission(s) via an RA procedure in an Non-RRC_CONNECTED state.

FIG. 17 illustrates uplink data transmission in an Non-RRC_CONNECTED state according to some embodiments. The wireless device may receive one or more messages comprising configuration parameters for the uplink data transmission. The wireless device may receive the one or more messages in the RRC_CONNECTED state. The wireless device may receive the one or more messages in the Non-RRC_CONNECTED state. The one or more messages may be broadcast, e.g., system information block. The one or more messages may be wireless-device-specific, e.g., an RRC message, MAC CE, and/or a DCI dedicated to the wireless device. The configuration parameters may indicate uplink resource(s) available, scheduled, and/or configured via an RA procedure during the Non-RRC_CONNECTED state. The wireless device may keep the RRC state as the Non-RRC_CONNECTED state, e.g., after or while performing the uplink data transmission.

In the present disclosure, uplink data transmission(s) in an Non-RRC_CONNECTED state may be interchangeable with uplink data transmission(s) in an RRC_INACTIVE state and/or in an RRC_IDLE state. For example, the procedure(s), configuration parameter(s), and/or feature description(s) that are related to uplink data transmission(s) in an Non-RRC_CONNECTED state may be applicable to and/or available to an RRC_INACTIVE state and/or an RRC_IDLE state, e.g., unless specify them for a particular RRC state. For example, if RRC_CONNECTED and/or RRC_IDLE state are RRC states that a wireless device has, the procedure(s), configuration parameter(s), and/or feature description(s) that are related to uplink data transmission(s) in an Non-RRC_CONNECTED state described in the present disclosure may be applicable to and/or available for an RRC_IDLE state of the wireless device. For example, if RRC_CONNECTED, RRC_INACTIVE, and/or RRC_IDLE state are RRC states that a wireless device has, the procedure(s), configuration parameter(s), and/or feature description(s) that are related to uplink data transmission(s) in an Non-RRC_CONNECTED described in the present disclosure may be applicable to and/or available for an RRC_INACTIVE and/or an RRC_IDLE state of the wireless device.

In the present disclosure, a four-step contention-based RA procedure (e.g., FIG. 13A) may be referred to as (and/or interchangeable with) a four-step RA type of an RA procedure, a four-step RA procedure, and/or an RA procedure with a four-step RA type. A two-step contention-free RA procedure (e.g., FIG. 13B) may be referred to as (and/or interchangeable with) a four-step RA type of an RA procedure, a four-step RA procedure, and/or an RA procedure with a four-step RA type. A two-step RA procedure (e.g., FIG. 13B) may be referred to as (and/or interchangeable with) a two-step RA type of an RA procedure, a two-step RA procedure, and/or an RA procedure with a two-step RA type.

A wireless device may initiate a random access (RA) procedure on a cell to transmit, via the cell, uplink data in an Non-RRC_CONNECTED state. The uplink data may be associated with a particular channel (e.g., logical channel). For example, the uplink data may be associated with (e.g., may be the one from) a particular logical channel (e.g., associated with DTCH). For example, the wireless device may initiate the RA procedure in the Non-RRC_CONNECTED state. The wireless device may keep its RRC state as the Non-RRC_CONNECTED state while performing the RA procedure and/or while transmitting the uplink data during the RA procedure. The wireless device may keep the

Non-RRC_CONNECTED state in response to or after completing the RA procedure and/or completing the transmission of the uplink data.

A network or a base station may indicate which cell is available for transmission (e.g., SDT and/or EDT) of uplink data (e.g., associated with DTCH) in an Non-RRC_CONNECTED state. For example, the base station may broadcast (multicast and/or unicast) system information block(s) of a cell. The system information block(s) may comprise one or more parameters indicating whether a wireless device performs, via the cell, the transmission of uplink data in an Non-RRC_CONNECTED state.

The one or more parameters may be a field indicating the wireless device is allowed to initiate an RA-based SDT on the cell. The indication may be true (e.g., initiating the RA-based SDT is allowed) or false (e.g., initiating the RA-based SDT is not allowed). The indication may be a presence of the field (e.g., initiating the RA-based SDT is allowed) or an absence of the field (e.g., initiating the RA-based SDT is not allowed).

The field may indicate that the wireless device is allowed to initiate RA-based SDT on the cell for transmission of a particular type of data. For example, the particular type of data may comprise control plane (CP) data, user plane (UP) data, mobile originating (MO) data (or call), and/or mobile terminating (MT) data (or call), and/or the like. Example formats of the field for CP and UP data may be:

cp-SDT ENUMERATED {true}	OPTIONAL, -- Need OR
up-SDT ENUMERATED {true}	OPTIONAL, -- Need OR.

cp-SDT (=true) and up-SDT (=true) may respectively indicate the wireless device is allowed to initiate SDT for transmission of CP data and UP data.

The field may indicate that the wireless device is allowed to initiate RA-based SDT on the cell when connected to a particular type of network. For example, the particular type of network may comprise an evolved packet core (EPC) network, a 5G core (5GC) network, and/or the like. The field may indicate that the wireless device is allowed to initiate an RA-based SDT on the cell for transmission of a particular type of data when connected to the particular type of network. Example formats of the field for transmission of CP data via EPC or 5GC may be:

cp-SDT-EPC	ENUMERATED {true}	OPTIONAL, -- Need OR
cp-SDT-5GC	ENUMERATED {true}	OPTIONAL, -- Need OR.

cp-SDT-EPC (=true) and cp-SDT-5GC (=true) may respectively indicate the wireless device is allowed to initiate the RA-based SDT for transmission of CP data via the EPC and 5GC.

The wireless device may initiate an RA-based SDT on a cell when one or more conditions are fulfilled. For example, the one or more conditions may be whether upper layer(s) request an establishment or resumption of an RRC connection, whether the wireless device supports the SDT for a particular type of data, whether one or more parameters (e.g., broadcast via system information block(s)) indicate that the wireless device the RA-based SDT for the particular type of data when connected to a particular type of network. For example, for CP-SDT when the wireless device is connected to 5GC, the wireless device may initiate the RA-based SDT for the CP data based on at least one of upper layer(s) requesting an establishment or resumption of an

RRC connection, CP-SDT available by the wireless device, and/or system information block(s) comprising cp-SDT=5GC=true.

For an RA-based SDT, the wireless device may determine a size of transport block (e.g., a size of message comprising uplink data). The transport block may comprise uplink data (e.g., associated with DTCH) that the wireless device transmits via the RA-based SDT. The transport block may comprise (e.g., further comprise) one or more MAC headers, e.g., if required, and/or one or more MAC CEs, e.g., if triggered. For example, the transport block that the wireless device transmits via the RA-based SDT may be a MAC PDU that comprises the uplink data, the one or more MAC headers, and/or the one or more MAC CEs.

A network or a base station may transmit (e.g., broadcast, multicast, and/or unicast) one or more message (e.g., system information block(s), RRC message(s), MAC CE(s), DCI(s) and/or any combination thereof) comprising one or more sdt-TBS values of a cell. For example, the one or more sdt-TBS values may indicate an amount of uplink data (e.g., associated with DTCH) that a wireless device transmits via an RA-based SDT on the cell. The wireless device that receives the one or more messages may determine, based on the one or more sdt-TBS values, whether the wireless device initiates an RA-based SDT on the cell. The wireless device may determine a size of transport block comprising uplink data. The wireless device may determine to transmit the uplink data via the RA-based SDT (or initiate the RA-based SDT for transmission of the uplink data), e.g., if the size is smaller than or equal to at least one of the one or more sdt-TBS values. For example, the wireless device may be allowed to initiate the RA-based SDT on the cell for transmission of the uplink data, e.g., if the size is smaller than or equal to at least one of the one or more sdt-TBS values. The wireless device may determine not to transmit the uplink data via the RA-based SDT, e.g., if the size is larger than at least one of the one or more sdt-TBS values (e.g., larger than all of the one or more sdt-TBS values). For example, the wireless device may not be allowed to initiate the RA-based SDT on the cell for transmission of the uplink data, e.g., if the size is larger than at least one of the one or more sdt-TBS values (e.g., larger than all of the one or more sdt-TBS more values).

The one or more sdt-TBS values may indicate whether the wireless device initiates the RA-based SDT for transmission of uplink data (e.g., associated with DTCH) or an RA procedure to make a connection to the network or the base station. For example, the wireless device may determine to transmit the uplink data via the RA-based SDT (or initiate the RA-based SDT for transmission of the uplink data), e.g., if the size is smaller than or equal to at least one of the one or more sdt-TBS values. The wireless device may keep its RRC state as an Non-RRC_CONNECTED state while the RA-based SDT and/or after completing the RA-based SDT. For example, the wireless device may determine not to perform (or initiate) the uplink data via the RA-based SDT, e.g., if the size is larger than at least one of the one or more sdt-TBS values (e.g., larger than all of the one or more sdt-TBS values). In this case, the wireless device may initiate the RA procedure to make the connection. The wireless device may transmit the uplink data, e.g., after or in response to determining that the RA procedure is successfully completed. The wireless device may transition its RRC state from an Non-RRC_CONNECTED state to an RRC_CONNECTED state after or in response to determining that the RA procedure is successfully completed. For

example, in this case, the wireless device may transmit the uplink data in the RRC_CONNECTED state.

A base station (or a network) may transmit (broadcast, multicast, and/or unicast) one or more messages (e.g., system information block(s), RRC message(s), MAC CE(s), DCI(s) and/or any combination thereof) comprising a sdt-TBS value of a cell. The one or more messages may comprise an sdt-TBS value per an RA type of an RA procedure of the cell. For example, one or more RA types of the RA procedure may be available on the cell. The one or more RA types may comprise a four-step contention-based RA procedure (e.g., FIG. 13A), a two-step contention-free RA procedure (e.g., FIG. 13A and/or FIG. 13B), and/or a two-step RA procedure (e.g., FIG. 13C). The sdt-TBS value may be a common parameters applied to one or more RA types of the RA procedure configured on the cell. A wireless device that receives the one or more messages may determine a particular RA type of RA procedure. The wireless device may determine (e.g., select) a particular sdt-TBS value of the particular RA type of RA procedure. The wireless device may determine, based on the particular sdt-TBS value, whether the wireless device transmits uplink data (e.g., associated with DTCH) via an RA-based SDT. The RA-based SDT may use one or more parameters (and/or procedures) of the particular RA procedure. For example, the wireless device may initiate, using the particular RA procedure, the RA-based SDT on the cell, e.g., if a size of transport block comprising the uplink data (e.g., a size of message comprising the uplink data) is smaller than or equal to the particular sdt-TBS value. For example, the wireless device may not initiate, using the particular RA procedure, the RA-based SDT, e.g., if the size of transport block is larger than the particular sdt-TBS value. The wireless device may select a different RA type of RA procedure of the cell and/or may initiate, using the different RA type of RA procedure, the RA-based SDT, e.g., if the size of transport block is larger than the particular sdt-TBS value. For example, an sdt-TBS value of the different RA type may be larger than the size of transport block.

An example configuration parameter of an sdt-TBS (e.g., or edt-TBS) value may be a value in bits. For example, an example format of the sdt-TBS value may be sdt-TBS-r15 ENUMERATED {b328, b408, b504, b600, b712, b808, b936, b1000or456}, where, for example, a value b328 may correspond to 328 bits, b408 may correspond to 408 bits and so on. For example, a value b1000or456 may correspond to 1000 bits for one or more first RA types of RA procedure, and 456 bits for one or more second RA types of RA procedure.

A base station (or a network) may transmit (e.g., broadcast, multicast, and/or unicast) one or more messages (e.g., system information block(s), RRC message(s), MAC CE(s), DCI(s) and/or any combination thereof) comprising one or more sdt-TBS values of a cell. The one or more sdt-TBS values may be per an RA type of an RA procedure of the cell. For example, one or more RA types of the RA procedure may be available on the cell. The one or more RA types may comprise a four-step contention-based RA procedure (e.g., FIG. 13A), a two-step contention-free RA procedure (e.g., FIG. 13A and/or FIG. 13B), and/or a two-step RA procedure (e.g., FIG. 13C). The one or more sdt-TBS values may be a common parameters applied to one or more RA types of the RA procedure configured on the cell. A wireless device that receives the one or more messages may determine a particular RA type of RA procedure. The wireless device may select a particular sdt-TBS value among the one or more sdt-TBS values for the particular RA type of RA procedure.

The one or more messages may indicate that the one or more sdt-TBS values are configured for the particular RA type of RA procedure. The wireless device may determine, based on the particular sdt-TBS value, whether the wireless device transmits uplink data (e.g., associated with DTCH) via an RA-based SDT. The RA-based SDT may use one or more parameters (and/or procedures) of the particular RA procedure. For example, the wireless device may initiate, using the particular RA procedure, the RA-based SDT on the cell, e.g., if a size of transport block comprising the uplink data (e.g., a size of message comprising the uplink data) is smaller than or equal to the particular sdt-TBS value. For example, the wireless device may not initiate, using the particular RA procedure, the RA-based SDT, e.g., if the size of transport block is larger than the particular sdt-TBS value. The wireless device may select a different RA type of RA procedure of the cell and/or may initiate, using the different RA type of RA procedure, the RA-based SDT, e.g., if the size of transport block is larger than the particular sdt-TBS value. For example, an sdt-TBS value of the different RA type may be larger than the size of transport block.

FIG. 18 illustrates one or more sdt-TBS values configured for an RA procedure according to some embodiments. A network or a base station may transmit (e.g., broadcast, multicast, or unicast) one or more messages (e.g., system information block(s), RRC message(s), MAC CE(s), DCI(s) and/or any combination thereof). The one or more messages may configure one or more RA types of an RA procedure for an RA-based SDT. For example, in FIG. 18, the one or more messages may configure K (e.g., $K \geq 1$) RA types of RA procedure for the RA-based SDT. Each RA type of the RA procedure may comprise one or more sdt-TBS values. A number of the one or more sdt-TBS values configured for a first RA type of the one or more RA types may be different from that configured for a second RA type of the one or more RA types. A number of the one or more sdt-TBS values may be the same between two or more of the one or more RA types. For example, in FIG. 18, the first RA type and the K-th RA type of RA procedure may comprise M sdt-TBS values (e.g., $M \geq 1$) and N sdt-TBS values (e.g., $N \geq 1$), respectively. A wireless device that receives the one or more messages may determine whether the wireless device transmits a TB (and/or a MAC PDU of the TB) comprising uplink data (e.g., associated with DTCH) via the K-th RA type of RA procedure. For example, the wireless device may determine to initiate the K-th RA type of RA procedure, e.g., if a size of the TB is smaller than or equal to at least one of N sdt-TBS values of the K-th RA type of RA procedure. For example, in FIG. 18, a size of a MAC subPDU(s) comprising the uplink data is larger than a first sdt-TBS value (e.g., 1st sdt-TBS of the K-th RA type in FIG. 18) and smaller than a second sdt-TBS value (e.g., 2nd sdt-TBS of the K-th RA type in FIG. 18). The wireless device may generate (e.g., multiplex, assemble, construct, and/or obtain) a TB comprising the uplink data (e.g., comprising the MAC subPDU(s)). The TB may comprise one or more padding bit(s). The wireless device may determine a number of one or more padding bit(s) based on a difference between the second sdt-TBS value and the size of the MAC subPDU(s).

A wireless device may initiate an RA-based SDT on a cell. The wireless device may determine a particular RA type of an RA procedure for the RA-based SDT. For example, a network or a base station may transmit (e.g., broadcast, multicast, and/or unicast) one or more messages (e.g., system information block(s), RRC message(s), MAC CE(s), DCI(s) and/or any combination thereof), comprising configuration parameters of the RA-based SDT for the cell. The

configuration parameters may indicate, among one or more RA types, at least one RA type of the RA procedure is available for the RA-based SDT on the cell. The wireless device may initiate the at least one RA type of the RA procedure for the RA-based SDT on the cell, e.g., based on one or more conditions (e.g., a type of uplink data that the wireless device transmits, a type of network that the wireless device connects, and/or sdt-TBS value(s) of the at least one RA type) described in the present disclosure.

An RA procedure may comprise one or more procedures and/or steps. For example, the RA procedure may comprise an RA procedure initialization. The wireless device may select an RA type of the RA procedure in the RA procedure initialization. For example, The RA procedure may comprise an initialization of variable(s) specific to the RA type that the wireless device may select in the RA procedure initialization. After or in response to the initialization of variable(s) specific to the RA type, the wireless device may perform one or more procedure(s) and/or step(s) based on the RA type of the RA procedure. The wireless device may switch the RA type during the RA procedure from one to another.

FIG. 19 illustrates an RA procedure may comprise one or more procedures and/or steps according to some embodiments. FIG. 19 is an example of selecting an RA type among a two-step RA type (e.g., FIG. 13C) and a four-step RA type (e.g., FIG. 13A and/or FIG. 13B). A wireless device may receive one or more messages comprising configuration parameters of an RA procedure. The configuration parameters may indicate that the two-step RA type (e.g., FIG. 13C) and/or the four-step RA type (e.g., FIG. 13A and/or FIG. 13B) are available on a cell. at least one RA type among the two-step RA (e.g., FIG. 13C) and/or the four-step RA (e.g., FIG. 13A and/or FIG. 13B) may be available for an RA-based SDT. None of the two-step RA (e.g., FIG. 13C) and/or the four-step RA (e.g., FIG. 13A and/or FIG. 13B) may be available for an RA-based SDT. The wireless device may perform an RA type selection. The wireless device may select an RA type of the RA procedure. For example, the wireless device may use an RSRP value to select the RA type during the RA type selection. For example, the wireless device may use, to select the RA type during the RA type selection, one or more conditions of initiating the RA-based SDT (e.g., a type of uplink data that the wireless device transmits, a type of network that the wireless device connects, and/or sdt-TBS value(s) of the at least one RA type).

FIG. 20A illustrates an RA-based SDT with a four-step RA procedure according to some embodiments. A wireless device may receive configuration parameters for the RA-based SDT according to some embodiments. The wireless device may initiate the four-step RA procedure for the RA-based SDT. The wireless device may determine to transmit a preamble (e.g., Msg1) via PRACH resource(s) (e.g., RA preamble transmission in FIG. 19). The wireless device may determine the preamble and/or the PRACH resource(s) to indicate, to a base station, a request of a transmission of uplink data (e.g., associated with DTCH) via Msg3 (e.g., RA resource selection in FIG. 19). The request may be an indication of triggering and/or initiating the RA-based SDT. The request may indicate a size (e.g., expected, measured, determined size) of a TB comprising the uplink data. The wireless device may receive a response to the preamble (e.g., Msg1). The response may indicate whether the wireless device is allowed to transmit the uplink data via Msg 3 transmission. If the response indicates that the wireless device is not allowed to transmit the uplink data, the wireless device may cancel the RA-based SDT. The wireless device may transmit Msg3 without the uplink data,

e.g., after or in response to canceling the RA-based SDT. If the response indicates that the wireless device is allowed to transmit the uplink data, the wireless device transmit the TB comprising the uplink data via Msg3 transmission. The wireless device may receive a response (e.g., Msg 4) to the Msg3 transmission.

FIG. 20B illustrates an RA-based SDT with a two-step RA procedure according to some embodiments. A wireless device may receive configuration parameters for the RA-based SDT according to some embodiments. The wireless device may initiate the two-step RA procedure for the RA-based SDT. The wireless device may determine to transmit a preamble (e.g., in MsgA) via PRACH resource(s) (e.g., MsgA transmission in FIG. 19). The wireless device may determine to transmit a TB (e.g., in MsgA) comprising uplink data (e.g., associated with DTCH) via PUSCH resource(s) (e.g., MsgA transmission in FIG. 19). The wireless device may determine the preamble, the PRACH resource(s), and/or PUSCH resource(s) to indicate, to a base station, a request of a transmission of uplink data via MsgA (e.g., RA resource selection for two-step RA type in FIG. 19). The request may be an indication of triggering and/or initiating the RA-based SDT. The request may indicate a size (e.g., expected, measured, determined size) of the TB comprising the uplink data. The wireless device may receive a response (e.g., MsgB reception and contention resolution for two-step RA type in FIG. 19) to the MsgA. The response may indicate a success (e.g., successRAR) of the MsgA transmission. The response may indicate a fallback (e.g., fallbackRAR) to a contention resolution of the four-step RA procedure. The wireless device may (re)transmit the TB via Msg 3 transmission of the contention resolution. The response may indicate that the wireless device is not allowed to perform the RA-based SDT. In this case, the wireless device may cancel the RA-based SDT.

A wireless device may initiate an RA procedure on a cell for transmission of uplink data (e.g., associated with DTCH) via an RA-based SDT. The wireless device may select an RA type of the RA procedure among a four-step RA type and a two-step RA type. The RA type may be associated with at least one sdt-TBS value. The wireless device may determine a TBS of a TB based on the at least one sdt-TBS value. For example, the TB may comprise a MAC PDU that comprises the uplink data and/or one or more padding bits. The wireless device may append the one or more padding bits to the MAC PDU, e.g., if a size of the uplink data (e.g., expected message comprising the uplink data) is smaller than the TBS.

FIG. 21 illustrates a MAC PDU comprising uplink data and/or padding bit(s) according to some embodiments. The wireless device may select an RA type of an RA procedure for an RA-based SDT to transmit uplink data (e.g., associated with DTCH). The wireless device may generate (e.g., multiplex, assemble, construct, and/or obtain) a TB comprising a MAC PDU for the transmission of the uplink data via the RA type. The MAC PDU may comprise a MAC SDU that comprises the uplink data. For example, the wireless device may determine a size of the TB based on at least one sdt-TBS value (e.g., sdt-TBS in FIG. 21) associated with the RA type. For example, the size may be smaller than or equal to the at least one sdt-TBS value. The wireless device may append the padding bit(s) to the MAC PDU, e.g., if a size of the uplink data is smaller than the at least one sdt-TBS value. For example, the wireless device may append the padding bit(s) to the MAC PDU to make the size of the TB be equal to the at least one sdt-TBS value, e.g., if a size of the uplink data is smaller than the at least one sdt-TBS value.

FIG. 22 illustrates a MAC PDU comprising uplink data and/or padding bit(s) for one or more sdt-TBS values according to some embodiments. The wireless device may select an RA type of an RA procedure for an RA-based SDT to transmit uplink data (e.g., associated with DTCH). The wireless device may generate (e.g., multiplex, assemble, construct, and/or obtain) a TB comprising a MAC PDU for the transmission of the uplink data via the RA type. The MAC PDU may comprise a MAC SDU that comprises the uplink data. For example, the wireless device may determine a size of the TB based on at least one sdt-TBS value (e.g., 2nd sdt-TBS in FIG. 22) associated with the RA type. The at least one sdt-TBS value may be one of the one or more sdt-TBS values associated with the RA type. For example, the size may be smaller than or equal to the at least one sdt-TBS value. The wireless device may append the padding bit(s) to the MAC PDU, e.g., if a size of the uplink data is smaller than the at least one sdt-TBS. For example, the wireless device may append the padding bit(s) to the MAC PDU to make the size of the TB be equal to the at least one sdt-TBS value, e.g., if a size of the uplink data is smaller than the at least one sdt-TBS value. For example, the at least one sdt-TBS value may be the smallest value among one or more second sdt-TBS values. The one or more second sdt-TBS values may be selected among the one or more sdt-TBS values and/or may be larger than or equal to the size of the uplink data.

In an existing technology, a wireless device initiating an RA procedure on a cell may select an RA type based on an RSRP of the downlink pathloss reference of the cell. For example, the wireless device may select a two-step RA type, e.g., at least based on the RSRP being larger than a threshold value (e.g., msgA-RSRP-Threshold). For example, the wireless device may select a four-step RA type, e.g., at least based on the RSRP being smaller than or equal to the threshold value.

In some embodiments, a wireless device may initiate an RA procedure on a cell for transmission of uplink data (e.g., associated with DTCH) via an RA-based SDT. The wireless device initiating the RA procedure may select an RA type of the RA procedure. The wireless device may select the RA type among one or more RA types. The wireless device may generate (e.g., multiplex, assemble, construct, and/or obtain) a TB comprising a MAC PDU that comprises the uplink data. The wireless device may determine a TBS of the TB based on a sdt-TBS value associated with the selected RA type. For example, one or more sdt-TBS values may be configured for the RA-based SDT on the cell. At least one of the one or more sdt-TBS values may be associated with at least one of the one or more RA types. For example, a first RA type of the one or more RA types is associated with a first sdt-TBS value of the one or more sdt-TBS values. A second RA type of the one or more RA types is associated with a second sdt-TBS of the one or more sdt-TBS values.

A problem may arise when the wireless device selects the first RA type associated with the first sdt-TBS that may be larger than the second sdt-TBS value. For example, a first number of padding bit(s) required for the first sdt-TBS may be larger than a second number of padding bit(s) required for the second sdt-TBS, e.g., if the first sdt-TBS that may be larger than the second sdt-TBS value. For example, the wireless device may determine a size of message comprising uplink data (e.g., associated with DTCH). The message may comprise one or more MAC headers and/or one or more MAC CEs. If the size of the message is x bits, the first number and the second number may be (the first sdt-TBS value- x) and (the second sdt-TBS value- x), respectively. If

the wireless device may select the first RA type, e.g., based on an RSRP of a downlink pathloss reference, the wireless device may append a larger number of padding bit(s) to the TB (e.g., MAC PDU of the TB). This result in an inefficient use of radio resource(s) and reducing a spectral efficiency of the radio resource(s).

In the present disclosure, a wireless device may select an RA type based on a number of (expected) padding bits to be appended to a TB (e.g., a MAC PDU of the TB) for an RA-based SDT. For example, if one or more RA types are available for the RA-based SDT, the wireless device may select, among the one or more RA types, an RA type that requires a smaller number of padding bits in the TB than the other RA types of the one or more RA types. For example, the wireless device may select the RA type in a way to reduce (e.g., or to minimize) a number of padding bit(s) appended to the TB (e.g., the MAC PDU of the TB). This result in an inefficient use of radio resource(s) and a decrease of spectral efficiency. The selection of the RA type based on a number of padding bits may result in an efficient use of radio resource(s) and improve a spectral efficiency of the radio resource(s).

FIG. 23 illustrates an RA type selection for an RA-based SDT according to some embodiments. A wireless device may initiate an RA procedure on a cell for transmission of uplink data (e.g., associated with DTCH) in Non-RRC_CONNECTED state. A first RA type of the RA procedure and a second RA type of the RA procedure may be configured (available) for the RA-based SDT on the cell. A first RA type and the second RA types may be the ones among a plurality of RA types comprising a two-step RA type and a four-step RA type. For example, the first RA type may be associated with a first sdt-TBS value (1st sdt-TBS in FIG. 21). For example, the wireless device may generate (e.g., multiplex, assemble, construct, and/or obtain), based on the first sdt-TBS value, a first TB comprising a first MAC PDU that comprises the uplink data, e.g., if the wireless device selects the first RA type. For example, a first TBS of the first TB generated based on the first sdt-TBS value may be the first sdt-TBS value. For example, the second RA type may be associated with a second sdt-TBS value (2nd sdt-TBS in FIG. 21). For example, the wireless device may generate (e.g., multiplex, assemble, construct, and/or obtain), based on the second sdt-TBS value, a second TB comprising a second MAC PDU that comprises the uplink data, e.g., if the wireless device selects the second RA type. For example, a second TBS of the second TB generated based on the second sdt-TBS value may be the second sdt-TBS value.

In FIG. 23, the first sdt-TBS value may be larger than the second sdt-TBS value. The wireless device may determine a size (e.g., expected size) of message comprising the uplink data. The message may further comprise one or more MAC headers, one or more MAC CEs, and/or one or more MAC SDUs. The size may be smaller than or equal to the first sdt-TBS value and the second sdt-TBS value as shown in FIG. 23. The wireless device may determine a first number of padding bits required for a first TB (e.g., a first MAC PDU of the first TB) generated based on the first sdt-TBS value. The wireless device may determine a second number of padding bits required for a second TB (e.g., a second MAC PDU of the TB) generated based on the second sdt-TBS value. The wireless device may select the second RA type, e.g., based on the second number is smaller than or equal to the first number. The wireless device may reduce one or more padding bits (e.g., unnecessary padding bit(s) in FIG. 23) by selecting the second RA type.

For example, the wireless device may select an RA type based on an RSRP of a downlink pathloss reference on the cell and/or a number of padding bits. For example, an RA type selection based on the RSRP may have a higher priority than an RA type selection based on a number of padding bits. For example, if the RSRP is larger than a threshold value (e.g., msgA-RSRP-Threshold), the wireless device may determine that the first RA type and the second RA type are available to select. In this case, the wireless device may determine an RA type of the RA-based SDT based on a number of padding bits. For example, if the RSRP is smaller than or equal to the threshold value (e.g., msgA-RSRP-Threshold), the wireless device may determine that one of the first RA type and the second RA type is available to select. For example, an available RA type for the RSRP being smaller than or equal to the threshold value may be configured or predefined. For example, a four-step RA type may be available to select (e.g., a two-step RA type may be unavailable to select), e.g., if the RSRP is smaller than or equal to the threshold value. In this case, the wireless device may determine the available RA type for the RA-based SDT.

In FIG. 23, for example, the first RA type and the second RA type may be a two-step RA type and a four-step RA type, respectively. An RSRP of a downlink pathloss reference may be larger than a threshold value (e.g., msgA-RSRP-Threshold). The wireless device may select the four-step RA type for the RA-based SDT, e.g., based on the second number being smaller than or equal to the first number and/or the RSRP being larger than a threshold value.

In FIG. 23, for example, the first RA type and the second RA type may be a four-step RA type and a two-step RA type, respectively. An RSRP of a downlink pathloss reference may be larger than a threshold value (e.g., msgA-RSRP-Threshold). The wireless device may select the two-step RA type for the RA-based SDT, e.g., based on the second number being smaller than or equal to the first number and/or the RSRP being larger than the threshold value.

In FIG. 23, for example, the first RA type and the second RA type may be a two-step RA type and a four-step RA type, respectively. An RSRP of a downlink pathloss reference may be smaller than or equal to a threshold value (e.g., msgA-RSRP-Threshold). The wireless device may select the four-step RA type for the RA-based SDT, e.g., based on the second number being smaller than or equal to the first number and/or the RSRP being smaller than or equal to the threshold value.

In FIG. 23, for example, the first RA type and the second RA type may be a four-step RA type and a two-step RA type, respectively. An RSRP of a downlink pathloss reference may be smaller than or equal to a threshold value (e.g., msgA-RSRP-Threshold). In this case, an RA type selection based on the RSRP may have a higher priority than an RA type selection based on a number of padding bits. For example, the wireless device may select the four-step RA type for the RA-based SDT, e.g., based on the RSRP being smaller than or equal to the threshold value.

In an example, as shown in FIG. 18, a network or a base station may transmit (e.g., broadcast, multicast, or unicast) one or more messages (e.g., system information block(s), RRC message(s), MAC CE(s), DCI(s) and/or any combination thereof). The one or more messages may configure one or more RA types of an RA procedure for an RA-based SDT. For example, in FIG. 18, the one or more messages may configure K (e.g., $K \geq 1$) RA types of RA procedure for the RA-based SDT. Each RA type of the RA procedure may comprise one or more sdt-TBS values. A number of the one or more sdt-TBS values configured for a first RA type of the

one or more RA types may be different from that configured for a second RA type of the one or more RA types. A number of the one or more sdt-TBS values may be the same between two or more of the one or more RA types. A wireless device that receives the one or more messages may determine an RA type among the one or more RA types based on a number of padding bits appended to a TB (e.g., a MAC PDU of the TB) comprising uplink data (e.g., associated with DTCH). For example, the wireless device may select the RA type among the one or more RA types in a way to reduce (e.g., minimize) the number of padding bits. For example, the wireless device may determine a size (e.g., expected size) of a message comprising the uplink data. The message may further comprise one or more MAC headers, one or more MAC CEs, and/or one or more MAC SDUs. The wireless device may determine a group of sdt-TBS values that are larger than or equal to the size for the one or more RA types. For example, in FIG. 18, the group may comprise sdt-TBS value(s) from the second sdt-TBS to the M-th sdt-TBS of the first RA type and from the second sdt-TBS to the N-th sdt-TBS of the K-th RA type. The wireless device may select one of the sdt-TBS value(s) among the sdt-TBS value(s). For example, the wireless device may select the smallest sdt-TBS value among the sdt-TBS value(s). The wireless device may select an RA type configured with the selected sdt-TBS among the sdt-TBS value(s). In FIG. 18, the wireless device may select the K-th RA type, e.g., if the 2nd sdt-TBS of the K-th RA type is the smallest value among the sdt-TBS value(s) in the group. The wireless device may determine a number of one or more padding bit(s) based on a difference between the second sdt-TBS value of the K-th RA type and the size of the message.

In an example, a network or a base station may transmit (e.g., broadcast, multicast, or unicast) one or more messages (e.g., system information block(s), RRC message(s), MAC CE(s), DCI(s) and/or any combination thereof). The one or more messages may comprise configuration parameter(s) indicating whether the selection based on a number of padding bit(s) presented in the present disclosure enabled or disabled.

In an example, the selection based on a number of padding bit(s) presented in the present disclosure may not be configured with one or more mechanisms (processes and/or procedure) that reduce a number of padding bit(s). For example, the one or more mechanisms may comprise a selection of smaller TBS for SDT. For example, the wireless device may receive a configuration parameter (e.g., sdt-SmallTBS) indicating whether the selection of smaller TBS for SDT is enabled. For example, the wireless device may determine a sdt-TBS value for an RA-based SDT. If the selection of smaller TBS is enabled, the wireless device may determine a TBS value smaller than the sdt-TBS value. For example, one or more TBS values smaller than or equal to the sdt-TBS value may be configured or predefined (e.g., as a table). For the case that one or more sdt-TBS values are configured the one or more TBS values may be configured or predefined per each of one or more sdt-TBS values. For example, the selection based on a number of padding bit(s) may not be enabled with the selection of smaller TBS for SDT. For example, a base station and/or a network may enable either one of the selection based on a number of padding bit(s) or the selection of smaller TBS for SDT.

In an example, after or in response to selecting a particular RA type with a particular sdt-TBS value, the wireless device may transmit a preamble via a PRACH resource (occasion). The preamble and/or the PRACH resource (occasion) may be associated the particular RA type and/or the particular

sdt-TBS value. For example, the preamble via the PRACH resource may be an indication, to a base station and/or a network, that the wireless device requests an RA-based SDT with the particular RA type for the transmission of uplink data with a size of the particular sdt-TBS value. For example, the wireless device may transmit, e.g., using a four-step RA type, the preamble via Msg1 (e.g., FIG. 13A and/or FIG. 13B). In this case, the wireless device may transmit the uplink data via Msg3 transmission. For example, the wireless device may transmit, using a two-step RA type, the preamble via MsgA (e.g., FIG. 13C). In this case, the wireless device may transmit the uplink data via TB of MsgA.

FIG. 24 illustrates an RA type selection for an RA-based SDT according to some embodiments. A wireless device may initiate a random access (RA) procedure on a cell for transmission of uplink data in a radio resource control (RRC) inactive state. The wireless device may determine that a size of a message comprising the uplink data is smaller than or equal to at least one first transport block (TB) size of the RA procedure of a first type; and at least one second TB size of the RA procedure of a second type. The wireless device may select the first type based on the at least one first TB size is smaller than the at least one second TB size. The wireless device may transmit, based on the first type, a TB comprising the message.

According to an example embodiment, the RA procedure of the first type may be a two-step RA procedure and the RA procedure of the second type may be a four-step RA procedure. According to an example embodiment, the wireless device may transmit a preamble. For example, the transport block and the preamble is transmitted as an MsgA transmission of the two-step RA procedure. According to an example embodiment, the RA procedure of the first type may be a four-step RA procedure, and the RA procedure of the second type may be a two-step RA procedure. According to an example embodiment, the wireless device may transmit a preamble. The wireless device may receive an RA response to the preamble, the RA response comprising an uplink grant for the transmitting the TB. According to an example embodiment, the wireless device may receive configuration parameters indicating one or more first TB sizes of the RA procedure of the first type. For example, the one or more first TB sizes are larger than the size. According to an example embodiment, the at least one first TB size may be the smallest TB size among the one or more first TB sizes. According to an example embodiment, the wireless device may receive configuration parameters indicating one or more second TB sizes of the RA procedure of the second type. For example, the one or more second TB sizes may be larger than the size. For example, the at least one second TB size may be the smallest TB size among the one or more second TB sizes. According to an example embodiment, the selecting the first type may be further based on a reference signal received power of the cell is higher than an RA type selection power value. According to an example embodiment, the wireless device may multiplex one or more medium access control (MAC) service data units (SDUs) comprising the uplink data onto the TB. According to an example embodiment, the TB may further comprise one or more padding bits. For example, the wireless device may determine a number of the one or more padding bits based on the size and the at least one first TB size. For example, the number may be smaller than or equal to a difference between the size and the at least one first TB size. According to an example embodiment, the wireless device may keep an RRC state of the wireless device as the RRC inactive state

in response to receiving the response to the TB. For example, the wireless device may determine that the RA procedure is unsuccessfully completed in response to failing to receive the response to the TB. For example, the wireless device may determine to cancel the transmission of uplink data in response to failing to receive the response to the TB. According to an example embodiment, the wireless device may initiate a second RA procedure in response to failing to receive the response to the TB. For example, the wireless device may transmit the uplink data in response to determining that the second RA procedure is successfully completed. For example, the wireless device may determine to transition an RRC state to an RRC connected state in response to determining the second RA procedure is successfully completed the second RA procedure. For example, the uplink data may be associated with a dedicated traffic channel (DTCH).

In present embodiments, a wireless device may initiate a random access (RA) procedure on a cell for transmission of uplink data in a radio resource control (RRC) inactive state. The wireless device may determine that a size of a message comprising the uplink data is smaller than or equal to at least one first transport block (TB) size of the RA procedure of a four-step RA type; and at least one second TB size of the RA procedure of a two-step RA type. The wireless device may select the first type based on the at least one first TB size is smaller than the at least one second TB size. The wireless device may transmit a preamble of the four-step RA type. The wireless device may transmit, based on receiving a random access response of the preamble, a TB comprising the message.

According to an example embodiment, the wireless device may determine a reference signal received power of a downlink pathloss reference of the cell is larger than a RA type selection threshold value. According to an example embodiment, the wireless device may receive one or more configuration parameters indicating a selection of the RA type is based on the selecting. According to an example embodiment, the wireless device may receive a response to the TB. According to an example embodiment, the wireless device may determine to keep an RRC state of the wireless device as the RRC inactive state in response to receiving the response to the TB.

In present embodiments, a wireless device may select a random access (RA) procedure of a first type based on at least one first TB size of the RA procedure of first type is smaller than at least one second TB size of the RA procedure of the second type. The wireless device may transmit, based on the RA procedure of the first type, a TB comprising the uplink data.

According to an example embodiment, the TB may comprise a message comprising the uplink data. According to an example embodiment, a size of the message is smaller than or equal to the at least one first TB size; and the at least one second TB size. According to an example embodiment, the wireless device may initiate the RA procedure on a cell for transmission of the uplink data in a radio resource control (RRC) inactive state.

FIG. 25 illustrates an RA-based SDT according to some embodiments. A base station may transmit configuration parameters of a random access (RA) procedure on a cell for a wireless device to transmit uplink data in a radio resource control (RRC) inactive state. For example, the configuration parameters indicate at least one first transport block (TB) size of the RA procedure of a first type. For example, the configuration parameters indicate at least one second TB size of the RA procedure of a second type. The base station

may receive a TB comprising a message via a radio resource of the RA procedure of the first type. For example, a size of the message may be smaller than or equal to the at least one first TB size. For example, the at least one first TB size may be smaller than the at least one second TB size.

In the present disclosure, for an RA-based SDT, a wireless device may initiate an RA procedure. The wireless device may initialize variable(s) specific to the RA type, e.g., after or in response to the RA procedure initialization. If the selected RA type is an RA procedure with a two-step RA type, the wireless device may perform an RA resource selection for the two-step RA type. The wireless device may transmit an MsgA using the RA resource (e.g., a preamble of a preamble transmission, and/or time and frequency resource for preamble transmission and PUSCH transmission) selected by the RA resource selection. The wireless device may monitor a control channel for an RAR (e.g., MsgB) to the MsgA and a contention resolution. The wireless device may determine that the RA procedure with the two-step RA type is successfully completed, e.g., after or in response to receiving successRAR (e.g., MsgB indicating a success) of the MsgA. The wireless device may receive a fallbackRAR (e.g., MsgB indicating a fallback) of the MsgA. The wireless device may transmit an Msg3 and perform a contention resolution, e.g., after or in response to receiving the fallbackRAR. The wireless device may switch an RA type from the two-step RA type to a four-step RA type, e.g., after or in response to failing to receive a response to the MsgA and/or a response to the Msg3.

The wireless device may initialize variable(s) specific to an RA type, e.g., after or in response to the RA procedure initialization during which the wireless device selects the RA type. If the selected RA type is an RA procedure with a four-step RA type, the wireless device may perform an RA resource selection for the four-step RA type. The wireless device may transmit a preamble using the RA resource (e.g., a preamble of a preamble transmission, and/or time and frequency resource for preamble transmission) selected by the RA resource selection. The wireless device may monitor a control channel for an RAR (e.g., a response to the preamble). The wireless device may determine that the RA procedure with the four-step RA type is successfully completed, e.g., after or in response to receiving the RAR of the preamble. The wireless device may continue to perform the RA procedure with the four-step RA type, e.g., after or in response to receiving the RAR. For example, the wireless device may transmit an Msg3 via a radio resource indicated by an uplink grant of the RAR. The Msg3 may be for a contention resolution. The wireless device may determine that the RA procedure with the four-step RA type is successfully completed, e.g., after or in response to receiving a response to the Msg3.

A wireless device may perform an RA procedure initialization based on one or more messages (e.g., system information block(s), RRC message(s), MAC CEs, DCIs and/or any combination thereof) that the wireless device receives, e.g., from a base station. The one or more messages may configure one or more parameters for an RA procedure. The one or more parameters may comprise at least one of following parameters (e.g., names of the parameters may vary depending on a system and/or technology):

prach-ConfigurationIndex indicating that an available set of PRACH occasions for the transmission of the RA preamble for Msg1. PRACH configuration index may be applicable to the MsgA PRACH, e.g., if the PRACH occasions are shared between two-step and four-step RA types.

prach-ConfigurationPeriodScaling-IAB indicating a scaling factor (that may be applicable to a particular case, e.g., integrated access and backhaul (IAB)-MTs). The scaling factor may extend a periodicity of the PRACH occasions baseline configuration indicated by prach-ConfigurationIndex.

prach-ConfigurationFrameOffset-IAB indicating a frame offset (that may be applicable to a particular case, e.g., IAB-MTs) altering the ROs frame defined in the baseline configuration indicated by prach-ConfigurationIndex.

prach-ConfigurationSOffset-IAB indicating a subframe and/or slot offset (that may be applicable to a particular case, e.g., IAB-MTs) altering the ROs subframe and/or slot defined in the baseline configuration indicated by prach-ConfigurationIndex.

msgA-prach-ConfigurationIndex indicating an available set of PRACH occasions for the transmission of the RA preamble for MsgA in two-step RA type.

preambleReceivedTargetPower indicating an initial RA preamble power for four-step RA type.

msgA-PreambleReceivedTargetPower indicating an initial RA preamble power for two-step RA type.

rsrp-ThresholdSSB indicating an RSRP threshold for a selection of the SSB for four-step RA type. If the Random Access procedure is initiated for a particular case (e.g., beam failure recovery), RARP threshold SSB used for a selection of the SSB within a candidate beam RS list (e.g., candidateBeamRSList) may refer to RSRP threshold SSB (e.g., rsrp-ThresholdSSB) in Beam failure recovery configuration (e.g., BeamFailureRecoveryConfig) information element (IE).

rsrp-ThresholdCSI-RS indicating an RSRP threshold for a selection of CSI-RS for four-step RA type. If the Random Access procedure is initiated for a particular case (e.g., beam failure recovery), RSRP threshold CSI-RS (e.g., rsrp-ThresholdCSI-RS) may be equal to RSRP threshold SSB (e.g., rsrp-ThresholdSSB) in a beam failure recovery configuration (e.g., BeamFailureRecoveryConfig) IE.

msgA-RSRP-ThresholdSSB indicating an RSRP threshold for the selection of the SSB for two-step RA type.

rsrp-ThresholdSSB-SUL indicating an RSRP threshold for the selection between an NUL carrier and an SUL carrier.

msgA-RSRP-Threshold indicating an RSRP threshold for selection between two-step RA type and four-step RA type, e.g., if both two-step and four-step RA type Random Access Resources are configured in a UL BWP of a cell.

msgA-TransMax indicating a maximum number of MsgA transmissions when both four-step and two-step RA type Random Access Resources are configured.

candidateBeamRSList indicating a list of reference signals (CSI-RS and/or SSB) identifying the candidate beams for recovery and the associated Random Access parameters.

recoverySearchSpaceId indicating an identity of a search space for monitoring the response of the beam failure recovery request.

powerRampingStep indicating a power-ramping factor.

msgA-PreamblePowerRampingStep indicating a power ramping factor for MsgA preamble.

powerRampingStepHighPriority indicating a power-ramping factor in case of prioritized Random Access procedure.

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scalingFactorBI indicating a scaling factor for prioritized Random Access procedure.

ra-PreambleIndex indicating an index of a RA preamble.

ra-ssb-OccasionMaskIndex indicating PRACH occasion(s) associated with an SSB in which the MAC entity of a wireless device may transmit a RA preamble.

msgA-SSB-SharedRO-MaskIndex indicating a subset of four-step RA type PRACH occasions shared with two-step RA type PRACH occasions for each SSB. If two-step RA type PRACH occasions are shared with four-step RA type PRACH occasions and msgA-SSB-SharedRO-MaskIndex may not be configured. In this case, all four-step RA type PRACH occasions may be available for two-step RA type.

ra-OccasionList indicating PRACH occasion(s) associated with a CSI-RS in which the MAC entity may transmit a RA preamble.

ra-PreambleStartIndex indicating a starting index of RA preamble(s) for on-demand SI request.

preambleTransMax indicating a maximum number of RA preamble transmission.

ssb-perRACH-OccasionAndCB-PreamblesPerSSB indicating a number of SSBs mapped to each PRACH occasion for four-step RA type and a number of contention-based RA preambles mapped to each SSB;

msgA-CB-PreamblesPerSSB-PerSharedRO indicating a number of contention-based RA preambles for two-step RA type mapped to each SSB when the PRACH occasions are shared between two-step and four-step RA types.

msgA-SSB-PerRACH-OccasionAndCB-Preambles-PerSSB indicating a number of SSBs mapped to each PRACH occasion for two-step RA type and a number of contention-based RA preambles mapped to each SSB;

msgA-PUSCH-ResourceGroupA indicating MsgA PUSCH resources that a wireless device may use, e.g., when performing MsgA transmission using RA preambles group A.

msgA-PUSCH-ResourceGroupB indicating MsgA PUSCH resources that a wireless device may use, e.g., when performing MsgA transmission using RA preambles group B.

msgA-PUSCH-resource-Index indicating an index of the PUSCH resource used for MsgA in case of contention-free Random Access with two-step RA type;

In an example, during an RA procedure initialization, a wireless device may select a RA preamble group among one or more groups (e.g., group A and group B). For example, if a groupB configured is configured (by a message that the wireless device receives), an RA preambles group B is configured for a four-step RA type. For example, amongst the contention-based RA preambles associated with an SSB, the first numberOfRA-PreamblesGroupA RA preambles may belong to an RA preambles group A. The remaining RA preambles associated with the SSB may belong to the RA preambles group B (e.g., if configured). For example, if a groupB-ConfiguredTwoStepRA is configured (by a message that the wireless device receives), the RA preambles group B is configured for two-step RA type. For example, amongst the contention-based RA preambles for two-step RA type associated with an SSB, the first msgA-numberOfRA-PreamblesGroupA RA preambles may belong to the RA preambles group A. The remaining RA preambles associated with the SSB may belong to the RA preambles group B (if configured).

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For example, if an RA preambles group B is configured for a four-step RA type, a wireless device may determine at least one of following: ra-Msg3SizeGroupA indicating a threshold to determine the groups of RA preambles for the four-step RA type, msg3-DeltaPreamble (e.g., $\Delta_{PREAMBLE_Msg3}$) indicating a power offset for an Msg 3 transmission, messagePowerOffsetGroupB indicating a power offset for a preamble selection, and/or numberOfRA-PreamblesGroupA indicating a number of RA preambles in an RA preamble group A for each SSB. The wireless device may receive a message comprising the at least one.

For example, if an RA preambles group B is configured for a two-step RA type, a wireless device may determine msgA-DeltaPreamble (e.g., Δ_{MsgA_PUSCH}) indicating a power offset for MsgA transmission, msgA-messagePowerOffsetGroupB indicating a power offset for preamble selection configured as message PowerOffsetGroupB of GroupB-ConfiguredTwoStepRA, msgA-numberOfRA-PreamblesGroupA indicating a number of RA preambles in RA preamble group A for each SSB configured as numberOfRA-PreamblesGroupA of GroupB-ConfiguredTwoStepRA, and/or ra-MsgA-SizeGroupA indicating a threshold to determine the groups of RA preambles for the two-step RA type.

For example, if an RA preambles group B is configured for a two-step RA type, a wireless device may determine a set of RA preambles and/or PRACH occasions for SI request, if any, a set of RA preambles and/or PRACH occasions for beam failure recovery request, if any, a set of RA preambles and/or PRACH occasions for reconfiguration with sync, if any, ra-ResponseWindow indicating a time window (e.g., a time duration, a size of time window, a time interval, and/or the like) to monitor RA response(s), ra-ContentionResolutionTimer indicating a Contention Resolution Timer, msgB-ResponseWindow indicating a time window (e.g., a time duration, a size of time window, a time interval, and/or the like) to monitor RA response(s) for the two-step RA type.

In an example, a wireless device may receive a message indicating one or more RA preamble groups. The wireless device may select a particular RA preamble group among the one or more RA preamble groups, e.g., based on a size of TB that the wireless device transmits during an RA procedure. The wireless device may transmit one of preamble(s) associated with the particular RA preamble group. Selecting and/or transmitting the one of the preamble(s) may indicate, to a base station, a size of the TB.

In an example, a wireless device may receive a message indicating one or more RA preamble groups for an RA-based SDT. For example, each of the one or more RA preamble groups may be associated with at least one of sdt-TBS values in FIG. 18. For example, a first RA preamble group is associated with a first sdt-TBS value (e.g., 1st sdt-TBS of the K-th (or 1st) RA type in FIG. 18), a second RA preamble group is associated with a second sdt-TBS value (e.g., 2nd sdt-TBS of the K-th (or 1st) RA type in FIG. 18), and so on. An RA preamble group of the one or more RA preamble groups may be associated with one or more sdt-TBS values. The wireless device may determine a particular RA preamble group based on a sdt-TBS value that the wireless device generate a TB (e.g., MAC PDU) comprising uplink data (e.g., associated with DTCH). For example, in FIG. 18, the wireless device may generate the TB (e.g., MAC PDU) comprising the uplink data based on the second sdt-TBS value. The wireless device may select a particular RA preamble group (e.g., the second RA preamble group)

associated with the second sdt-TBS value and transmit one of preamble(s) in the particular RA preamble group.

In an example, a base station may configure one or more sdt-TBS values based on a size of (expected) message comprising uplink data that the base station allows the wireless device to transmit via an RA-based SDT. For example, the base station may determine the one or more sdt-TBS values based a size of uplink data, one or more MAC headers, and/or one or more MAC CEs. The uplink data may be associated with (e.g., may be received via) DTCH. The uplink data may not be associated with (e.g., may be received via) CCCH. The (expected) message comprising the uplink data may further comprise the one or more MAC headers, the one or more MAC CEs, and/or second uplink data associated with (e.g., may be received via) CCCH.

In example, a base station may configure one or more message sizes associated with one or more RA preamble groups for an RA procedure without SDT. For example, a wireless device may receive a message, from the base station, that configure the one or more RA preamble groups. The one or more RA preamble groups may comprise an RA preamble group A and an RA preamble group B. The wireless device may determine one of the one or more RA preamble groups based on a size of an Msg3 or MsgA payload (e.g., transport block and/or MAC PDU). For the RA procedure without SDT, the Msg3 or MsgA payload comprise uplink data, one or more MAC headers, and/or one or more MAC CEs. The uplink data may not be associated with (e.g., may not be received via) DTCH. The uplink data may be associated with (e.g., may be received via) CCCH. For example, the uplink data may comprise control message(s) of a higher layer (RRC layer, and/or NAS). For example, the uplink data may not comprise user data (e.g., MO data and/or MT data).

In an example, a base station may configure a plurality of RA preamble groups. The plurality of RA preamble groups may comprise first RA preamble groups for an RA-based SDT. The plurality of RA preamble groups may comprise second RA preamble groups for an RA procedure without SDT. The base station may determine the first RA preamble groups and the second RA preamble groups based on uplink data (e.g., associated with DTCH) that the wireless device transmits via the RA procedure. For example, the wireless device may transmit control message(s) of a higher layer (RRC layer and/or NAS) via the RA procedure without SDT. For example, the wireless device may transmit user data (MO and/or MT data) via the RA procedure with SDT (e.g., via the RA-based SDT). A size of TB transmitted by the wireless device via the RA-based SDT may be larger than that of TB transmitted via the RA procedure without SDT.

In an example, a wireless device may select a particular RA group among one or more RA groups (e.g., configured for an RA-based SDT). The wireless device may select one of RA preambles in the particular RA group in an RA resource selection for an two-step RA type and/or in an RA resource selection for a four-step RA type. For example, the wireless device may select a particular sdt-TBS value, e.g., among one or more sdt-TBS values, for the RA-based SDT. The wireless device may select a particular RA group associated with the particular sdt-TBS value. The association between the one or more RA groups and the one or more sdt-TBS values may be configured by a message (e.g., SIB) that the wireless device receives from a base station. The one of RA preambles may be selected from the particular RA group associated with the particular sdt-TBS value. The one or more RA groups configured for the Ra-based SDT may be

different between a two-step RA type and a four-step RA type. The one or more RA groups configured for the Ra-based SDT may be shared between a two-step RA type and a four-step RA type.

The RA procedure initialization may comprise initializing one or more variables and/or buffers. For example, a wireless device may initiate an RA procedure on a cell (e.g., for an RA-based SDT). The wireless device may flush the Msg3 buffer, and/or flush the MsgA buffer, e.g., after or in response to initiating the RA procedure. The wireless device may determine one or more initial values of variables used for the RA procedure, e.g., after or in response to initiating the RA procedure. For example, the wireless device may set the PREAMBLE_TRANSMISSION_COUNTER to 1, set the PREAMBLE_POWER_RAMPING_COUNTER to 1, set the PREAMBLE_BACKOFF to 0 ms, and/or set POWER_OFFSET_2_STEP_RA to 0 dB.

A wireless device may initiate an RA procedure (e.g., for an RA-based SDT) on a particular carrier of a cell. For example, the cell may comprise an NUL carrier and an SUL carrier. The wireless device may receive an indication or configuration parameter(s) indicating a particular carrier to use for the RA procedure. The wireless device may select the particular carrier for the RA procedure. The wireless device may not receive the indication or the configuration parameter(s). In this case, the wireless device may determine a carrier to use for the RA procedure based on an RSRP of a downlink pathloss reference. For example, if the RSRP of the downlink pathloss reference is less than $\text{rsrp-Threshold-SSB-SUL}$, the wireless device may select the SUL carrier for performing Random Access procedure, and/or set the PCMAX to $P_{\text{CMAX},fc}$ of the SUL carrier. For example, if the RSRP of the downlink pathloss reference is larger than or equal to $\text{rsrp-ThresholdSSB-SUL}$, the wireless device may select the NUL carrier for performing Random Access procedure.

A wireless device may determine an RA type of an RA procedure (e.g., for an RA-based SDT) in an RA procedure initialization, according to some embodiments. For example, the wireless device may use one or more conditions to select the RA type. For example, the one or more conditions comprise an RSRP value of a downlink pathloss reference. For example, the one or more conditions comprise a size to a TB (e.g., a MAC PDU of the TB) comprising uplink data (e.g., associated with DTCH) that the wireless device transmits via the RA-based SDT. For example, the one or more conditions comprise a number of padding bit(s) to be appended to the TB.

For example, a wireless device may select a two-step RA type, e.g., if a BWP (of a cell) selected for Random Access procedure is configured with Random Access Resources for the two-step RA type. For example, a wireless device may select a two-step RA type, e.g., an RSRP value of the downlink pathloss reference is above $\text{msgA-RSRP-Threshold}$. For example, a wireless device may select the two-step RA type, e.g., if the wireless device may receive an indication to initiate the two-step RA type. The wireless device may use one or more conditions for the RA type selection. For example, the wireless device may select the two-step RA type, e.g., if a BWP selected for RA procedure is configured with RA resource(s) for the two-step RA type, and/or if the RSRP value of the downlink pathloss reference is above $\text{msgA-RSRP-Threshold}$.

For example, a wireless device may select a four-step RA type, e.g., if one or more conditions for selecting a two-step RA type do not satisfy. For example, a wireless device may select a four-step RA type, e.g., if a BWP (of a cell) selected

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for Random Access procedure is configured with Random Access Resources for four-step RA type. For example, a wireless device may select a four-step RA type, e.g., if a BWP (of a cell) selected for Random Access procedure is not configured with Random Access Resources for two-step RA type. For example, a wireless device may select a four-step RA type, e.g., an RSRP value of the downlink pathloss reference is smaller than or equal to msgA-RSRP-Threshold. For example, a wireless device may select a four-step RA type, e.g., if the wireless device may receive an indication to initiate the four-step RA type. The wireless device may use one or more conditions for the RA type selection. For example, the wireless device may select a four-step RA type, e.g., if a BWP selected for RA procedure is configured with RA resource(s) for the four-step RA type, and/or if the RSRP value of the downlink pathloss reference is smaller than or equal to msgA-RSRP-Threshold.

The wireless device may perform an initialization of variables specific to Random Access type. For example, if the wireless device selects a two-step RA type, the wireless device may perform a Random Access Resource selection procedure for the two-step RA type. For example, if the wireless device selects a four-step RA type, the wireless device may perform a Random Access Resource selection procedure for the four-step RA type.

In an example, in an initialization of variables specific to Random Access type, a wireless device may determine one or more values of variables for a selected RA type of an RA procedure. For example, the one or more values may be initial values required for the selected RA type of the RA procedure. For example, the wireless device may determine at least one value of the one or more values as a predefined value. For example, the wireless device may receive a message comprising at least one value of the one or more values.

In an example, a wireless device may initiate an RA procedure with a two-step RA type. The wireless device may switch an RA type of the RA procedure from the two-step RA type to a four-step RA type. The wireless device may perform the initialization of variables specific to the four-step RA type, e.g., after or in response to switching to the four-step RA type. For example, if a wireless device determines to switch an RA type from a two-step RA type to a four-step RA type, during the RA procedure, the wireless device may determine set $\text{POWER_OFFSET_2_STEP_RA to (PREAMBLE_POWER_RAMPING_COUNTER-1) \times (MsgA_PREAMBLE_POWER_RAMPING_STEP-PRE-AMBLE_POWER_RAMPING_STEP)}$.

One or more RA resources may be configured for the selected RA type for the RA-based SDT. Each of the one or more RA resources may comprise particular preamble(s), particular PRACH resource(s) (e.g., occasions(s)), and/or particular PUSCH resource(s) (e.g., occasions(s)). At least one of the one or more RA resources may be configured for the selected RA type of the RA-based SDT. For example, the at least one may be associated with at least one sdt-TBS value of the selected RA type of the RA-based SDT.

A transport block (TB) may be a payload comprising data to be transmitted or to be received. For an uplink transmission, a MAC layer of a wireless device may send a TB comprising data to a physical layer of the wireless device. The physical layer may attach a CRC to the TB. The physical layer may segment the TB into one or more code blocks. The physical layer may map the TB (e.g., with the attached CRC) or the segmented one or more code blocks to resource(s) or onto a channel (e.g., PUSCH) for the uplink transmission. For a downlink transmission, a wireless device

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may receive data via a channel (e.g., PDSCH). The physical layer of the wireless device may send the received data (or data decoded) to the MAC layer of the wireless device in the form of the TB.

In an example, a TB may comprise a MAC PDU. The MAC packet data unit (PDU) may comprise a MAC service data unit (SDU). The MAC SDU may comprise data that a wireless device transmits or receives. For example, for an uplink transmission, the wireless device may generate (e.g., multiplex, assemble, construct, and/or obtain) a MAC SDU comprising uplink data (e.g., associated with DTCH). A MAC layer of the wireless device may generate a MAC PDU comprising the MAC SDU that comprises the uplink data. The MAC layer may send a TB comprising the MAC PDU that comprises the uplink data to a physical layer of the wireless device. The physical layer may add (or attach) a CRC to the TB. The physical layer may transmit the TB via PUSCH. A processing for downlink transmission may be a reverse order of the one for the uplink transmission. For example, the physical layer may receive the TB via PDSCH. The physical layer may send the received TB to the MAC layer. The received TB may comprise at least one MAC PDU. The MAC layer may demultiplex the at least one MAC PDU into one or more MAC SDUs. At least one of the one or more MAC SDUs may comprise downlink data.

The wireless device may determine a size of the TB to transmit the TB (e.g., via PUSCH) or to receive the TB (via PDSCH). The size of the transport block may be referred to as a transport block size (TBS). A wireless device may receive an (explicit or implicit) indication of a size of a transport block for transmission of uplink data and/or signal (s).

For example, a wireless device may receive one or more messages (e.g., SIB and/or RRC message) comprising configuration parameter(s). For example, the configuration parameter(s) may indicate a TBS (e.g., an explicit indication). For example, the wireless device may determine the TBS based on the configuration parameters (e.g., implicit indication). For example, the configuration parameters may indicate at least one of a modulation order, code rate, number of RBs, a transmission duration, and/or number of transmission layers. One or more of modulation order, code rate, number of RBs, a transmission duration, and/or number of transmission layers may be predefined.

An indication of a TBS may be based on a table. For example, the TBS may be tabulated as a function of one or more transmission parameters (e.g., MCS). For example, the wireless device may receive an uplink grant comprising the one or more transmission parameters (e.g., MCS). The wireless device may determine the TBS based on the one or more transmission parameters using the table. For example, the table may map the one or more transmission parameters to a particular value of TBS. The table may be configured by message(s) received from a base station. The table may be predefined as a function of the one or more transmission parameters.

An indication of a TBS may be based on a formula. For example, a wireless device may determine the TBS as a function of one or more transmission parameters (e.g., MCS). For example, the wireless device may receive an uplink grant comprising the one or more transmission parameters. The wireless device may determine the TBS based on the one or more transmission parameters using the formula. For example, the formula may be a function of the one or more transmission parameters. The output of the formula may be a particular value of TBS. The formula may be configured by message(s) received from a base

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station. The formular may be predefined as a function of the one or more transmission parameters.

A wireless device may determine a modulation order and/or a code rate. The wireless device may receive configuration message (e.g., RRC message) and/or DCI that comprise an MCS field. For example, the MCS fields is a x-bits fields (e.g. x=5). For example, the wireless device may use a table (predefined, configured, or combination thereof), to map a value of the MCS fields to a particular combination of modulation scheme and channel-coding rate. Of the 2^x combination of the x-bit MCS fields, at least one may be used to signal a modulation-and-coding scheme. For example, each of the at least one may represent a particular combination of modulation scheme and channel-coding rate and/or a certain spectral efficiency measured in the number of information bits per modulation symbol.

A wireless device may determine a TBS based on a table and/or a formular. The wireless device may receive one or more message (RRC message, MAC CE, DCI, and/or any combination thereof) that indicate parameters for uplink transmission. For example, the parameters comprise a modulation order (indicated by an MCS field) for the uplink transmission, a number of resource blocks scheduled for the uplink transmission, and/or transmission duration scheduled for the uplink transmission. For example, the wireless device may determine a number of available resource elements (REs) for the uplink transmission based on the parameters. The wireless device may determine a resulting estimate of REs available for data by subtracting overhead from the number of available REs. For example, from the number of available REs, the wireless device may subtract the resource element(s) used for DM-RS (CSI-RS and/or SRS). The wireless device may use the resulting estimate of REs available for data to determine an intermediate number of information bits. For example, the wireless device may use the resulting estimate of REs, a number of transmission layers, the modulation order, and the code rate to determine the TBs.

FIG. 26 illustrates a determination of TBS according to some embodiments. The wireless may receive one or more configuration parameters for uplink transmission. The one or more configuration parameters may comprise one or more fields indicating a modulation order, code rate, a number of RBs, and/or a number of transmission layers for the uplink transmission. The wireless device may receive the configuration parameters via one or more messages (e.g., RRC message, MAC CE, DCI, or any combination thereof). One or more of a modulation order, code rate, a number of RBs, and/or a number of transmission layers may be predefined. The wireless device may determine, based on at least the modulation order and code rate, number of RBs and transmission duration, and/or number of transmission layers, an intermediate number of information bits. The wireless device may determine the TBS by quantizing the intermediate number of information bits.

What is claimed is:

1. A method comprising:

receiving, by a wireless device, configuration parameters indicating:

a first transport block (TB) size threshold of a first random access (RA) type; and

a second TB size threshold of a second RA type;

selecting, for transmitting a data unit and among the first RA type and the second RA type, an RA type based on a comparison among a size of the data unit, the first TB size threshold of the first RA type, and the second TB size threshold of the second RA type;

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multiplexing the data unit with one or more padding bits of the RA type onto a TB, wherein a quantity of the one or more padding bits of the RA type is determined based on a difference between the size of the data unit and a TB size threshold corresponding to the RA type; and

transmitting, using an RA procedure associated with the RA type, the TB comprising the data unit.

2. The method of claim 1, wherein:

the first RA type comprises a two-step RA type; and

the second RA type comprises a four-step RA type.

3. The method of claim 1, wherein:

the first RA type comprises a four-step RA type; and

the second RA type comprises a two-step RA type.

4. The method of claim 1, wherein the configuration parameters further indicate that the first RA type and the second RA type are available for a small data transmission in a radio resource control (RRC) inactive state, and wherein the selecting the RA type is in response to the first RA type and the second RA type being available for the small data transmission.

5. The method of claim 1, further comprising initiating, for a small data transmission in a radio resource control (RRC) inactive state, the RA procedure associated with the RA type.

6. The method of claim 1, wherein the second TB size threshold is larger than the first TB size threshold, wherein the RA type comprises the first RA type in response to the size of the data unit being smaller than or equal to the first TB size threshold, wherein the RA type comprises the second RA type in response to the size of the data unit being: larger than the first TB size threshold; and smaller than or equal to the second TB size threshold.

7. The method of claim 1, further comprising:

in response to the RA type being a two-step RA type, transmitting the TB that the data unit is multiplexed onto as Message A of the two-step RA type; and

in response to the RA type being a four-step RA type, transmitting the TB that the data unit is multiplexed onto as Message 3 of the four-step RA type.

8. The method of claim 1, wherein the selecting is further based on:

a first quantity, of one or more padding bits of the first RA type, determined based on the size of the data unit and the first TB size threshold; and

a second quantity, of one or more padding bits of the second RA type, determined based on the size of the data unit and the second TB size threshold.

9. The method of claim 1, wherein transmitting the TB is based on multiplexing the data unit with the one or more padding bits onto the TB; and wherein the TB size threshold being among the first TB size threshold and the second TB size threshold.

10. A wireless device comprising:

one or more processors; and

memory storing instructions, that when executed by the one or more processors, cause the wireless device to: receive configuration parameters indicating:

a first transport block (TB) size threshold of a first random access (RA) type; and

a second TB size threshold of a second RA type;

select, for transmitting a data unit and among the first RA type and the second RA type, an RA type based on a comparison among a size of the data unit, the first TB size threshold of the first RA type, and the second TB size threshold of the second RA type;

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5 multiplex the data unit with one or more padding bits of the RA type onto a TB, wherein a quantity of the one or more padding bits of the RA type is determined based on a difference between the size of the data unit and a TB size threshold corresponding to the RA type; and

transmit, using an RA procedure associated with the RA type, the TB comprising the data unit.

11. The wireless device of claim 10, wherein:

the first RA type comprises a two-step RA type; and

the second RA type comprises a four-step RA type.

12. The wireless device of claim 10, wherein:

the first RA type comprises a four-step RA type; and

the second RA type comprises a two-step RA type.

13. The wireless device of claim 10, wherein the configuration parameters further indicate that the first RA type and the second RA type are available for a small data transmission in a radio resource control (RRC) inactive state, and wherein the RA type is selected in response to the first RA type and the second RA type being available for the small data transmission.

14. The wireless device of claim 10, wherein the instructions further cause the wireless device to initiate, for a small data transmission in a radio resource control (RRC) inactive state, the RA procedure associated with the RA type.

15. The wireless device of claim 10, wherein the instructions further cause the wireless device to determine a size of a message comprising the data unit is smaller than or equal to at least one of:

the first TB size threshold; and

the second TB size threshold.

16. The wireless device of claim 10, wherein the second TB size threshold is larger than the first TB size threshold, wherein the RA type comprises the first RA type in response to the size of the data unit being smaller than or equal to the first TB size threshold, wherein the RA type comprises the second RA type in response to the size of the data unit being:

larger than the first TB size threshold; and

smaller than or equal to the second TB size threshold.

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17. The wireless device of claim 10, wherein the instructions further cause the wireless device to:

in response to the RA type being a two-step RA type, transmit the TB that the data unit is multiplexed onto as Message A of the two-step RA type; and

in response to the RA type being a four-step RA type, transmit the TB that the data unit is multiplexed onto as Message 3 of the four-step RA type.

18. The wireless device of claim 10, wherein the instructions further cause the wireless device to select the RA type further based on:

a first quantity, of one or more padding bits of the first RA type, determined based on the size of the data unit and the first TB size threshold; and

a second quantity, of one or more padding bits of the second RA type, determined based on the size of the data unit and the second TB size threshold.

19. The wireless device of claim 10, wherein to transmit the TB is based on the multiplexing the data unit with the one or more padding bits onto the TB; and wherein the TB size threshold being among the first TB size threshold and the second TB size threshold.

20. A non-transitory computer-readable medium comprising instructions that, when executed by one or more processors of a wireless device, cause the wireless device to:

receive configuration parameters indicating:

a first transport block (TB) size threshold of a first random access (RA) type; and

a second TB size threshold of a second RA type;

select, for transmitting a data unit and among the first RA type and the second RA type, an RA type based on a comparison among a size of the data unit, the first TB size threshold of the first RA type, and the second TB size threshold of the second RA type;

multiplex the data unit with one or more padding bits of the RA type onto a TB, wherein a quantity of the one or more padding bits of the RA type is determined based on a difference between the size of the data unit and a TB size threshold corresponding to the RA type; and

transmit, using an RA procedure associated with the RA type, the TB comprising the data unit.

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